7 Natural Selection MCQ

1. All of the following are examples of prezygotic genetic isolating mechanisms EXCEPT:

(A) Male fireflies of different species have differing flash patterns.

(B) Three closely related orchid species flower on different days.

(C) The progeny of a cross between two different lizard species fail to develop properly. ✓

(D) One species of snake lives in water and another is terrestrial.

(E) Peeper frogs mate in April, and tree frogs mate in June.

2. All of the following were likely present on the primitive Earth during the evolution of self-replicating molecules EXCEPT

(A) amino acids and nucleotides

(B) nitrogen

(C) simple carbohydrates

(D) freestanding liquid water

(E) an O_2-rich atmosphere ✓

3.

TABLE 1. PERCENT HOMOLOGY OF $MEF2A$ NUCLEOTIDE SEQUENCES

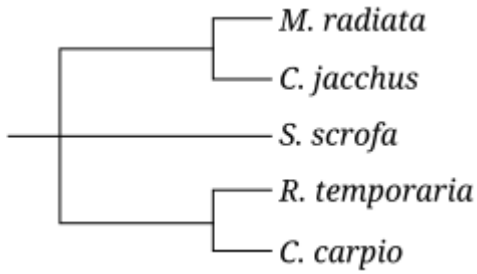
Species	*M. radiata*	*C. jacchus*	*S. scrofa*	*C. carpio*	*R. temporaria*
M. radiata		90.3	71.5	42.7	50.9
C. jacchus			70.6	42.1	50.5
S. scrofa				40.8	48.9
C. carpio					44.5
R. temporaria					

The $MEF2A$ gene is critical for muscle development in many vertebrates. A researcher sequenced the $MEF2A$ gene from five vertebrate species and calculated the percent homology between the nucleotide sequences from each pair of species (Table 1).

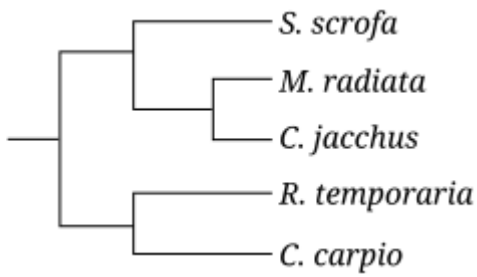
Which of the following cladograms is the most accurate representation of the data in Table 1 ?

7 Natural Selection MCQ

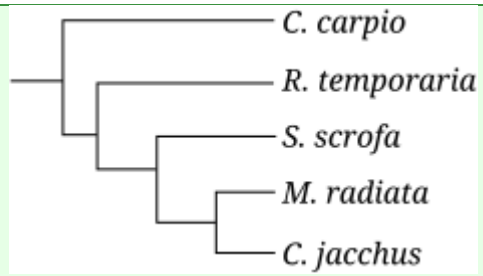
(A)



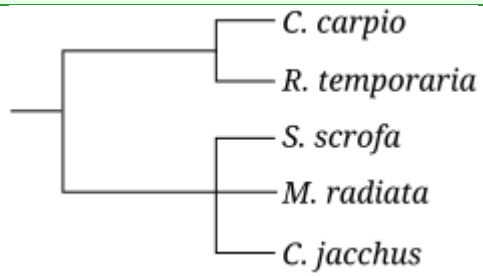
(B)



(C) ✓



(D)



Answer C

Correct. This cladogram assumes that *M. radiata* is the most recently evolved species and has thus undergone the most genetic changes. *M. radiata* also shares the most sequence homologies with *C. jacchus*, and this is indicated in the cladogram. As other species are compared with *M. radiata* and the level of homology decreases, it is assumed that these species are more closely related to the most recent common ancestor.

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4. A small number of lizards from a mainland population have been deposited on four isolated islands because of the effects of a rare strong storm.

Which of the following best predicts the outcome of these lizards reproducing for many generations on the islands?

(A) Courtship rituals specific to each island lizard species prevent the lizards from interbreeding.

(B) Speciation results from bottleneck events that happened before the ancestral species reached the islands.

(C) The different species that currently exist are the result of hybridization between lizards from different islands.

(D) The isolation prevents gene flow; thus, the lizards on different islands experience prezygotic isolation. ✓

Answer D

Correct. The primary cause of speciation is the prevention of gene flow between different populations. In this case, the original lizard populations were geographically isolated because they were on different islands, which prevented migration. Each island's *Anolis* species radiated from a similar-appearing common ancestor that underwent speciation in isolation.

5. A number of different phylogenies (evolutionary trees) have been proposed by scientists. These phylogenies are useful because they can be used to

(A) determine when two similar populations of a species evolved into two separate species

(B) evaluate which groups of organisms may be most closely related ✓

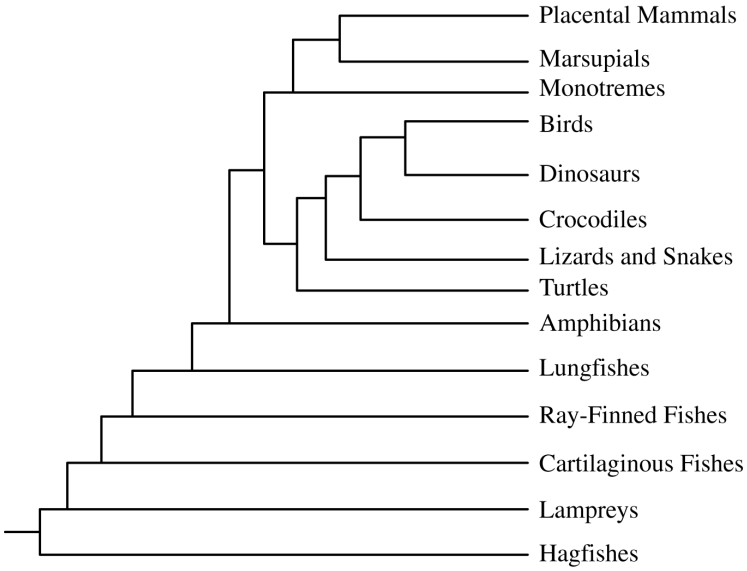
(C) demonstrate that all photosynthetic organ-isms are members of the Kingdom Plantae

(D) demonstrate that natural selection is the prevailing force in evolution

(E) demonstrate which taxa (groups of organisms) contain the most highly evolved species

7 Natural Selection MCQ

6.



To determine the evolutionary history and relationships among organisms, scientists gather evidence from a wide variety of sources including paleontology, embryology, morphology, behavior, and molecular biology. A phylogenetic tree of vertebrates is shown.

Which of the following statements is most consistent with the phylogenetic tree shown?

(A) Birds and turtles evolved their own means of gas exchange independently of the other vertebrates.

(B) Mammals are most closely related to birds because they share a direct common ancestor.

(C) The common ancestor of reptiles, birds, and mammals produced amniotic eggs. ✓

(D) Crocodiles are direct descendents of ray-finned fishes since they live in the same environment.

7. A researcher hypothesizes that RNA molecules were present in the most recent common ancestor of all living organisms. Which of the following scientific questions would best test the hypothesis?

(A) Is it possible to produce an RNA polymer in a laboratory setting?

(B) How many distinct functions can a particular RNA molecule perform in a cell?

(C) How many different monomers of RNA are found in a eukaryotic cell?

(D) Do any known organisms function entirely without RNA? ✓

7 Natural Selection MCQ

Ascorbic acid (vitamin C) is an organic molecule necessary for the health of plants and animals. The majority of animals, including most mammals, synthesize ascorbic acid from organic precursors, but some primates are unable to synthesize ascorbic acid and must instead acquire it from dietary sources, such as certain fruits and vegetables.

The *L-gulonolactone oxidase (GULO)* gene encodes an enzyme that catalyzes a required step in the biosynthesis of ascorbic acid. Most mammals carry a functional copy of the *GULO* gene, but some primates carry only a *GULO* pseudogene, which is a nonfunctional variant.

A comparison of *GULO* genes and *GULO* pseudogenes from different animals can provide insight into the evolutionary relatedness of the animals. In Table I, selected members of some mammalian groups are listed, along with an indication of their ability to synthesize ascorbic acid. Table II shows an alignment of amino acid coding sequences from homologous regions of the *GULO* genes and *GULO* pseudogenes of the organisms listed in Table I. Figure 1 represents the universal genetic code.

TABLE I: SELECTED MAMMALIAN GROUPS

Group	Selected Members	Biosynthesis of Ascorbic Acid
Nonprimate mammals	Elephant, mouse	Yes
Primate mammals	Lemur	Yes
	Orangutan, chimpanzee	No
	Human	No

TABLE II: DNA SEQUENCE ALIGNMENT*

	Relative Positions of Nucleotides in Nontemplate (Coding) Sequence																									
	1 (5′)	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27 (3′)
Elephant	G	A	C	A	C	C	C	A	T	C	T	G	A	A	G	A	A	G	T	C	G	G	A	A	T	A	C
Mouse	G	A	C	A	G	C	C	A	C	C	T	G	A	A	G	A	A	G	T	C	T	G	A	G	T	A	C
Lemur	G	A	C	A	G	C	C	A	C	C	T	G	A	A	G	A	G	G	T	C	C	G	A	G	T	A	C
Orangutan	G	A	C	A	G	C	-	A	T	T	G	G	A	A	G	A	A	A	T	C	T	G	A	G	G	A	C
Chimp	G	A	C	A	G	C	-	A	T	T	G	G	A	A	G	A	A	A	T	C	T	G	A	G	G	A	C
Human	G	A	C	A	G	C	-	A	T	T	G	G	A	A	G	A	A	A	T	C	T	G	A	G	G	A	C

*For each DNA segment, the alternating shaded and unshaded nucleotides indicate the triplet codons of the open reading frame, shown from left (5′) to right (3′) as the nontemplate (coding) strand. An "–" indicates the absence of a nucleotide.

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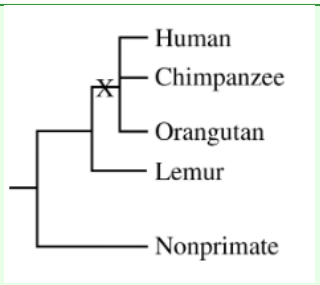
Second Base

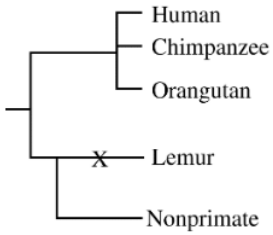
5′ First Base	U	C	A	G	3′ Third Base
U	UUU Phe	UCU Ser	UAU Tyr	UGU Cys	U
	UUC Phe	UCC Ser	UAC Tyr	UGC Cys	C
	UUA Leu	UCA Ser	UAA stop	UGA stop	A
	UUG Leu	UCG Ser	UAG stop	UGG Trp	G
C	CUU Leu	CCU Pro	CAU His	CGU Arg	U
	CUC Leu	CCC Pro	CAC His	CGC Arg	C
	CUA Leu	CCA Pro	CAA Gln	CGA Arg	A
	CUG Leu	CCG Pro	CAG Gln	CGG Arg	G
A	AUU Ile	ACU Thr	AAU Asn	AGU Ser	U
	AUC Ile	ACC Thr	AAC Asn	AGC Ser	C
	AUA Ile	ACA Thr	AAA Lys	AGA Arg	A
	AUG Met	ACG Thr	AAG Lys	AGG Arg	G
G	GUU Val	GCU Ala	GAU Asp	GGU Gly	U
	GUC Val	GCC Ala	GAC Asp	GGC Gly	C
	GUA Val	GCA Ala	GAA Glu	GGA Gly	A
	GUG Val	GCG Ala	GAG Glu	GGG Gly	G

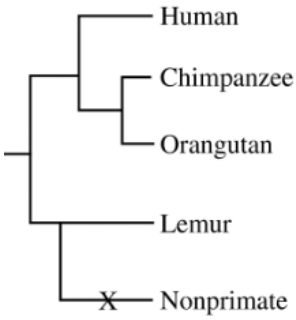
Figure 1. Universal genetic code.

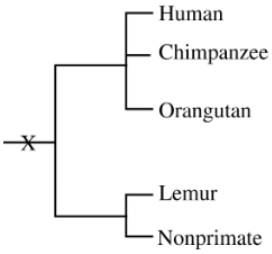
8. Which of the following phylogenetic trees best illustrates (with the X) the point at which the mutation in the *GULO* gene most likely occurred during the evolutionary history of these organisms?

7 Natural Selection MCQ

(A) 

(B) 

(C) 

(D) 

A biologist spent many years researching the rate of evolutionary change in the finch populations of a group of islands. It was determined that the average beak size (both length and mass) of finches in a certain population increased dramatically during an intense drought between 1981 and 1987. During the drought, there was a reduction in the number of plants producing thin-walled seeds.

9. Which of the following procedures was most likely followed to determine the change in beak size?

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(A) A few finches were trapped in 1981 and again in 1987, and their beak sizes were compared.

(B) The beak size in fifteen finches was measured in 1987, and the beak size in the original finches was determined by estimation.

(C) The beak size in a large number of finches was measured every year from 1981 to 1987. ✓

(D) Finches were captured and bred in 1981, and the beak size of the offspring was measured.

Answer C

This option is correct. It demonstrates ability to apply the scientific method to evaluate data that illustrate evolution as an ongoing process. Based on observation, the beak sizes of a large number of individuals were measured between the time interval (1981 to 1987) correlating with the period of drought.

10. The biologist discovered that from 1988 to 1993, the average beak size declined to pre-1981 levels. The reversal in beak size from 1988 to 1993 was most likely related to which of the following events?

(A) A loss of food supply for the finches

(B) The end of the drought ✓

(C) An increase in drought conditions

(D) An increase in predators consuming finches

Answer B

This option is correct. It demonstrates an ability to evaluate data providing evidence for evolution as an ongoing process. At the end of the drought period, there was a natural return to a variety of seeds available — including a decrease in thick-walled seeds. Thus, allele frequencies shifted to pre-drought numbers because no longer was there selective pressure favoring larger beak sizes.

11. Which of the following statements might best explain the increase in average beak size in the finch population during the drought?

(A) Finches with bigger beaks are better able to crack thick-walled seeds and produce more surviving offspring. ✓

(B) Finches with bigger beaks can attack and kill finches with smaller beaks.

(C) Finches with bigger beaks possess more powerful flight muscles and are able to find more food.

(D) Finches that crack large seeds develop larger beaks over time.

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Answer A

This option is correct. It demonstrates understanding of evolution through natural selection and explains the relationship between changes in genotype, phenotype, and environment. During the drought period, plants with thick- walled seeds were favored; in turn, finches with larger beaks that could crack the seeds to obtain nourishment were selected for, thus increasing chances for survival.

12. Which of the following best explains the mechanism behind the change in beak size in the finch population?

(A) The formation of two new finch species from a single parent species

(B) A change in gene frequencies in the finch population due to selective pressure from the environmental change ✓

(C) A new allele appearing in the finch population as a result of mutation

(D) The achievement of dynamic equilibrium in the finch population as a result of homeostasis

Answer B

This option is correct. It demonstrates an understanding of the role of natural selection in evolution. Natural selection acts on phenotypic variations in populations. During the drought period the environment favored plants with thick-walled seeds, and finches with larger beaks had a selective advantage over finches with smaller beaks in using the thick-walled seeds as food. More food led to greater reproductive success for the large-beaked finches. Thus, allele frequencies within the finch population changed, with an increase in alleles conferring the more advantageous phenotype — larger beaks. Changes in allele frequencies in a population provide evidence for evolution.

---

13. A widely accepted hypothesis about the origin of life on Earth is that life arose approximately 3.5 billion years ago as the result of a complex sequence of chemical reactions that took place spontaneously in Earth's atmosphere. Another hypothesis about the origin of life suggests that life began somewhere else in the universe and arrived on Earth by chance.

Which of the following questions might scientists ask to most reliably determine if there has ever been life on Mars?

(A) How far from Earth was Mars 3.5 billion years ago?

(B) Is there scientific proof that Earth ever had an oxygen-free atmosphere?

(C) Did Mars have the elements present in its atmosphere to support life? ✓

(D) Was there oxygen gas on Mars 3.5 billion years ago?

7 Natural Selection MCQ

14. Commercial bananas are grown as a monoculture, with all banana plants cloned from one original banana plant. The commercial strains of bananas are seedless, so each new banana plant has to be manually planted from a cutting of an existing banana root. In the 1950s, the Gros Michel banana strain, the dominant export banana at that time, was destroyed by the fungus *Fusarium oxysporum*. A new *Fusarium* resistant variety, the Cavendish banana, was developed and is currently the banana strain grown for export. Recently, a *Fusarium* strain that successfully attacks the Cavendish strain has been documented.

Which of the following best provides reasoning supporting a method that would help protect commercial banana crops from infection by pathogenic organisms such as *Fusarium* fungi?

(A) The commercial banana strains should be exposed to X-rays to encourage random mutations that will then be passed to offspring, producing resistance to pathogenic organisms.

(B) The Cavendish banana plants should be exposed to pathogenic organisms under controlled conditions, so the plants can be encouraged to mutate and develop resistance to the pathogens.

(C) The commercial banana strains should not be grown in monocultures, since many pathogenic organisms are able to evolve rapidly in response to a single selective pressure. ✓

(D) Growing the Cavendish strain under different conditions will allow natural selection to produce the variation needed to resist infection by pathogenic organisms.

Answer C

Correct. The commercial banana strains are clones that lack both genetic diversity and the ability to generate diversity through sexual reproduction. Any single clone grown in a monoculture will not prevent selection for a pathogen able to successfully infect the plants.

15. The condition in which there are barriers to successful interbreeding between individuals of different species in the same community is referred to as

(A) latent variations

(B) sterility

(C) structural differences

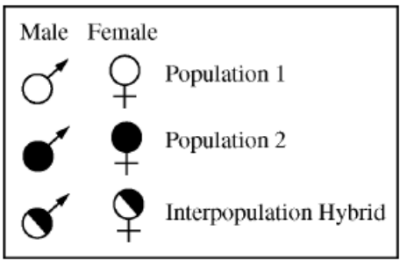
(D) geographic isolation

(E) reproductive isolation ✓

7 Natural Selection MCQ

Over many generations, two populations of the same species can diverge into separate species through reproductive isolation. The figures below represent a model of speciation and show the results of matings between individuals from two diverging populations at four different stages of speciation. The males represented in the model are heterogametic, which means they have two different sex-determining chromosomes, (e.g., XY). The females are homogametic, which means they have two similar sex-determining chromosomes, (e.g., XX). The offspring from each mating are labeled interpopulation hybrids.

Figure 1 shows the results of a mating between a male and a female from the two populations. In each subsequent figure, the males are from one of the diverging populations and the females are from the other population. The fertility and viability of the offspring from each mating are indicated in the figures.



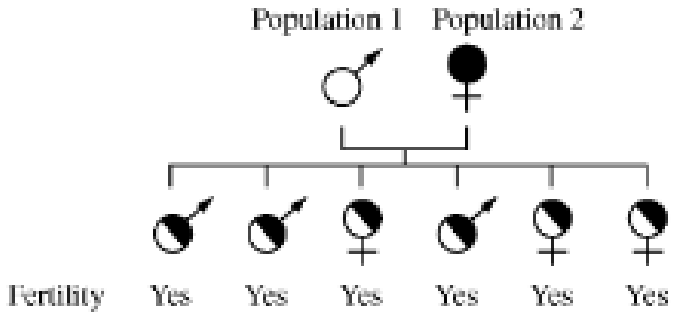


Figure 1. Mating between individuals from the initial populations

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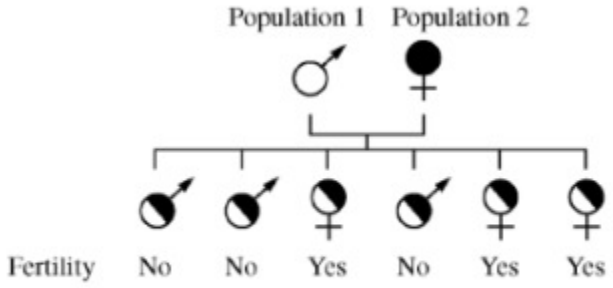


Figure 2. Mating between individuals from diverging populations at an intermediate stage of speciation

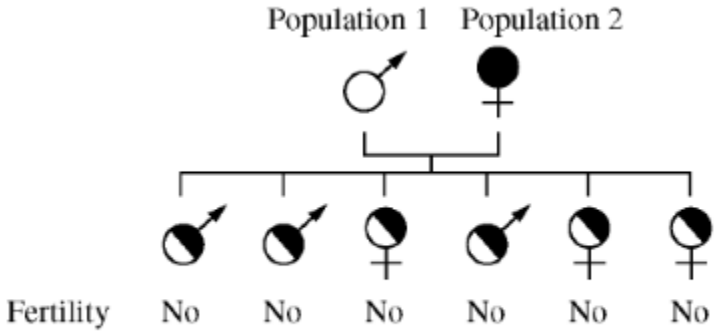


Figure 3. Mating between individuals from diverging populations at a late stage of speciation

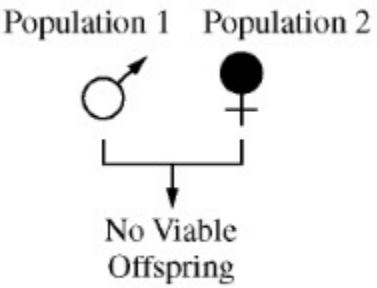


Figure 4. Mating between individuals from divergent populations after speciation is complete

16. Based on the model of speciation presented, which of the following describes the most likely consequence to the populations over time?

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(A) Hybrid individuals are less likely to pass their genetic information on to subsequent generations. ✓

(B) Hybrid individuals living together are physically unable to mate with each other.

(C) Geographically isolated populations gradually acquire similar heritable traits.

(D) Differences between hybrid individuals of a species disappear over time.

17. In a separate investigation, individual mice from two populations that in nature are geographically isolated from each other are mated in the laboratory. The hybrid offspring were then mated with individuals from either of the original populations. Only the female hybrid offspring were fertile. The experimental results are most consistent with which of the stages that are depicted in the model?

(A) Initial population (Figure 1)

(B) Intermediate stage (Figure 2) ✓

(C) Late stage (Figure 3)

(D) Terminal stage (Figure 4)

18. Which of the following best describes the reason for excluding hybrid males when calculating the allele frequencies of two interbreeding populations at the intermediate stage of speciation (Figure 2) ?

(A) The process of mate selection in large groups often favors females over males.

(B) The frequency of sex-determining chromosomes is usually equal to 0.5.

(C) Sterile individuals make no genetic contribution to the next generation. ✓

(D) The chance of inheriting a recessive allele from a male is too small to calculate.

19. Using the model of speciation and applying it to a different population, which of the following outcomes is most consistent for a different species in which the males are homogametic and the females are heterogametic?

(A) Sterility would appear in females before appearing in males. ✓

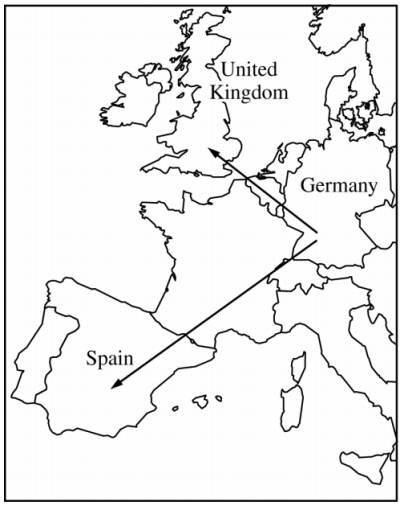
(B) Speciation would occur more rapidly because females would produce more offspring.

(C) Behavioral isolation would occur sooner in species exhibiting nonrandom mating.

(D) The population would reach Hardy-Weinberg equilibrium at an accelerated rate.

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20.



Blackcap birds (*Sylvia atricapilla*) migrate out of Germany before wintertime. Prior to the 1960s, all members of a particular blackcap population flew to Spain, which had an abundant natural food source. Now, some members of the same blackcap population fly to the United Kingdom, where food placed in feeders by humans is abundant. The blackcaps return to the same forests in Germany to nest during the breeding season.

Some blackcaps that migrate to the United Kingdom have become distinguishable by certain physical and behavioral traits from blackcaps that migrate to Spain. Which of the following best predicts the effect on the blackcap population if humans in the United Kingdom continue to place food in feeders during the winter?

(A) The blackcaps that migrate to Spain will selectively mate with the blackcaps that migrate to the United Kingdom, resulting in increased genetic variation in the blackcap population.

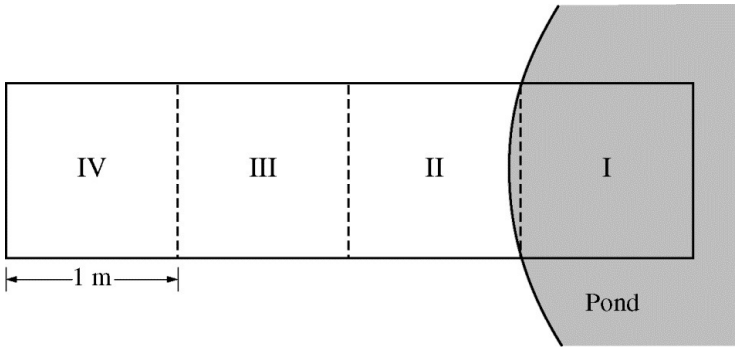
(B) The blackcaps that migrate to the United Kingdom will become reproductively isolated from the blackcaps that migrate to Spain, resulting in speciation in the blackcap population. ✓

(C) The blackcaps in Spain and the United Kingdom will migrate to other geographical locations in search of potential mating partners, resulting in a smaller blackcap population in Germany.

(D) The blackcaps will begin nesting at their wintering sites in Spain or the United Kingdom, resulting in a larger blackcap population migrating back to Germany after the breeding season has ended.

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Researchers investigated the habitat preferences of two species of garter snakes, *Thamnophis sirtalis* and *Thamnophis atratus*. To create a choice chamber, the researchers built a meshed enclosure and positioned one end of the enclosure at the edge of a small pond. Zone I of the enclosure was located in the water, whereas zone IV of the enclosure was located 2–3 meters away from the water, as represented in the figure below. Snakes inside the enclosure were able to move freely between zones.



In a series of experiments, the researchers introduced a single snake into zone IV of the enclosure at 7:00 A.M. The researchers recorded the location of the snake at six time points throughout the day. In a related experiment, the researchers introduced two snakes, one of each species, into the enclosure at the same time and observed the location of each of the two snakes at the same six time points as before. The researchers repeated both the one-snake and two-snake experiments using different individual snakes of each species. The results are presented in the table.

ZONES MOST FREQUENTLY OCCUPIED BY GARTER SNAKES IN A MESHED ENCLOSURE

	Species Introduced Inside the Meshed Enclosure			
	T. atratus Only	*T. sirtalis* Only	*T. atratus* and *T. sirtalis*	
Time of Day	Zone Most Frequently Occupied by *T. atratus*	Zone Most Frequently Occupied by *T. sirtalis*	Zone Most Frequently Occupied by *T. atratus*	Zone Most Frequently Occupied by *T. sirtalis*
8:00 A.M.	I	II	III	II
10:00 A.M.	II	I	II	II
12:00 P.M.	I	I	III	I
2:00 P.M.	I	I	IV	I
4:00 P.M.	II	I	IV	I
6:00 P.M.	II	II	IV	I

21. Both species of garter snakes prey on the California newt, *Taricha torosa*, a small amphibian that produces a potent neurotoxin (TTX) in its skin. However, neither species of garter snake is affected by TTX. The resistance to TTX is associated with mutations in the *SCN4A* gene. Which of the following best supports a claim that TTX resistance arose independently in *T. atratus* and *T. sirtalis* ?

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(A) The stomach contents in fossils of both species include the remains of food items that contained TTX.

(B) Both species of snakes possess the *SCN4A* gene.

(C) The two species of snakes have different genetic mutations in the *SCN4A* gene. ✓

(D) *T. atratus* and *T. sirtalis* are sister species that share many of the same morphological features.

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22. By discharging electric sparks into a laboratory chamber atmosphere that consisted of water vapor, hydrogen gas, methane, and ammonia, Stanley Miller obtained data that showed that a number of organic molecules, including many amino acids, could be synthesized. Miller was attempting to model early Earth conditions as understood in the 1950s. The results of Miller's experiments best support which of the following hypotheses?

(A) The molecules essential to life today did not exist at the time Earth was first formed.

(B) The molecules essential to life today could not have been carried to the primordial Earth by a comet or meteorite.

(C) The molecules essential to life today could have formed under early Earth conditions. ✓

(D) The molecules essential to life today were initially self-replicating proteins that were synthesized approximately four billion years ago.

Answer C

This option is correct. It demonstrates the ability to evaluate scientific questions about the origin of life on Earth by recognizing that the Miller experiment modeled the presumed early atmospheric conditions and, under laboratory conditions, produced biological molecules, such as amino acids, that were required for early life.

23. By discharging electric sparks into a laboratory chamber atmosphere that consisted of water vapor, hydrogen gas, methane, and ammonia, Stanley Miller obtained data that showed that a number of organic molecules, including many amino acids, could be synthesized. Miller was attempting to model early Earth conditions as understood in the 1950s. The results of Miller's experiments best support which of the following hypotheses?

(A) The molecules essential to life today did not exist at the time Earth was first formed.

(B) The molecules essential to life today could not have been carried to the primordial Earth by a comet or meteorite.

(C) The molecules essential to life today could have formed under early Earth conditions. ✓

(D) The molecules essential to life today were initially self-replicating proteins that were synthesized approximately four billion years ago.

7 Natural Selection MCQ

24. A group of researchers noticed that using a particular drug to treat cancer in mice would reduce the size of the tumor, but some of the tumor cells appeared to be resistant to the drug since some of these cells would increase in number over time. The researchers wish to increase the reliability of their experimental observations before investigating new claims that chemotherapy-resistant tumors are an example of continuing evolution.

Of the following, which is the best next step the researchers might take to increase the reliability of their evidence?

(A) Increasing the dose of chemotherapy being administered to the mice

(B) Increasing the number of trials and the number of mice in each treatment group ✓

(C) Plotting the tumor volume data as a graph rather than as a table

(D) Isolating resistant tumor cells and looking for novel mutations in proto-oncogenes

Answer B

Correct. The statistical power and reliability of the experimental results will increase if more replicates with increased sample sizes are used. Experimental findings must be consistently reproducible for them to be reliable.

25. Researchers observed selected internal structures of four different microscopic organisms as part of a larger study on the divergence between eukaryotes and prokaryotes. Their observations are recorded in Figure 1.

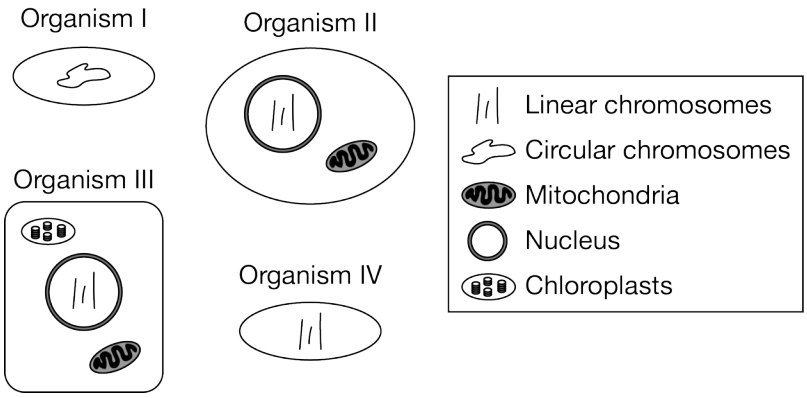


Figure 1. Selected cellular structures in four organisms

Which organism would the researchers most likely predict to be the most distantly related to eukaryotes?

7 Natural Selection MCQ

(A) Organism I ✓

(B) Organism II

(C) Organism III

(D) Organism IV

Answer A

Correct. The circular chromosome in Organism I identifies it as a prokaryote.

26. Because cheetahs, *Acinonyx jubatus*, experienced a bottleneck event about ten thousand years ago, living cheetahs exhibit very little genetic variation compared to other living species of large cats.

Which of the following principles best justifies a claim that cheetahs, compared to other large cats, have a relatively low resilience to environmental perturbations?

(A) A captive population of cheetahs kept in a confined area will have a greater risk of contracting and passing on a virus.

(B) When a population is reduced, new niches become available in the environment, which can trigger adaptive radiation.

(C) Populations with little genetic diversity are less likely to contain individuals that can withstand different selective pressures. ✓

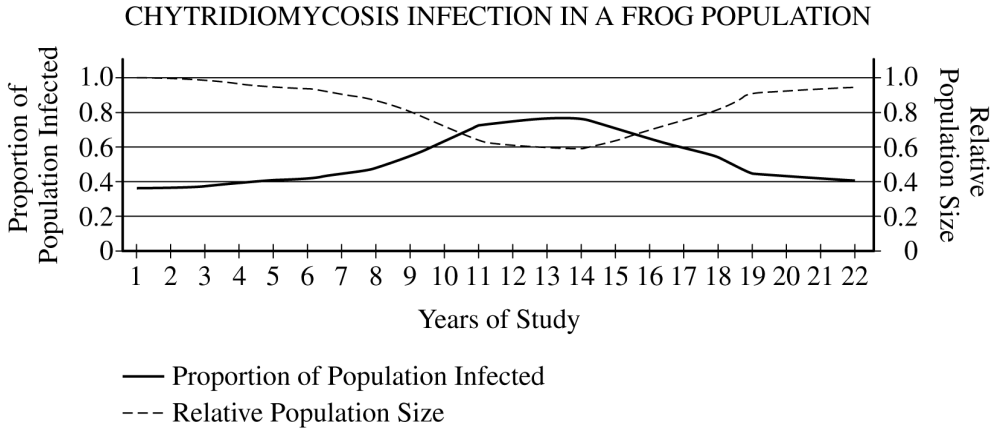
(D) Populations with low genetic diversity are more likely to contain individuals that can withstand different selective pressures.

Answer C

Correct. Cheetahs exhibit less genetic diversity than other cats, which is why they are at an increased risk of decline and extinction. The ability of a population to respond to changes in the environment is influenced by the level of genetic diversity among the individuals.

7 Natural Selection MCQ

27.



Chytridiomycosis is a potentially lethal fungal infection that adversely affects some frog populations. The incidence and severity of the fungal infection can vary over time. Which of the following statements best describes the changes in the frog population that are depicted in the model above?

(A) The fungus that infected the frog population was initially virulent but it later became harmless. Based on the trend over the past few years of the study, the fungus has become extinct.

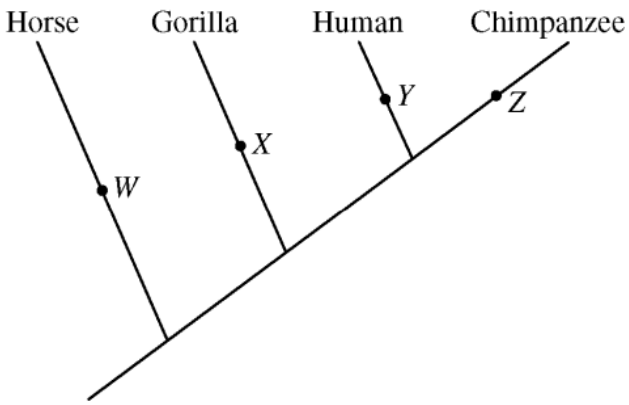
(B) Infected individuals gradually died out, and genetically resistant individuals became more common. The frog population recovered because of the increased frequency of resistant individuals. ✓

(C) The patterns of the curves on the graph indicate a sudden increase in the severity of the fungal infection. A large proportion of the frog population died because the individuals could not evolve fast enough.

(D) The size of the frog population decreased sharply upon initial contact with the fungus. The fungus eventually became symbiotic with individual frogs, and the frog population began to recover.

7 Natural Selection MCQ

28. The cladogram shown below depicts an accepted model of the evolutionary relationships among selected species.



The amino acid at position 104 in the beta-hemoglobin protein for each of these four organisms is listed below.

Species	Amino Acid 104
Horse	Arginine
Gorilla	Leucine
Human	Arginine
Chimpanzee	Arginine

The validity of the cladogram is best supported by molecular evidence for which of the following changes in the amino acid composition of the beta-hemoglobin protein during the evolution of these species?

(A) Arginine to leucine at position *X* on the cladogram ✓

(B) Arginine to leucine at position *Y* on the cladogram

(C) Leucine to arginine at position *W* on the cladogram

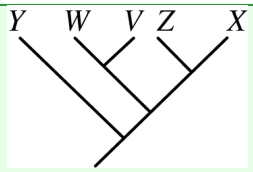
(D) Leucine to arginine at position *Z* on the cladogram

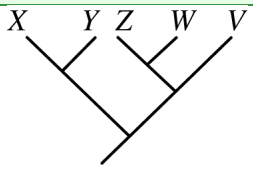
7 Natural Selection MCQ

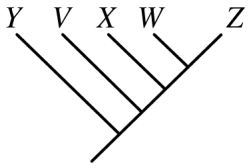
29. Data regarding the presence (+) or absence (-) of five derived traits in several different species are shown in the table below.

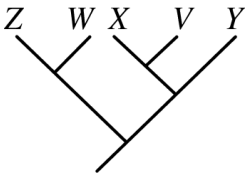
	Trait				
Species	1	2	3	4	5
V	+	+	+	–	–
W	+	+	–	–	–
X	+	–	–	+	+
Y	–	–	–	–	–
Z	+	–	–	–	+

Which of the following cladograms provides the simplest and most accurate representation of the data in the table?

(A) 

(B) 

(C) 

(D) 

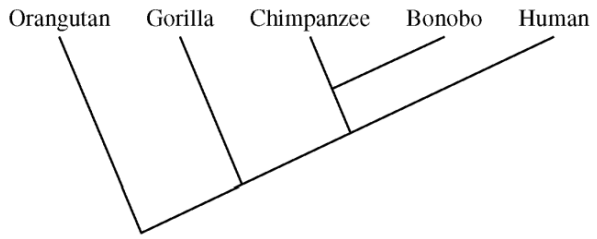
7 Natural Selection MCQ

The *TAS2R38* gene encodes a receptor protein that influences the ability to taste bitterness. The gene has two alleles: a dominant, wild-type allele that enables an individual (taster) to taste bitterness and a recessive, mutant allele that interferes with the ability of an individual (nontaster) to taste bitterness. Three single nucleotide mutations in the coding region of the *TAS2R38* gene are associated with the nontaster allele. The nucleotides present at the three positions are shown in the table below.

NUCLEOTIDE VARIATION IN THE *TAS2R38* GENE

	Position in the Nucleotide Sequence		
	145	785	886
Human nontaster	G	T	A
Human taster	C	C	G
Bonobo	C	C	G
Chimpanzee	C	C	G
Gorilla	C	C	G

A cladogram representing the evolutionary relatedness of selected primates is shown below.



30. Which of the following conclusions is supported by the data?

(A) Humans and bonobos belong to the same genus because the nucleotide sequence of the *TAS2R38* gene is the same

(B) Chimpanzees are equally related to humans and bonobos because all three groups possess variants of the *TAS2R38* gene.

(C) Humans are more closely related to bonobos than to chimpanzees because the branch of the tree with humans is closest to the branch of the tree with bonobos.

(D) Bonobos are more closely related to humans than to gorillas because bonobos and humans share a more recent common ancestor than bonobos and gorillas do. ✓

31. Toxic substances often have a bitter taste that causes animals who try to eat such substances to spit them out rather than swallow them. Additional data suggest that gorilla populations have a very low frequency of nontasters. Which of the following best predicts the likely evolution of the *TAS2R38* locus in the gorilla population?

7 Natural Selection MCQ

(A) Gorillas who could taste bitter toxins were more likely to survive and reproduce than nontasters. ✓

(B) Gorillas who could taste bitter toxins became resistant to the toxins, so the taster gene was no longer under selective pressure.

(C) Gorillas acquired the taster allele by horizontal gene transfer from chimpanzees.

(D) Gorillas who could taste bitter toxins could utilize food resources that nontasters could not.

32. Which of the following scientific questions will best help researchers determine when the nontaster allele arose in the evolutionary history of the selected primates?

(A) How does the nontaster allele contribute to individual fitness?

(B) Does the nontaster allele exist in any nonhuman primate populations? ✓

(C) What caused the mutations in the *TAS2R38* gene?

(D) How do the mutations in the nontaster allele affect the structure of the resulting protein?

33. In a sample of 2,400 people, 1,482 were found to have the dominant (taster) phenotype. Assuming that the population is in Hardy-Weinberg equilibrium, approximately how many individuals in the sample are expected to be heterozygous for *TAS2R38*?

(A) 741

(B) 918

(C) 1,133 ✓

(D) 1,482

34.

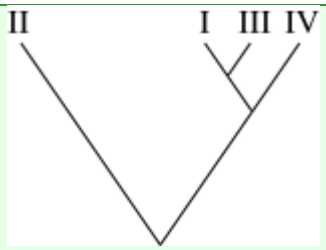
TABLE 1. PERCENT DNA SEQUENCE DIFFERENCES IN THE *MT-CO1* GENE

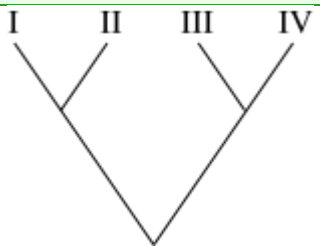
Species	I	II	III	IV
I		13.5	1.5	3.0
II			12.7	12.2
III				3.8
IV				

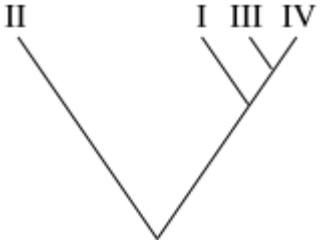
The gene encoding the catalytic subunit of cytochrome c oxidase (*MT-CO1*) is found in the mitochondria of all animals. The gene evolves rapidly and is often used to determine evolutionary relationships among species. Table 1 shows DNA sequence differences among four animal species (I, II, III, and IV).

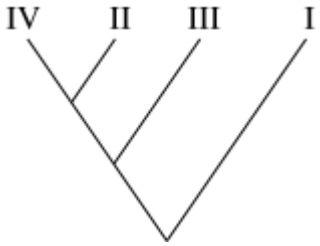
Which of the following cladograms best represents the evolutionary relationships among the four species?

7 Natural Selection MCQ

(A) 

(B) 

(C) 

(D) 

Answer A

Correct. The cladogram indicates that species II split off early in the evolutionary lineage and that species IV split off prior to the separation of species I from species III. The cladogram is supported by the data in Table 1 because the greatest percent difference in the sequence of the $MT-CO1$ gene is between species II and each of the other species. Additionally, the lowest percent difference is between species I and III.

7 Natural Selection MCQ

35. In a particular region, a species of lizard lives in the lowlands. Even though there are numerous mountains in the region, the climate on the mountains is too cold for the lizards to survive. Scientists are concerned that the lowlands will become much warmer by the end of the twenty-first century and will become unsuitable for this lowland lizard species.

Which of the following best predicts the impact on the lizard species if there is a gradual increase in the average temperatures found in both the lowlands and mountains in this region?

(A) Some lizards will migrate to new habitats in the mountains and become isolated, leading to allopatric speciation. ✓

(B) The species will undergo artificial selection, as the genes will adapt to warmer temperatures.

(C) The species will experience a genetic bottleneck, and sympatric speciation will result in the development of several new species.

(D) The lizards will hybridize with a different lizard species that is adapted to a warmer environment, leading to increased genetic variation in the species.

Answer A

Correct. In allopatric speciation, a new species forms in geographic isolation from its parent population. In this case, the lizards would be migrating to the mountainsides in small groups, each group isolated from the others by the now inhospitable lowlands.

36. Two species of fish that live in extremely cold environments produce near-identical antifreeze glycoproteins, called AFGPs. Scientists have determined the origins of the AFGP gene in both species. Selected characteristics of the two fish species are provided in Table 1.

Table 1. Characteristics of two fish species producing nearly identical antifreeze glycoproteins

Fish Species	Order	Location	Evolutionary origin of the *AFGP* Gene
Boreogadus saida	Gadiformes	Arctic	No sequence identity with trypsinogen
Dissostichus mawsoni	Perciformes	Antarctic	Pancreatic trypsinogen

Based on the information in Table 1, which of the following statements best describes the production of nearly identical AFGPs in these two species of fish?

(A) The fish eat the same type of food, which resulted in the evolution of similar digestive proteins.

(B) The fish live in environments with similar selective pressures, and those that produce AFGPs are better able to survive. ✓

(C) The Antarctic fish species evolved into a separate species after being geographically isolated from the Arctic population because of commercial fishing.

(D) The production of similar AFGPs was due to random splicing of exons in both species.

7 Natural Selection MCQ

Answer B

Correct. This is an example of convergent evolution. This occurs when similar selective pressures lead to similar phenotypic adaptations in different populations or species.

37. Whales, bony fishes, and cartilaginous fishes all have similar adaptations that enable them to live in the ocean. Figure 1 shows the phylogenetic relationships of selected vertebrates.

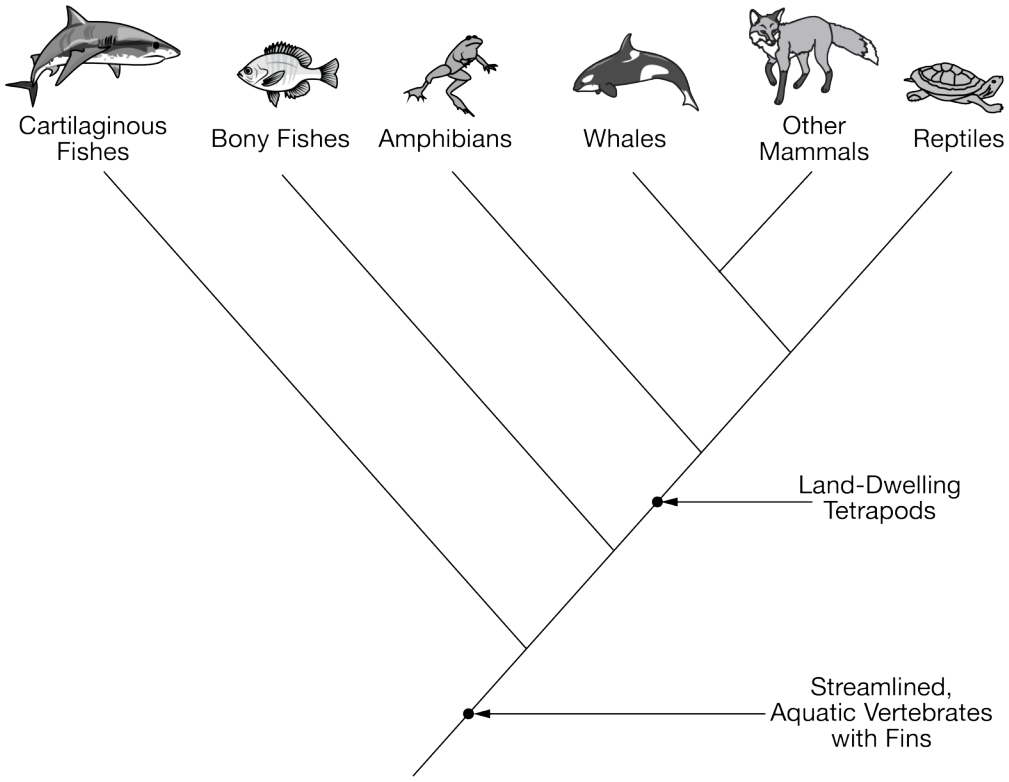


Figure 1. Cladogram of selected vertebrate groups

Based on the phylogenetic relationships shown in Figure 1, which of the following best explains the similarities exhibited by whales, bony fishes, and cartilaginous fishes?

(A) Convergent evolution, because these organisms do not share a recent common ancestor. ✓

(B) Adaptive radiation, because all three groups evolved from a unique, shared common ancestor from the cartilaginous fishes group.

(C) Mutation, because the shared traits can be traced back to a shared recent common ancestor.

(D) Founder effect, because all three groups formerly existed together as one population that then was disrupted, causing speciation.

7 Natural Selection MCQ

Answer A

Correct. Convergent evolution states that similar physical traits in unrelated groups are the result of the same selective pressures experienced by each group.

38. Female European corn borer moths (*Ostrinia nubilalis*) produce the sex pheromone 11−tetradecenyl acetate ($11-\mathrm{TDA}$), which attracts males of their species. $11-\mathrm{TDA}$ can be produced in two forms, E and Z, each with a different three-dimensional shape. Researchers have discovered two different strains of the European corn borer moth living in the same area. Females of the E strain produce primarily the E form of $11-\mathrm{TDA}$ and females of the Z strain produce primarily the Z form of $11-\mathrm{TDA}$. Males of both strains are more attracted to the form of $11-\mathrm{TDA}$ produced by the females of the same strain.

Which of the following best predicts a long-term effect of the differences between the E and Z strains?

(A) The difference between the forms of the sex pheromone $11-\mathrm{TDA}$ produced by E and Z strains will act as a prezygotic reproductive barrier leading to sympatric speciation. ✓

(B) The difference between the forms of the sex pheromone $11-\mathrm{TDA}$ produced by E and Z strains will act as a postzygotic reproductive barrier, leading to allopatric speciation.

(C) Since the E and Z strains are both part of the same species, they will produce fertile offspring, leading to a single strain that produces both forms of $11-\mathrm{TDA}$.

(D) Since the E and Z strains are found in the same geographic area, they will hybridize, leading to a single strain that produces the dominant form of $11-\mathrm{TDA}$.

Answer A

Correct. There are many different mechanisms that can provide reproductive isolation within a population leading to speciation without a geographic barrier. Barriers that prevent fertilization are called prezygotic barriers and those that act after fertilization are called postzygotic barriers. Sex pheromone preferences affecting mate choice would reduce the likelihood of mating between members of the E and Z European corn borer strains, acting as a prezygotic barrier by preventing mating that could lead to fertilization. Speciation that occurs without geographic isolation is called sympatric speciation.

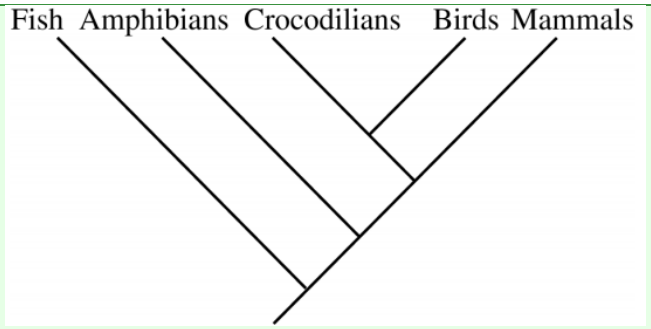
7 Natural Selection MCQ

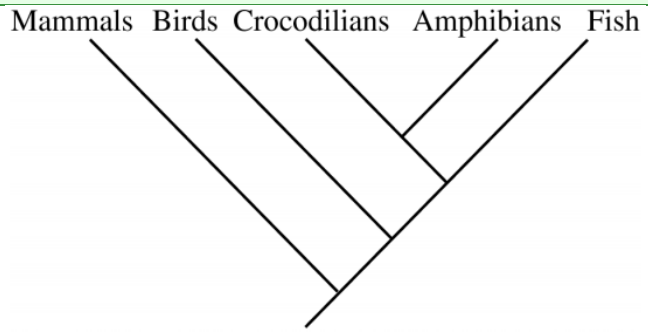
39.

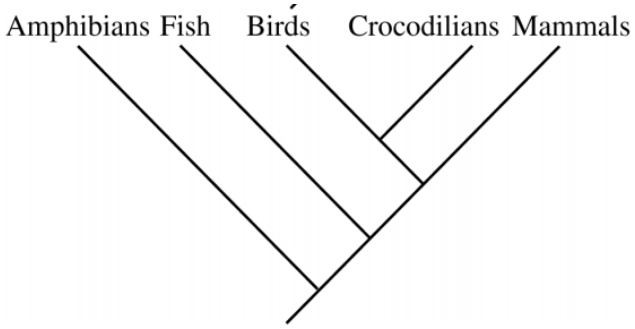
Organisms	Derived Characters (+ indicates the character is present)					
	Hair	Jaws	Feathers	Amniotic Eggs	Two Pairs of Limbs	β-Keratin Scales
Mammals	+	+		+	+	
Crocodilians		+		+	+	+
Fish		+				
Amphibians		+			+	
Birds		+	+	+	+	+

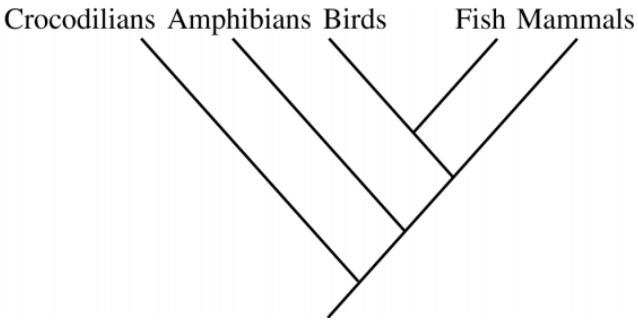
The table above shows derived characters for selected organisms. Based on the information in the table, which of the following cladograms best represents the evolutionary relationships among the organisms?

7 Natural Selection MCQ

(A) 

(B) 

(C) 

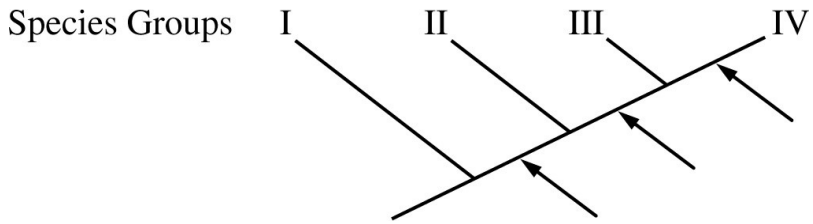
(D) 

7 Natural Selection MCQ

40.

	Derived Characters		
Plant Species	**Vascular Tissue**	**Seeds**	**Flowers**
P	+	+	+
Q	–	–	–
R	+	+	–
S	+	+	+
T	–	–	–
U	+	–	–
W	+	–	–

The table above shows the presence (+) or absence (-) of three different derived characters (vascular tissue, seeds, and flowers) for several selected plant species. The cladogram below was created based on the information in the table to represent groups of related species. The presence of the derived characters is indicated with arrows.



Group IV of the cladogram most likely includes which of the following species?

(A) P and S only ✓

(B) Q and T only

(C) R and W only

(D) U and W only

41. The apple maggot fly, *Rhagoletis pomonella*, is native to North America and originally fed on fruit of the wild hawthorn. Since the mid-1800s, a population of flies has emerged that instead feed on domesticated apples. Apple maggot flies typically mate on or near the fruit of their host plants. Many varieties of apples ripen three to four weeks before the hawthorn fruits do.

The different fruit preferences of the two fly populations will most likely have which of the following effects?

7 Natural Selection MCQ

(A) The flies that eat hawthorn fruit will increase in number, while the flies that eat apples will decrease in number because of the use of insecticides on apple trees.

(B) The single fly species will evolve into two distinct species because of the lack of gene flow between the two populations. ✓

(C) The ability to survive on a diet of two different fruits will help the flies learn to eat many more types of fruit.

(D) The flies that eat hawthorn fruit will lay some of their eggs on the earlier-ripening apples to minimize competition among the larvae.

42. One of the oldest known mammalian fossils belongs to a small furry species, *Hadrocodium wui,* that lived 195 million years ago. For the following 100 million years, only a small number of groups of mammalian fossils are found in the fossil record. By 65 million years ago, the nonavian dinosaurs were extinct. By 55 million years ago, there were 130 mammal genera, including 4,000 different species that occupied a wide variety of habitats.

Which of the following best describes the cause of the rapid increase in the number of mammalian species between 65 and 55 million years ago?

(A) It took over 100 million years for enough random mutations to occur in the genomes of early mammals to provide the variation needed for the observed speciation.

(B) It took over 100 million years for the early mammals to disperse over a wide enough geographic area to allow allopatric speciation to occur.

(C) After the dinosaur extinction, many ecological niches became available, leading to the adaptive radiation of mammals. ✓

(D) After the dinosaur extinction, mammal population sizes increased, allowing species to become more easily detectable in the fossil record.

Answer C

Correct. From 245 to 65 million years ago, dinosaurs were the dominant land animals and occupied a wide range of ecological roles. Early mammals were nocturnal, allowing them to avoid predation and competition with dinosaurs. With the dinosaur extinction 65 million years ago, many ecological niches were suddenly left empty. Speciation can occur rapidly when new niches become available, and a small number of ancestral species rapidly diversify into many new species, each adapted to a different role in the environment. This process is called adaptive radiation and has occurred many times on many scales throughout evolutionary history.

7 Natural Selection MCQ

43. A group of birds is found in an island chain that have all descended from a single ancestral species. Figure 1 shows the phylogeny of the birds.

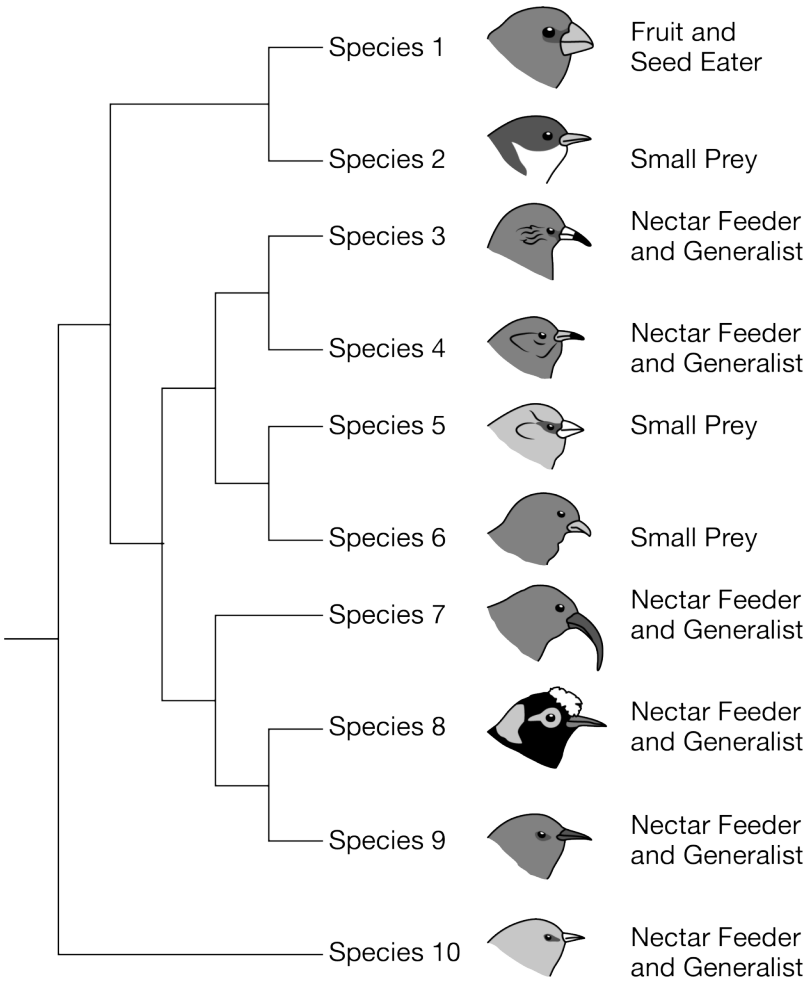


Figure 1. Phylogeny of island birds

Which of the following best explains the divergence in morphology in the birds?

(A) The original habitat contained mutagens, leading to the rise of new species as a result of accumulated mutations.

(B) The original habitat had many available unoccupied ecological niches that were colonized by populations of the original species. ✓

(C) The original habitat contained many predators, and the species diversified in order to avoid extinction.

(D) The original habitat was tropical, which increased the rate of mutation and led to rapid diversification.

7 Natural Selection MCQ

Answer B

Correct. This is an example of adaptive radiation and divergent evolution.

44. In plants, members of the euphorbia family and the cactus family have separate origins, with the euphorbia evolving in the desert regions of Asia and Africa and the cacti in the desert regions of the Americas. Both have fleshy stems adapted for water storage, protective spines, and leaves that are greatly reduced or absent. The similarities between these two families of plants represent an example of

(A) coevolution

(B) convergent evolution ✓

(C) genetic drift

(D) sympatric speciation

(E) balanced polymorphism

45. A scientist is attempting to provide support for the hypothesis that RNA was the first genetic material. Which of the following would be a workable alternate hypothesis?

(A) RNA, like DNA, is a carrier of genetic information and contains the nitrogen bases adenine, guanine, and cytosine.

(B) RNA can be observed to self-replicate without the assistance of proteins, while DNA always requires protein-based enzymes to replicate. ✓

(C) RNA exists in both single-stranded and double-stranded configurations; however, it is less stable compared with DNA.

(D) Transfer RNA and ribosomal RNA are products of transcription, but they are not translated as is mRNA.

Answer B

Correct. Both DNA and RNA can serve as genetic storage, but only RNA can also catalyze enzymatic reactions needed for replication and metabolic activities, so it may have been the earliest genetic material. Stating that there is a difference between the two types of nucleic acids is a characteristic of an alternate hypothesis.

7 Natural Selection MCQ

46. A group of researchers claim that chance events play an important role in the evolution of populations. To test their claim, they monitored 12 populations of *E. coli* for 10,000 generations under the same environmental conditions. They found that although all 12 populations were exposed to the same environmental pressures, each population had differentiated from one another genetically after 10,000 generations.

Which of the following statements includes a reasonable refinement that researchers could use in repeating the experiment?

(A) Start each population with genetically identical bacterial cells from a single culture to make sure that the initial genetic variation in all of the populations was exactly the same. ✓

(B) Start each population with a different species of bacteria to determine whether the same environment has an effect on genetic differentiation.

(C) Start each population with bacterial cells that have different genotypes to make sure there is enough genetic variation for selection to occur.

(D) Start each population with bacteria that differ from one another by a single mutation to ensure that the initial genetic variation in all populations is known.

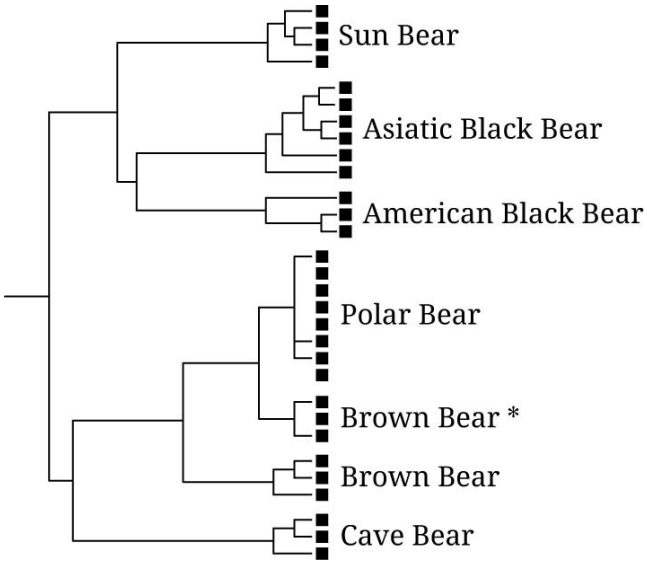
Answer A

Correct. The only way to know whether chance events were influencing the evolution of the bacteria would be if all of the bacteria had the exact genetic makeup at the beginning of the experiment.

7 Natural Selection MCQ

47. Scientists studying the evolution of bears analyzed the phylogenetic relationship among several extant bear species using 27 mitochondrial DNA (mtDNA) genomes. In addition to the five extant bear species, three genomes from an extinct species of cave bear were examined. The scientists constructed the phylogenetic tree (Figure 1).

Figure 1. Phylogenetic tree of bears based on 27 mtDNA genomes. Each square represents one mtDNA genome. The asterisk symbol (*) denotes brown bears from Admiralty, Baranof, and Chichagof Islands.



Which of the following statements best explains the characteristics of two species that are on the same branch of the phylogenetic tree in Figure 1?

(A) Their mtDNA genomes are identical in sequence.

(B) Their mtDNA genomes share more similarities with each other than with the mtDNA genomes of other species on the tree. ✓

(C) Their mtDNA genomes have the same number of genes and are the same length.

(D) Their mtDNA genomes must have evolved at exactly the same rate since their divergence from a common ancestor.

48. Geologic events such as mountain building can directly affect biodiversity. The Sino-Himalayan fern, *Lepisorus clathratus* is widely distributed on the Qinghai-Tibetan plateau.

If geologic events lead to a series of new mountain ranges that divide the plateau into several discrete valleys, which of the following best describes how the *L. clathratus* population would be affected?

7 Natural Selection MCQ

(A) Geographic isolation will create separate gene pools, leading to speciation over time. ✓

(B) Gene flow will be reduced by sexual selection and polyploidy.

(C) Chance events will cause allele frequencies to fluctuate unpredictably from one generation to the next.

(D) Ferns reproduce asexually, so there will be no impact on the gene pool of the species.

Answer A

Correct. A geographic barrier will likely lead to allopatric speciation because the gene pools of the species will be separated.

49. The appearance of a fertile, polyploid individual within a population of diploid organisms is a possible source of a new species. If this individual is capable of reproducing to form a new population, scientists would consider this to be an example of

(A) allopatric speciation

(B) sympatric speciation ✓

(C) polygenic inheritance

(D) genetic drift

(E) Hardy-Weinberg equilibrium

50. Researchers examined the ability of cultures of the bacterium *Pseudomonas aeruginosa* to adapt to the antibiotics ceftazidime and avibactam when they are administered together. The researchers grew four replicate cultures in growth medium with no antibiotics added (control) and four additional replicate cultures in the same growth medium with added ceftazidime and avibactam. At the conclusion of the experiment, samples of the replicates exposed to the antibiotics were removed and their genomes were sequenced. All of the antibiotic-resistant mutants were missing three genes ($hmgA$, $galU$, $mexXY$). It is known that one of these three genes makes wild-type *P. aeruginosa* susceptible to the two antibiotics. However, the researchers do not know which gene it is.

Which of the following experiments would specifically determine which gene is responsible for ceftazidime-avibactam sensitivity in *P. aeruginosa* ?

(A) Clone the three deleted genes from a wild-type *P. aeruginosa* strain, and insert these sequences into the mutant bacterial strains to restore sensitivity to ceftazidime-avibactam.

(B) Delete genes that are homologous to the three deleted genes in other bacterial species, and determine if doing so also confers ceftazidime-avibactam resistance in those bacteria.

(C) Perform gene knockout (targeted mutation) of either the $hmgA$, $galU$, or $mexXY$ gene on separate cultures of the wild-type *P. aeruginosa*. For each individual mutant, determine whether or not ceftazidime-avibactam resistance has occurred. ✓

(D) Perform gene knockout (targeted mutation) of both the $hmgA$ and $galU$ genes of the wild-type *P. aeruginosa* in a single culture, and then determine if ceftazidime-avibactam resistance has occurred in the mutant strain.

7 Natural Selection MCQ

Answer C

Correct. After specifically mutating the three genes separately, a check for ceftazidime-avibactam resistance in each different mutant strain would indicate which of the three genes is required for ceftazidime-avibactam sensitivity.

51.

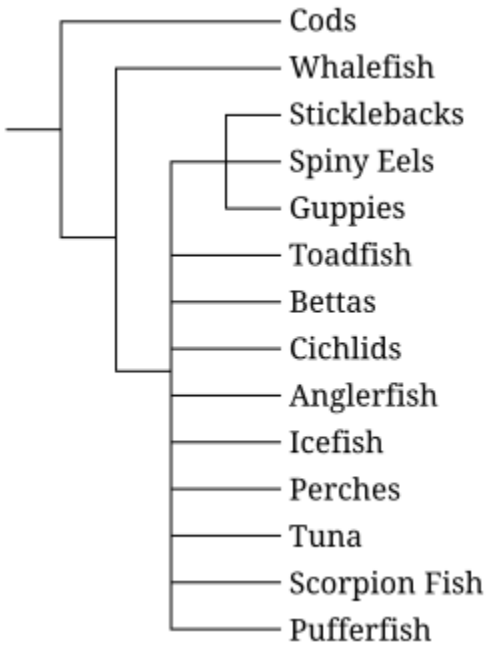


Figure 1. Simplified cladogram of several orders of fish

Cods and icefish are the only orders of fish that include species that produce antifreeze proteins in their blood. Based on the cladogram in Figure 1, which of the following scenarios describes the most likely evolutionary history of antifreeze protein synthesis in fish?

(A) Antifreeze proteins arose twice, once in the cod lineage and once in the icefish lineage. ✓

(B) Antifreeze proteins arose once in a common ancestor of cods and icefish.

(C) Antifreeze proteins arose once in the most recent common ancestor of whalefish and icefish.

(D) Antifreeze proteins were lost repeatedly in fish lineages.

Answer A

Correct. This is an example of convergent evolution. Cods and icefish do not share a recent common ancestor. Additionally, none of the other fish to which icefish are more closely related produce antifreeze

7 Natural Selection MCQ

proteins. This suggests that the presence of antifreeze proteins evolved separately in the cods and icefish lineages, and similar environmental conditions selected for the proteins to continue to be produced in future generations.

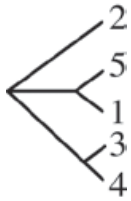
52. Five new species of bacteria were discovered in Antarctic ice core samples. The nucleotide (base) sequences of rRNA subunits were determined for the new species. The table below shows the number of nucleotide differences between the species.

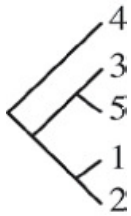
NUCLEOTIDE DIFFERENCES

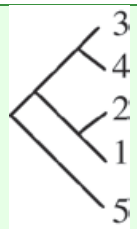
Species	1	2	3	4	5
1	-	3	19	18	27
2		-	19	18	26
3			-	1	27
4				-	27
5					-

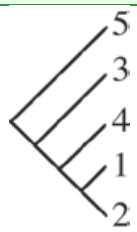
Which of the following phylogenetic trees is most consistent with the data?

7 Natural Selection MCQ

(A) 

(B) 

(C) 

(D) 

Answer C

This option is correct. It demonstrates the ability to select the phylogenetic tree that correctly represents evolutionary history and speciation from the data set. The data indicate that species 5 is not closely related to the others and that 3 and 4 are very closely related. This cladogram incorporates those differences as well as the close relationship between 1 and 2.

7 Natural Selection MCQ

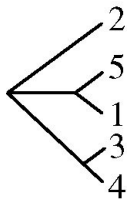
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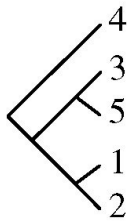
NUCLEOTIDE DIFFERENCES

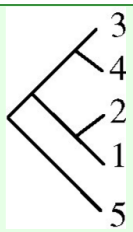
Species	1	2	3	4	5
1	--	3	19	18	27
2		--	19	18	26
3			--	1	27
4				--	27
5					--

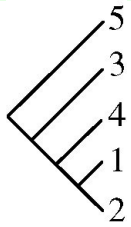
Which of the following phylogenetic trees is most consistent with the data?

7 Natural Selection MCQ

(A) 

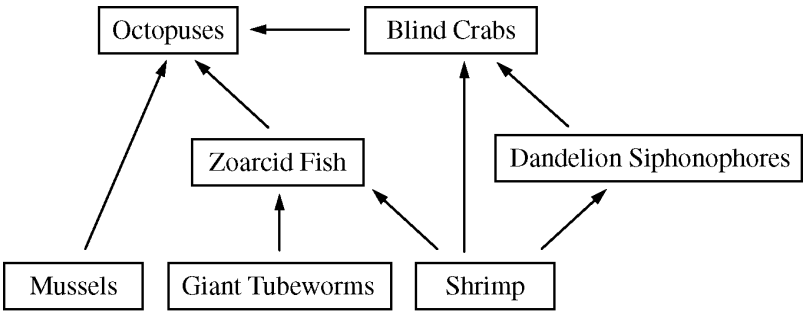
(B) 

(C) 

(D) 

7 Natural Selection MCQ

- **Directions:** Each of the questions or incomplete statements below is followed by four suggested answers or completions. Select the one that is best in each case and then fill in the corresponding circle on the answer sheet.



The food web above represents feeding relationships in a biological community near a deep-sea hydrothermal vent. Hydrothermal vents are geysers on the seafloor that gush super-heated, mineral-rich water. The seawater surrounding hydrothermal vents typically contains carbon dioxide (CO_2), molecular hydrogen (H_2), hydrogen sulfide (H_2S), and methane (CH_4). Sunlight, however, fails to reach the seafloor where deep-sea hydrothermal vents are located.

As part of an investigation, researchers collected living specimens from an area near a deep-sea hydrothermal vent. Mussels in the collection were found to be dependent on molecular hydrogen in seawater. Also, the researchers discovered multiple species of bacteria living in the gills of the mussels. Mussels use gills for filter-feeding and gas exchange with the surrounding seawater. On the basis of their experimental results, the researchers hypothesized that some bacteria living in the gills of the mussels are capable of chemosynthesis.

54. Researchers are investigating the evolutionary relationships among organisms found near deep-sea hydrothermal vents and similar organisms found closer to the ocean surface. Which of the following scientific questions is most relevant to the investigation?

(A) What are the nucleotide sequences of ribosomal RNA genes that are found in the genomes of the different species? ✓

(B) What large-scale geological events have occurred recently in the Mid-Atlantic Ocean?

(C) Does water temperature at different ocean depths affect the relative levels of dissolved oxygen?

(D) Do species found near deep-sea hydrothermal vents all have the same haploid number of chromosomes?

55. The fossil record indicates that in some cases reasonably well-defined species appear suddenly and remain unchanged for a long time before they become extinct. This phenomenon is referred to as

(A) allopatric speciation

(B) sympatric speciation

(C) phyletic gradualism

(D) punctuated equilibrium ✓

(E) parallel evolution

7 Natural Selection MCQ

56. Finch beak depth was measured before and after a severe drought in 1977. The averaged results are shown in Figure 1. After the drought, there were fewer seeds available for the finches to eat and the seeds that did remain were larger and harder to crack open. Researchers claim that when large, hard seeds are the predominate food source, the average beak depth of these finches tends to increase in subsequent generations.

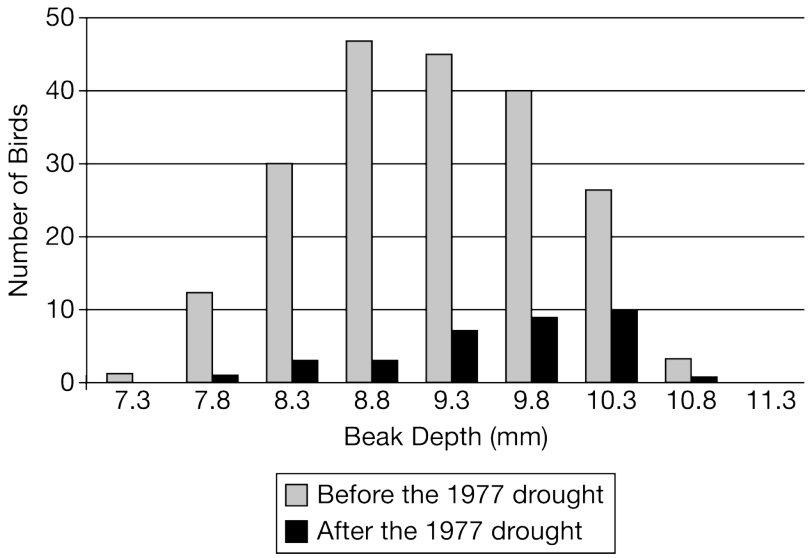


Figure 1. Finch beak depth before and after the 1977 drought

Using the evidence provided, which of the following best justifies the researchers' claim?

(A) Finch beak depth increased after the drought because the overall numbers of finches decreased, reducing competition for resources.

(B) Finch beak depth increased as a consequence of selective pressure after the drought because the remaining seeds were larger and harder. ✓

(C) Finch beak depth increased after the drought due to intraspecific competition since finches with deeper beaks attract more mates.

(D) Finch beak depth increased after the drought due to increased predation since finches with deeper beaks can fight off predators more effectively.

Answer B

Correct. Increased beak depth is associated with the ability to crack open harder and larger seeds. Seeds were harder and larger after the drought, so finches with increased beak depth could get more food and had a survival advantage after the drought.

7 Natural Selection MCQ

57. A researcher uses the protocol shown in Figure 1 to isolate a eukaryotic gene for the purpose of genetically engineering bacteria to synthesize the corresponding protein.

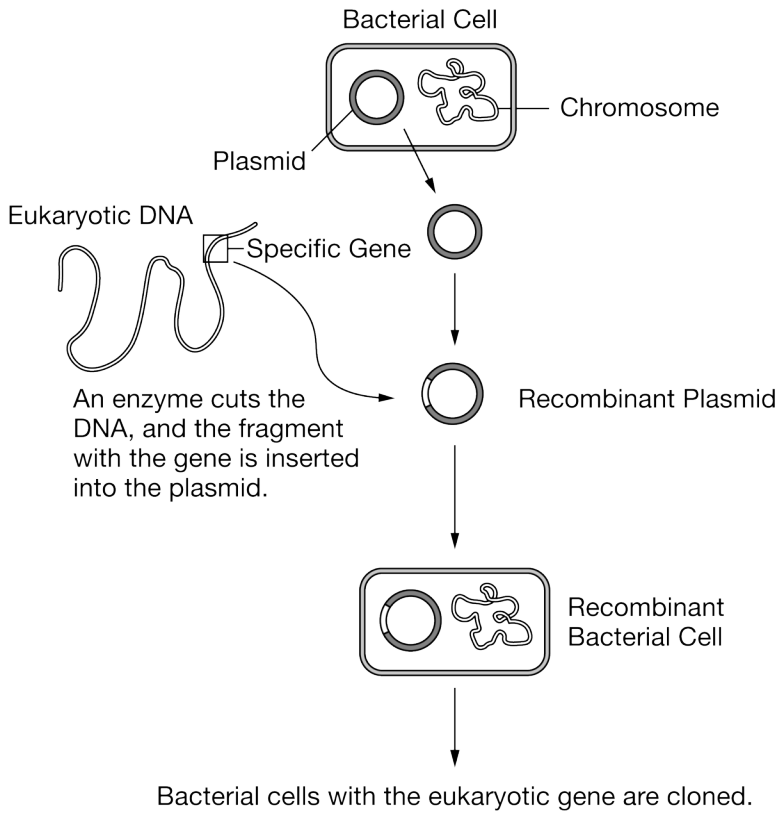


Figure 1. Creating recombinant bacteria that contain a specific gene

Which of the following best predicts why the recombinant bacteria will fail to produce the eukaryotic protein?

(A) Introns must be removed from eukaryotic DNA before the gene is inserted into the plasmid. ✓

(B) Bacterial ribosomes cannot recognize eukaryotic mRNA and will not translate the codon sequence.

(C) Restriction enzymes in the cytoplasm of the bacteria degrade linear DNA.

(D) Bacteria and eukaryotes use different mRNA codons for the same amino acids during translation.

Answer A

Correct. Because eukaryotic genes contain introns (noncoding segments), inserting eukaryotic genes directly into bacteria cells will not yield usable instructions for protein synthesis. Prokaryotic genes do not have introns, and bacteria are prokaryotic. The presence of introns is evidence at the molecular level that indicates common ancestry of all eukaryotes.

7 Natural Selection MCQ

58. Two populations of a species of squirrel are geographically isolated from each other. Although they have the same population density, one population is significantly larger in number than the other. A new bacterial disease, which is easily spread and extremely virulent, affects both populations at the same time. Which of the following is the best prediction of how the new disease will affect the two populations?

(A) The two populations will be equally affected, because the ability to trigger an immune response is randomly distributed among all squirrels of that species.

(B) The larger population will be less affected by the disease than will the smaller population, because the mutation rate of the larger population is higher than the mutation rate of the smaller population.

(C) The smaller population will be more affected than will the larger population, because the smaller population has less genetic variation than the larger population has. ✓

(D) The smaller population will be less affected than will the larger population, because the smaller population exhibits more genetic drift than the larger population exhibits.

59. In a certain small mammal, fur color is controlled by a pair of alleles, with B being dominant over b. The genotypes BB and Bb result in dark-colored fur, the genotype bb results in light-colored fur. A population of the species has the following allele frequencies: $B = 0.85$ and $b = 0.15$. If a fire caused the surface of the ground in the habitat of the species to become darker in color, which of the following claims about the impact on the allele frequencies in this population is most reasonable?

(A) The frequency of homozygous recessive individuals will increase because light-colored fur absorbs less heat.

(B) The frequency of the B allele will increase because light-colored individuals will be more visible to predators. ✓

(C) The frequency of the b allele will increase because there will be more heterozygous individuals.

(D) The frequency of both alleles will increase because the population will enter a period of rapid growth.

Answer B

Correct. A darker-colored habitat will give darker-colored individuals a selective advantage as they will be less visible to both predators and prey. More dark-colored individuals will survive and reproduce. This will result in an increase in the frequency of the B allele.

60. Dr. Robert Hazen has worked to develop a hypothesis that explains the origin of life on Earth. His work has focused on hydrothermal vents, which are cracks in the ocean floor. Water heated by molten rock beneath the crust escapes from these vents at very high temperatures, producing hot, high-pressure environments at the vents on the ocean floor.

In order to provide evidence that these vents may be areas where life originated, which of the following states a null hypothesis Hazen might have used to begin his research?

7 Natural Selection MCQ

(A) Organic molecules were transported to the hydrothermal vents via a meteorite, thereby seeding them with precursors for more complex molecules.

(B) The temperatures and pressures found in hydrothermal vents resemble conditions described in currently accepted origins of life hypotheses. ✓

(C) Organic waste sinking to the ocean floor would have used the unique hot water environment as an energy source to begin manufacturing food.

(D) Chemosynthetic organisms can best develop in the hydrothermal environment, manufacturing nutrients for other living things to feed on.

Answer B

Correct. This states the null hypothesis that the similarity between the conditions at the surface of the early Earth and the conditions of these deep-sea vents made them similarly eligible as the source of the first life.

61.

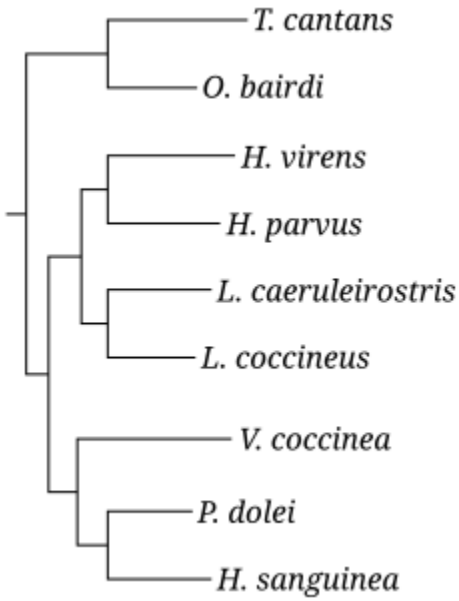


Figure 1. A hypothesized phylogenetic relationship among some members of the Hawaiian honeycreeper bird family

One hypothesized phylogenetic relationship among selected members of the Hawaiian honeycreeper bird family is shown in Figure 1. Based on the phylogenetic tree, birds in which of the following pairs share the most recent common ancestor?

7 Natural Selection MCQ

(A) *H. parvus* and *O. bairdi*

(B) *L. coccineus* and *V. coccinea*

(C) *L. caeruleirostris* and *H. virens* ✓

(D) *T. cantans* and *H. sanguinea*

Answer C

Correct. There are only two nodes separating *L. caeruleirostris* and *H. virens*. They share a more recent common ancestor compared with the other pairs listed.

Rhagoletis pomonella is a parasitic fly native to North America that infests fruit trees. The female fly lays her eggs in the fruit. The larvae hatch and burrow through the developing fruit. The next year, the adult flies emerge.

Prior to the European colonization of North America, the major host of *Rhagoletis* was a native species of hawthorn, *Crataegus marshallii*. The domestic apple tree, *Malus domestica*, is not native to North America, but was imported by European settlers in the late 1700s and early 1800s.

When apple trees were first imported into North America, there was no evidence that *Rhagoletis* could use them as hosts. Apples set fruit earlier in the season and develop faster, where hawthorns set later and develop more slowly.

Recent analysis of *Rhagoletis* populations has shown that two distinct populations of flies have evolved from the original ancestral population of flies that were parasitic on hawthorns. One population infests only apple trees, and the other infests only hawthorns. The life cycles of both fly populations are coordinated with those of their host trees. The flies of each population apparently can distinguish and select mates with similar host preferences and reject mates from the population specific to the other host tree. There is very little hybridization (only about 5 percent) between the two groups.

62. Initially, which of the following isolating mechanisms is likely to have been the most important in preventing gene flow between the two populations of *Rhagoletis*?

(A) Gamete incompatibility

(B) Temporal isolation ✓

(C) Mechanical isolation

(D) Reduced hybrid viability

63. Matings between individuals from the two populations of *Rhagoletis* produce hybrid flies that appear to be healthy and have normal life spans. The eggs laid by these hybrid flies, however, hatch less often than those of flies from either of the two populations. What isolating mechanism seems to be important in this hybrid population?

7 Natural Selection MCQ

(A) Prezygotic isolation
(B) Mechanical isolation
(C) Reduced hybrid fertility ✓
(D) Habitat isolation

64. The divergence between the two populations of *Rhagoletis* must have occurred very rapidly because

(A) the apple tree was imported into North America with European settlement approximately 200 years ago ✓
(B) flies were imported into North America with European settlement approximately 200 years ago
(C) long-distance rail transport of fruit increased only after the American Civil War (1861–1865)
(D) heavy use of gunpowder during the American Civil War (1861–1865) led to increased mutation rates in many natural populations of plants and animals

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65. Many scientists claim that the synthesis of the first organic molecules from inorganic precursors was possible because of the highly reducing atmosphere found on primitive Earth.

Which of the following is an appropriate null hypothesis that could be used when investigating the claim?

(A) The level of atmospheric oxygen on modern Earth is significantly higher than on primitive Earth.
(B) Organic molecules were transported to primitive Earth by a meteorite or other celestial event.
(C) The synthesis of organic molecules from inorganic molecules is possible under current atmospheric oxygen levels as well as those found on primitive Earth. ✓
(D) The absence of a significant quantity of atmospheric oxygen is required for the synthesis of organic molecules from inorganic precursors.

Answer C

Correct. A null hypothesis is a hypothesis of no difference. In this case, it should stipulate that the highly reducing atmosphere found on primitive Earth is not needed for the synthesis of organic molecules from inorganic precursors.

7 Natural Selection MCQ

66.

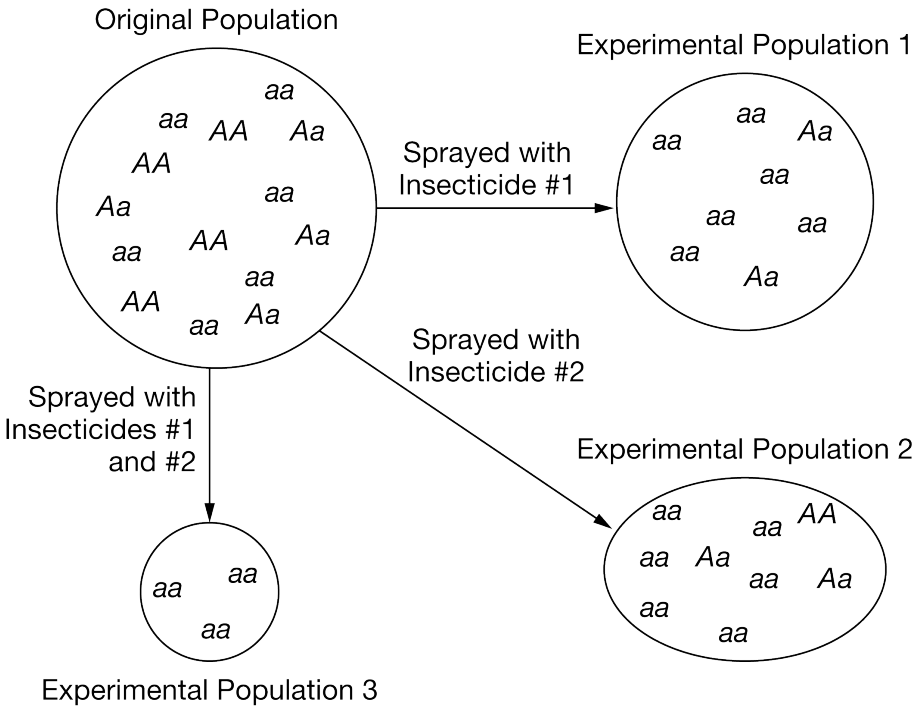


Figure 1. Effectiveness of Insecticides #1 and #2

Of the following, which is the best next step for the researchers to best determine the long-term effectiveness of these insecticides with regards to the development of resistance?

(A) Continue treating the three insect populations with the same insecticides and determine whether their efficiencies decline significantly after several generations. ✓

(B) Add an additional treatment group to the experiment to measure the effectiveness of another insecticide (Insecticide #3) over several generations.

(C) Set up a new experiment in which these insects are sprayed with two different insecticides to see if they are more effective than Insecticides #1 and #2.

(D) Spray these insecticides on a different species of insect (whose genotypes are also known) to determine its effectiveness on insects that are not pests.

Answer A

Correct. If the researchers continued the experiment for several generations and found that one of the insect population sizes started to increase, this would indicate that they have adapted and become resistant. Thus, this insecticide would probably not be useful in controlling the populations long-term.

7 Natural Selection MCQ

67. In the "RNA world" model for the origin of life, the first protocells (probionts) relied on RNA, not DNA, for information storage and transmission. Which of the following could best be considered evidence in support of the RNA world model?

(A) DNA is more stable than is RNA.

(B) Some RNA molecules in contemporary cells have catalytic function. ✓

(C) RNA is so complex in both structure and function that it must have preceded DNA.

(D) RNA has secondary structure whereas DNA does not.

68. Which of the following observations best supports the hypothesis that a large object collided with Earth in a period of time associated with a mass extinction?

(A) A species found only in Australia resembles a species found only in North America.

(B) An analysis of radiation reaching Earth from outer space suggests that water might have existed on distant planets.

(C) A fossil of an extinct species of mammal is found to have morphological characteristics that are similar to those of a living nocturnal mammal.

(D) Iridium, which is common in meteorites but rare on Earth, is found in sedimentary rock that contains the last appearance of many species in the fossil record. ✓

69. On a large volcanic island, researchers are studying a population of annual herbaceous plants. Which of the following observations best supports the prediction that speciation will occur within the existing plant population?

(A) Individuals of the species sometimes reproduce asexually by producing runners.

(B) Lava has separated the population into two areas: an upland forest and a lowland marsh. ✓

(C) Multiple groups of birds depend on the fruit produced by the plants as a source of food.

(D) The plants produce more seeds during warm summers than they do during cool summers.

7 Natural Selection MCQ

70. Figure 1 shows some major structural changes between a typical prokaryotic cell, the first eukaryotic common ancestor, and the last eukaryotic common ancestor.

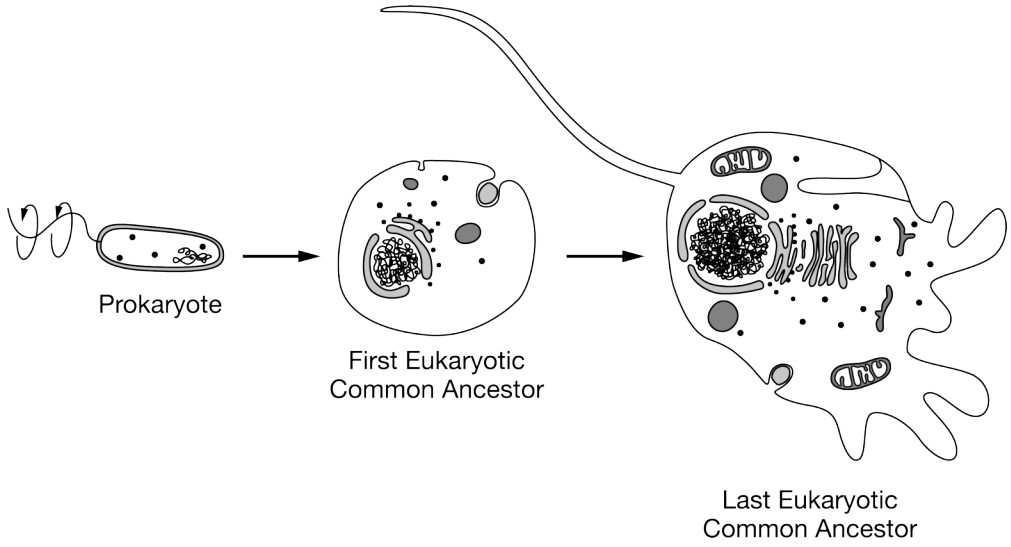


Figure 1. Models of a prokaryote, first eukaryotic common ancestor, and last eukaryotic common ancestor.

Based on the information provided in the figure, which of the following best predicts how membrane-bound organelles provided unique evolutionary advantages to the last eukaryotic common ancestor?

(A) Cells with nuclear membranes were better able to protect their genetic material against damage.

(B) Cells with flagella were better able to find food and escape predators.

(C) Cells with ribosomes were better able to produce proteins needed for a variety of metabolic functions.

(D) Cells with mitochondria were able to utilize metabolic energy more efficiently. ✓

Answer D

Correct. Eukaryotic cells acquired mitochondria via endosymbiosis. Most eukaryotic cells today have mitochondria because the last eukaryotic common ancestor had a selective advantage because it used mitochondria to perform aerobic cellular respiration. There are a few eukaryotic cell types (e.g., mature mammalian red blood cells) and a few parasitic single-celled eukaryotic organisms that have lost their mitochondria.

7 Natural Selection MCQ

71. Low doses of antibiotics are often added to livestock feed to increase production. Studies have shown that bacterial populations constantly exposed to the sublethal doses can evolve resistance to the antibiotics. A research group claimed that when a population of bacteria are constantly exposed to sublethal doses of streptomycin, their fitness declines due to the increased energy requirements for survival in the presence of the antibiotic. For subsequent studies, the researchers wish to determine whether adding a low, sublethal dose of an additional antibiotic causes further decline in the fitness of the bacteria.

Which of the following best represents the next step the researchers should take with respect to experimental design?

(A) Group 1: high dose of streptomycin + low dose of additional antibiotic
Group 2: low dose of streptomycin + high dose of additional antibiotic

(B) Group 1: low dose of streptomycin + no dose of additional antibiotic
Group 2: low dose of streptomycin + low dose of additional antibiotic ✓

(C) Group 1: no dose of streptomycin + no dose of additional antibiotic
Group 2: low dose of streptomycin + low dose of additional antibiotic

(D) Group 1: low dose of streptomycin + low dose of additional antibiotic
Group 2: high dose of streptomycin + high dose of additional antibiotic

Answer B

Correct. Since both groups have low doses of streptomycin, any deviation in the fitness of the bacteria in Group 2 may be attributed to the sublethal dose of the additional antibiotic. Group 1 bacteria serve as an appropriate control group for the experiment.

72. Malaria is caused by several different species of *Plasmodium*, a protozoan parasite. *Plasmodium* resistance to the common drugs used to treat malaria has increased in recent years.

In a scientific study, *Plasmodium* samples were analyzed in blood drawn from a large number of infected patients before drug treatment and subsequently from the subset of infected patients with drug-resistant *Plasmodium*. DNA sequences of four different *Plasmodium* genes thought to be involved in resistance were compared between samples from patients with drug-sensitive *Plasmodium* and patients with drug-resistant *Plasmodium*.

Which of the following best supports the hypothesis that preexisting mutations confer drug resistance?

7 Natural Selection MCQ

(A) Drug resistance can sometimes be reversed by simultaneous treatment with a drug that inhibits the drug-pumping mechanism of *Plasmodium*.

(B) Some of the mutations that occur most frequently in samples from patients treated with antimalarial drugs are detectable in *Plasmodium* samples collected prior to use of antimalarial drugs. ✓

(C) Continued use of the same antimalarial drugs leads to increased *Plasmodium* resistance.

(D) The same mechanism of drug resistance in *Plasmodia* isolated from separate populations of patients resulted from an accumulation of new genetic mutations.

73. Scientists claim that a population of mammals with high genetic diversity is better suited to survive and reproduce in a given geographic region. Which of the following reasonings best justifies the scientists' claim?

(A) High genetic diversity attracts more mates from other areas and increases the population size.

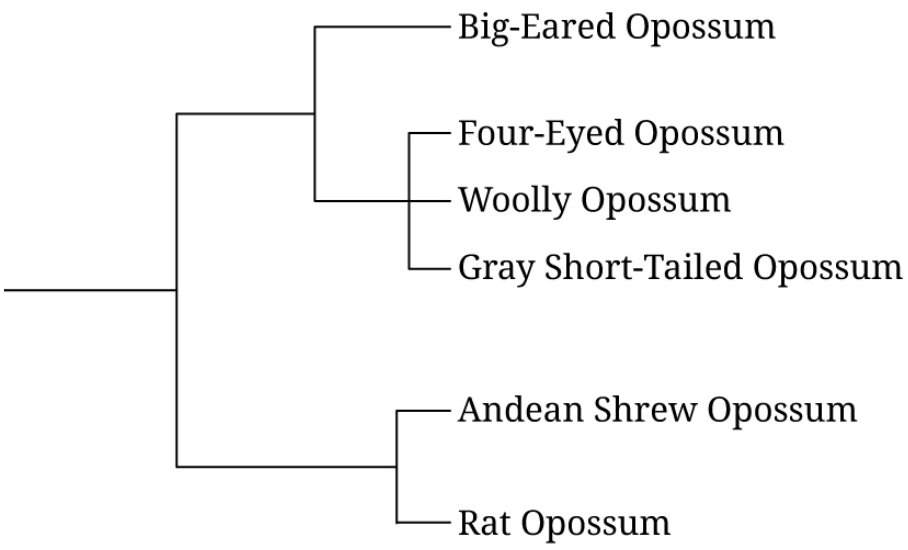
(B) High genetic diversity improves the ability of individuals in the population to hide from predators and survive.

(C) High genetic diversity increases the likelihood that individuals in the population possess traits that can withstand environmental changes. ✓

(D) High genetic diversity ensures that most individuals in the population have identical genetic makeup so that they can withstand environmental pressures.

Scientists studied the marsupial population in South America to determine species relatedness. The hypothetical relationships are presented in Figure 1.

Figure 1. Simplified cladogram of marsupial species in South America



74. Based on the cladogram in Figure 1, which of the following marsupials is most distantly related to the woolly opossum?

7 Natural Selection MCQ

(A) Big-eared opossum
(B) Four-eyed opossum
(C) Gray short-tailed opossum
(D) Rat opossum ✓

Answer D

Correct. The rat opossum branches off in the cladogram, indicating it shares a distant common ancestor with the woolly opossum compared to other species shown. This makes it the most distantly related species.

75. Which of the following best explains the relationships illustrated in Figure 1?

(A) The gray short-tailed opossum is more closely related to the Andean shrew opossum than to the other opossum species.
(B) The rat opossum and the gray short-tailed opossum share a more recent common ancestor than the rat opossum and the Andean shrew opossum do.
(C) The Andean shrew opossum and the rat opossum form a separate branch from the other opossum species. ✓
(D) The four-eyed opossum and the woolly opossum are more distantly related to each other than to the big-eared opossum.

Answer C

Correct. The Andean shrew opossum and the rat opossum form a distinct evolutionary branch separate from the other opossum species, indicating a different evolutionary lineage.

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76. Iridium is an element that is rare on Earth but commonly found in meteorites. A scientist believes that the first organic molecules may have come to Earth on meteorites 3.6 billion years ago. Which of these would be an appropriate null hypothesis to test related to whether meteorites hit the Earth 3.6 billion years ago?

(A) If sediment layers that are 3.6 billion years old are tested, they will contain relatively high iridium levels, because that is when many meteorites crashed into the Earth.
(B) There is no difference in iridium levels in sediments that were deposited on Earth 3.6 billion years old, compared to recent sediments. ✓
(C) If a sediment layer that is 3.6 billion years old is tested, it will contain organic material.
(D) If recently fallen meteorites and meteoroids collected in space are tested, they will not contain organic molecules.

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Answer B

Correct. This hypothesis states that there is no difference in iridium levels 3.6 billion years ago compared to other geological time periods.

77. MRSA is the acronym for methicillin-resistant *Staphylococcus aureus*. Many of the strains of the common bacterium are also resistant to other antibiotics in use today. The resistance is linked to a collection of genes carried on plasmids that are passed from one bacterium to another by conjugation. Suppose a newly discovered, chemically different antibiotic is used in place of methicillin. Which of the following would be the most likely effect on *Staphylococcus aureus* antibiotic resistance?

(A) The gene for methicillin resistance, no longer needed, would disappear entirely from *Staphylococcus aureus* populations within a few generations.

(B) Transmission of the methicillin-resistance plasmid by conjugation would increase among the *Staphylococcus aureus* population as the genes would confer resistance to the new antibiotic.

(C) Transmission of the methicillin-resistance plasmid would gradually decrease but the plasmid would not entirely disappear from the *Staphylococcus aureus* population. ✓

(D) Transmission of the methicillin-resistance plasmid by conjugation would increase among the *Staphylococcus aureus* population due to destruction of bacteria without the plasmid through use of the new antibiotic.

Answer C

This option is correct. It demonstrates an understanding that the plasmid would remain in the population even though it is not favored by selection and because it is not being selected against.

78. Mollusk fossils were excavated and collected from a coastal cliff. Scientists studying the fossils observed patterns of changing shell shape in the mollusks. The scientists identified a period of stability or stasis, followed by rapid change, and then stasis again with regard to shell shape.

Which of the following statements best predicts the result of a comparison between the fossils found before and after the period of rapid change?

(A) The same species will be found in both static periods, but they will have different shell shapes.

(B) The same species will be found in both static periods, but the species in the second static period will have smaller shells due to the stress of undergoing rapid change.

(C) Most of the species found in the first static period will be absent in the second static period. ✓

(D) Most of the species found in the second static period will resemble each other more than the species found in the first static period.

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Answer C

Correct. The relatively rapid changes occurring during the period of change indicate that both extinction and speciation were occurring. Most of the species observed in the second static period only date back to the period of rapid change. A majority of those species present before the rapid change will have become extinct or will have changed sufficiently to be considered a new species in the second static period. The scenario described exemplifies punctuated equilibrium.

Researchers were modeling the effects of repeated cycles of isolation with occasional interbreeding among five hypothetical fish species (species A, B, C, D, and E) found in two separate lakes (lakes I and II) that are occasionally joined by flooding, as shown in Figure 1. The research team indicated that species A would be considered to be the single common ancestor of species B, C, D, and E.

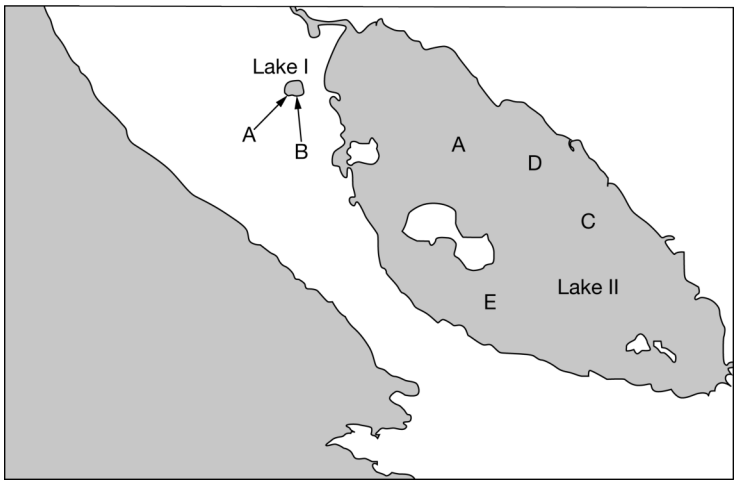


Figure 1. Current locations of populations of fish species A, B, C, D, and E

79. Which of the following mechanisms could most likely account for the origin of species B in Lake I?

(A) Allopatric speciation

(B) Directional selection

(C) Stabilizing selection

(D) Sympatric speciation ✓

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Answer D

Correct. Since species A and species B coexist, there must have been some isolating mechanism other than a physical barrier. The two species likely occupy separate niches, accounting for the lack of gene flow between the species.

80. Which of the following data could best support the claim that species B arose from a speciation event in Lake I?

(A) All of the fish species occupy the same niches in their respective lakes.

(B) DNA analysis shows fewer differences between species A and species B in Lake I than between species B and the populations in Lake II. ✓

(C) Species A and species B have similar body shapes due to convergent evolution in their habitats.

(D) Individuals of species A from Lake I can mate and produce viable offspring with individuals of species A from Lake II.

Answer B

Correct. Speciation results from accumulated genetic changes. DNA differences between the two lake populations would support the claim that a speciation event occurred.

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81. Old World vultures are birds that have digestive systems that enable them to feed mainly on carrion (dead animals). They have large wings used to soar as they search for dead animals with their vision. New World vultures have similar digestive systems and wingspans; however, they search for food using their sense of smell. Old World vultures and New World vultures are not closely related.

Which of the following best explains how Old World vultures and New World vultures both evolved to feed mainly on carrion?

(A) Old World vultures and New World vultures had a common ancestor that fed mainly on carrion, and this trait was selected for in populations of New World and Old World vultures.

(B) Old World vultures and New World vultures were exposed to similar selective pressures, which contributed to the independent evolution of carrion feeding in two separate lineages. ✓

(C) Old World vultures were introduced into areas with New World vultures, and the trait for carrion feeding was selected for in the population of New World vultures.

(D) Old World vultures and New World vultures have similar digestive systems, which drove them to consume a food source that was not available to other bird species.

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Answer B

Correct. This is an example of convergent evolution, where similar environmental pressures select for similar characteristics, including the ability to feed on carrion, in unrelated species.

82. Which of the following questions about the origin of life on Earth is most scientifically testable?

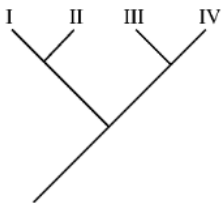
(A) Why were the earliest life-forms created using only twenty amino acids?

(B) Did life originate to make the universe a better place?

(C) Was catalytic RNA used by ancient organisms as a stepping stone to acquire protein enzymes?

(D) Can simple organic molecules form spontaneously in an oxygen-free atmosphere? ✓

83.



Based on morphological characteristics, the phylogenetic tree above has been developed for species I, II, III, and IV. DNA sequencing has recently been completed for a particular gene found in all four species. The sequencing will provide additional information for answering which of the following questions?

(A) Is species I the common ancestor of species II and III?

(B) Does species I contain the greatest number of vestigial structures?

(C) Are species III and IV the most closely related? ✓

(D) Should the positions of species III and IV be reversed?

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84. The degrees of relatedness in several native populations of prairie chickens in Marion and Jasper Counties in Illinois were determined. The degree of relatedness is inversely proportional to genetic diversity; that is, populations with a high degree of relatedness have less genetic diversity. Table 1 shows the collected data. Populations J1 through J5 are located in Jasper County. Populations M1 through M5 are located in Marion County. Numbers above 0.125 indicate a high degree of relatedness.

Table 1. RELATEDNESS OF PRAIRIE CHICKEN POPULATIONS

Population	Degree of Relatedness
J1	0.215
J2	0.215
J3	0.339
J4	0.280
J5	0.362
M1	0.305
M2	0.365
M3	0.199
M4	0.432
M5	0.405

Scientists claim that some of the populations in Marion County are at an increased risk of decline or extinction. Based on the data in table 1, which of the following best provides evidence to support this claim?

(A) The Marion County populations have a higher rate of mutation than the Jasper County populations, as evidenced by the higher degrees of relatedness.

(B) The Marion County populations are less stable because they are smaller than the Jasper County populations.

(C) The M2 population has a higher degree of relatedness than the M1 population, meaning that it has more genetic diversity.

(D) The M4 and M5 populations have the lowest genetic diversity of all populations. ✓

Answer D

Correct. Genetic diversity is inversely proportional to the degree of relatedness. The M4 and M5 populations have degrees of relatedness above 0.4; therefore, they have very low genetic diversity. Thus, these populations are at an increased risk of decline or extinction.

7 Natural Selection MCQ

85. In an exercise to understand how genetically controlled physical traits could affect speciation, a theoretical population was imagined that contained organisms with three shapes and three levels of pigmentation, as illustrated in Figure 1.

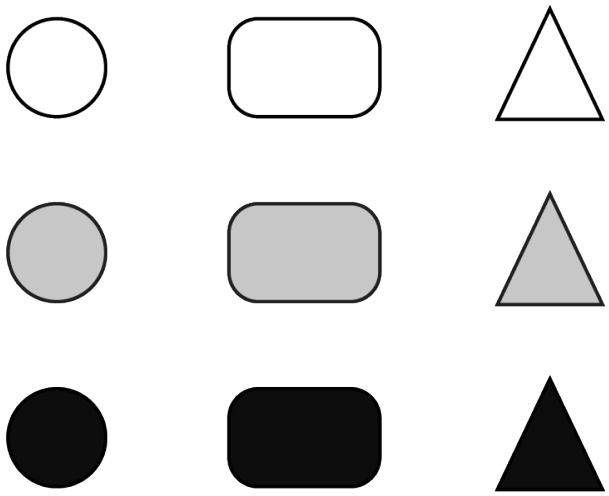
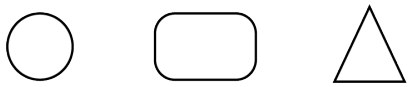
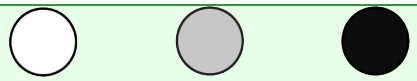


Figure 1. A single population with individuals of different shapes and pigmentation

Which of the following would most likely represent the types of individuals found in one of the species resulting from shape becoming a prezygotic isolating mechanism in the initial population represented in Figure 1 ?

(A) 

(B)  ✓

(C) 

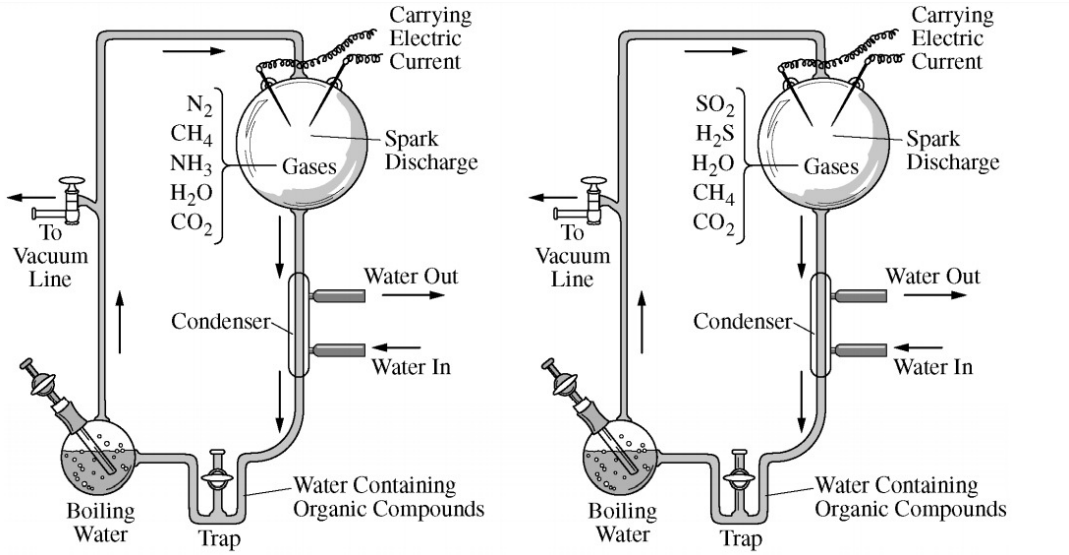
(D) 

Answer B

Correct. Since shape was the prezygotic isolating mechanism, individuals with different shapes would not be able to mate and produce viable offspring. This would lead to different species based on shape.

7 Natural Selection MCQ

86. Two groups of students attempted to re-create the primitive atmospheric conditions of early Earth using the apparatus represented below. Each group ran the experiment with different gas mixtures in the apparatus.



Which of the following statements best justifies the claim that the conditions in at least one of the experiments could generate the molecular building blocks essential for life?

(A) The carbon dioxide (CO_2) gas and water (H_2O) vapor in experiment 1 could react spontaneously to produce the phospholipids required by the first life-forms.

(B) The nitrogen (N_2) gas and ammonia (NH_3) gas in experiment 1 could provide the elemental nitrogen required for the formation of amino acids. ✓

(C) The sulfur dioxide (SO_2) gas in experiment 2 could donate the excited electrons required to drive the process of photosynthesis.

(D) The methane (CH_4) gas in experiment 2 could act as the electron acceptor required to complete the process of cellular respiration

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87. Figure 1 compares two models of speciation, A and B.

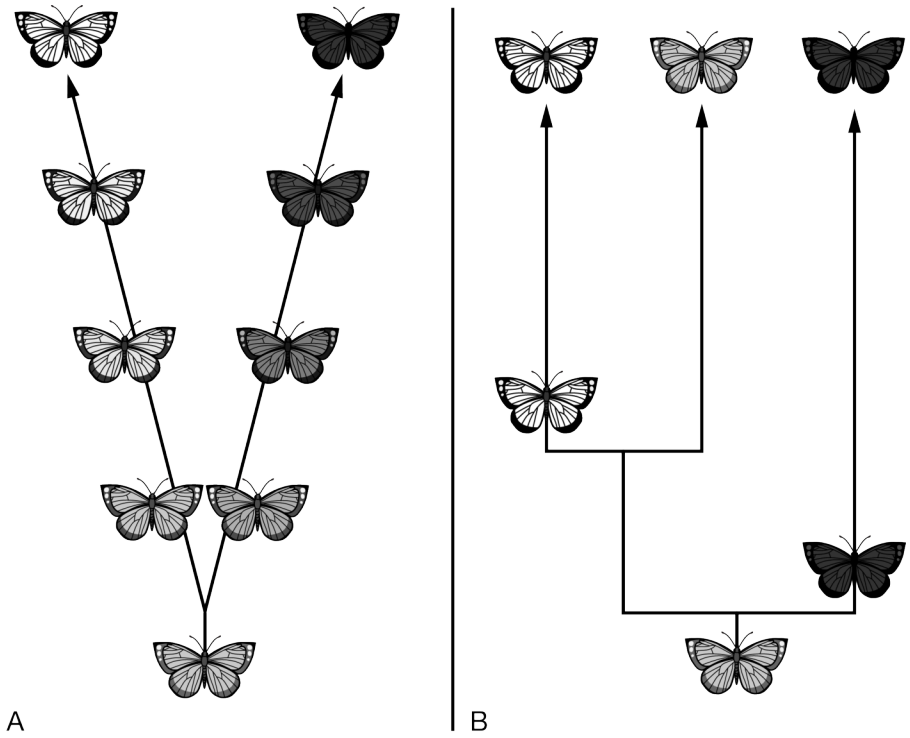


Figure 1. A comparison of two speciation models

Which of the following best explains how the ecological conditions are likely to be different in the two models?

(A) In model A the ecological conditions remain unchanged for long periods of time and then change drastically; in model B the ecological conditions change drastically from generation to generation.

(B) In model A the ecological conditions remain unchanged; in model B the ecological conditions change gradually over long periods of time.

(C) In model A the ecological conditions are changing drastically from generation to generation; in model B the ecological conditions remain unchanged.

(D) In model A the ecological conditions change gradually over a long period of time; in model B the ecological conditions remain unchanged for long periods of time and then change drastically. ✓

Answer D

Correct. Model A reflects gradualism while model B reflects punctuated equilibrium.

88. Which of the following statements best expresses the concept of punctuated equilibrium?

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(A) Small variations gradually accumulate in evolving lineages over periods of millions of years.

(B) Random mating ensures that the proportions of genotypes in a population remain unchanged from generation to generation.

(C) Stability is achieved when selection favors the heterozygote, while both types of homozygotes are at a relative disadvantage.

(D) Evolutionary changes consist of rapid bursts of speciation alternating with long periods in which species remain essentially unmodified. ✓

(E) Under competition for identical resources, one of the two competing species will be eliminated or excluded.

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89. The distribution of four populations of giant kangaroo rats, *Dipodomys ingens*, living on the same island when it was first surveyed in 1789 are shown in Figure 1.

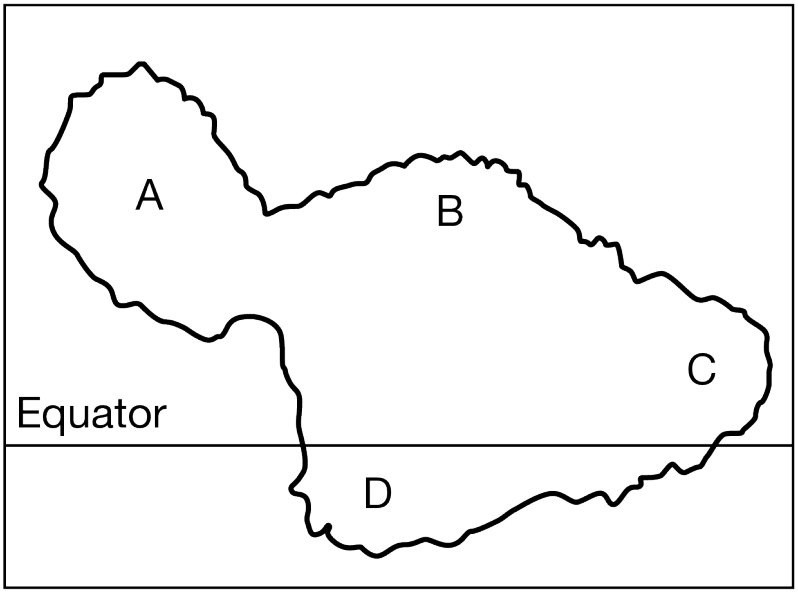


Figure 1. Four populations of *D. ingens* in 1789

In 1852, there was a large earthquake which altered the island's geography, as shown in Figure 2.

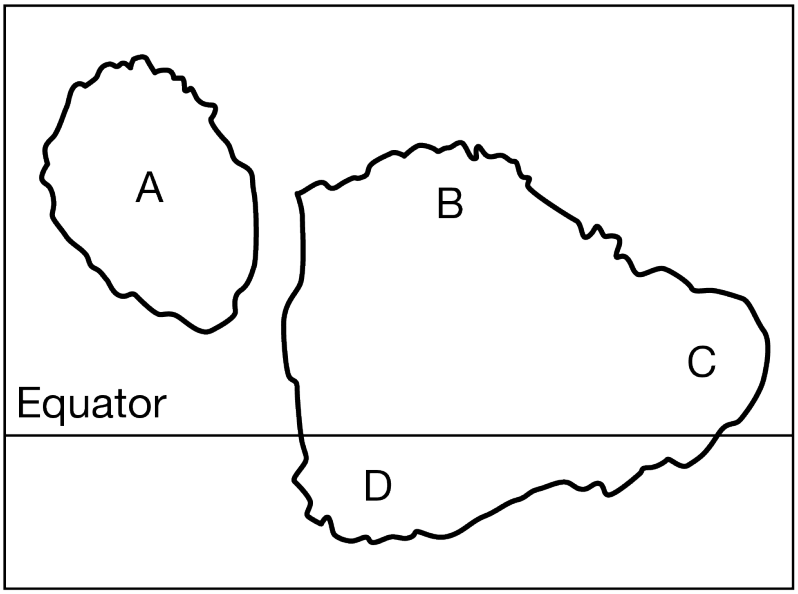


Figure 2. Map of the island following the 1852 earthquake

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The island was surveyed again in 2015, and scientists determined that the giant kangaroo rats originally in population A are now to be considered as a new species.

Which of the following best explains the mechanism that caused this speciation event in the giant kangaroo rats of population A?

(A) The animals in population A were exposed to different predators and had to modify their fur color to be better camouflaged.

(B) The animals in population A were reproductively isolated from those in populations B, C, and D. ✓

(C) The animals in population A accumulated fewer genetic mutations, adapting to their microclimate.

(D) The animals in population A mate at different times of year, causing their gametes to be incompatible with those from animals in populations B, C, and D.

Answer B

Correct. This is an example of reproductive isolation; one species diverges into two or more when some members of the species cannot reproduce with other members of the same species due to a physical barrier.

90. Rock pocket mice live in the deserts of the American southwest. Ancestral pocket mice all had light-colored coats that blended in with the region's rocks and sandy soil, keeping the mice hidden from predatory owls. About 1.7 million years ago, a series of volcanic eruptions spewed out wide trails of black lava into the middle of rock pocket mouse territory. Currently there are two color morphs of rock pocket mice: light-colored mice that are typically found in habitats with sandy soil, and dark-colored mice that are typically found in habitats with dark-colored lava rocks.

Which of the following best justifies the claim that alleles that may be adaptive in one type of environment can be deleterious in another because of different selective pressures?

(A) Light- and dark-colored mice survive in many environments because they are able to avoid predation by moving into tunnels.

(B) Light-colored mice are more likely to survive than dark-colored mice in habitats with lava rocks and sandy soil.

(C) Light-colored mice are significantly more susceptible to predation by owls in habitats with lava rocks than in habitats with sandy soil. ✓

(D) Dark-colored mice are more likely to survive than light-colored mice in habitats with lava rocks and sandy soil.

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Answer C

Correct. On the darker-colored lava rocks, alleles that confer light coat colors in mice are now deleterious because these mice are more easily identified by predators, such as owls. Thus, the dark lava rock environment places different selective pressures on the light-colored mice than the sandy soil environment does.

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Researchers have undertaken a long-term study of populations of rock pocket mice, *Chaetodipus intermedius*, in Arizona and New Mexico. Rock pocket mouse fur is either dark or light in color. Most of the populations studied live in areas with light-colored granite rocks; however, some areas are dominated by dark-colored rocks from ancient lava flows. Researchers collected mice from six sites in Arizona and recorded their coat colors. Figure 1 shows the study sites, the rock colors, and the distribution of coat colors.

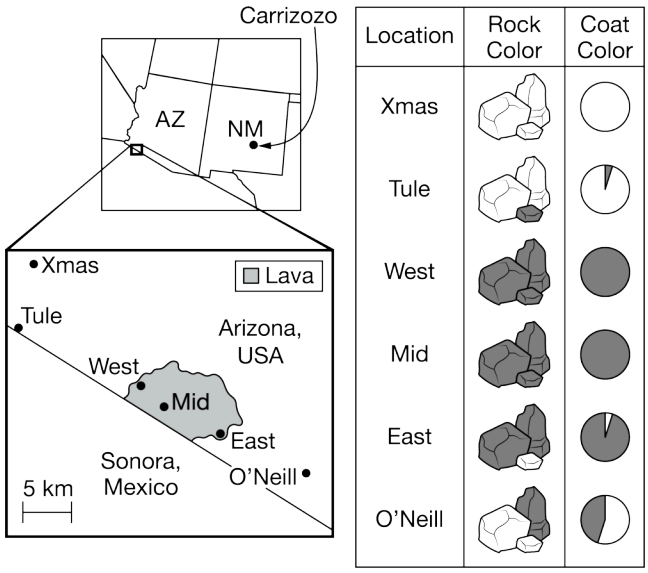


Figure 1. Rock color and coat color at three sites on a lava flow (West, Mid, and East), and three nearby sites: O'Neill Hills (O'Neill), the Tule Mountains (Tule), and Christmas Pass (Xmas)

In the Arizona populations, the melanocortin-1 receptor $(MC1R)$ gene is responsible for coat color phenotype (light or dark). The $MC1R$ D allele, which results in a dark coat, is dominant to the d allele. Researchers collected samples of mice from all six study sites and sequenced the $MC1R$ gene. They also collected mice with dark coats from a location in New Mexico, the Carrizozo lava flow, to determine if the $MC1R$gene was also responsible for the dark coat colors observed in that population. The rock pocket mouse population in Carrizozo is isolated from the populations in Arizona. Table 1 presents the genotypic data from sampled mice from all seven locations.

Table 1. $MC1R$ Genotypes of Rock Pocket Mice at Seven Collection Sites

State	Collection Site	$MC1R$ Genotype			Total Collected
		DD	Dd	dd	
Arizona	Christmas Pass	0	0	6	6
Arizona	Tule Mountains	0	5	80	85
Arizona	West	6	1	0	7
Arizona	Mid	3	2	0	5
Arizona	East	35	7	3	45
Arizona	O'Neill Hills	9	34	34	77
New Mexico	Carrizozo	0	0	9	9

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91. Based on the data in Table 1, which of the following best describes the relationship between the $MC1R$ gene and coat color in the Carrizozo, New Mexico, rock pocket mouse population?

(A) The $MC1R$ D allele is responsible for dark coat color in the Carrizozo population and likely spread from the Arizona population through migration.

(B) The $MC1R$ D allele is not responsible for dark coat color in the Carrizozo population, so the dark coat color is likely a result of convergent evolution. ✓

(C) The $MC1R$ D allele is absent from the Carrizozo population, so all the mice in this population must have coats that are light in color.

(D) The $MC1R$ D allele is absent from the Carrizozo population because of different environmental pressures in the Carrizozo population.

Answer B

Correct. In the Arizona rock pocket mouse populations, mice with dark coats have a homozygous dominant or heterozygous genotype for the $MC1R$ gene. However, according to the data table, the dark mice from the Carrizozo population are all homozygous recessive for the $MC1R$ gene. The Carrizozo population is also found on a dark lava flow. Convergent evolution occurs when different species or populations show a similar adaptation to a similar environment, but the adaptation is due to different genes. In this case, the Carrizozo mice population have a different mutation that results in a dark coat color.

92. Which of the following best describes the processes occurring between the mouse populations and their environments?

(A) Natural selection favors the mice with the dark coat color, causing the mice in all of the populations to be better adapted.

(B) Natural selection favors the mice with the light coat color, causing the mice in all of the populations to be better adapted.

(C) Dark color is selected for in mice from the West, Mid, and East populations. ✓

(D) Light color is selected for in mice from the West, Mid, and East populations.

Answer C

Correct. On the dark lava flow rocks (West, Mid, and East locations), most of the mice have dark coats. These dark-colored mice are better camouflaged, thus they are more likely to survive and reproduce. Therefore, on the dark lava flow rocks, natural selection favors the mice with dark coats.

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93. Owls (primary predators of rock pocket mice) are nocturnal visual predators that are able to distinguish effectively between dark and light colors in low-light conditions.

Which of the following best explains the relationship between coat color and predation rate in the rock pocket mouse population on the lava flow rocks with respect to the different coat color phenotypes?

(A) A dark coat color provides camouflage to mice on the lava flow rocks; therefore, mice with dark coats have a lower predation rate in that habitat. ✓

(B) At night, predators have reduced vision, so there is no relationship between coat color and predation rate on rock pocket mice.

(C) A dark coat color reflects less light and therefore mice with dark coats have a lower predation rate.

(D) A light coat color reflects more light and therefore mice with light coats have a lower predation rate.

Answer A

Correct. Many prey species have adaptations that allow them to blend in with their environment, thus escaping predators. In this case, the dark coat color of rock pocket mice blends in with the dark rocks on the lava flows but is more obvious on the light granite rocks. Therefore, mice with dark coats would have a lower predation rate on the lava flow rocks and a higher predation rate on the granite rocks.

94. The researchers built an enclosure to prevent migration into and out of the O'Neill rock pocket mouse population for a period of two years.

Which of the following best predicts the effect of the enclosure on the O'Neill population?

(A) The O'Neill population will rapidly evolve into a separate species.

(B) The O'Neill population will become less diverse due to the bottleneck effect.

(C) The frequency of the $MC1R\ D$ allele will decline due to natural selection. ✓

(D) One of the alleles for the $MC1R$ gene will become fixed due to genetic drift.

Answer C

Correct. The $MC1R\ D$ allele occurs at a higher frequency in the O'Neill population than in the other populations of rock pocket mice on the light granite rocks. This higher-than-expected frequency is likely due to immigration from the East rock pocket mouse population on the lava flow rocks. An enclosure that would prevent this gene flow would reinforce the selective pressures on the O'Neill population due to the light-colored granite rocks. Since mice with dark coats are more visible to predators on light

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granite rock, the mice with dark coats would be less likely to survive and reproduce, so the $MC1R\ D$ allele frequency would decrease over time.

95. The researchers determined that the East lava flow population is not in Hardy-Weinberg equilibrium.

Which of the following best explains why the population is not in Hardy-Weinberg equilibrium?

(A) The presence of heterozygotes for the $MC1R$ gene in the East lava flow population indicates that mating is occurring randomly between mice with different coat colors.

(B) The selective pressures on the East lava flow population are not strong enough to maintain Hardy-Weinberg equilibrium.

(C) The East lava flow population does not experience migration, which is a required condition for Hardy-Weinberg equilibrium.

(D) The presence of $MC1R\ d$ alleles in the East lava flow population is an indication of gene flow from mice populations living on light granite rocks. ✓

Answer D

Correct. One of the conditions of Hardy-Weinberg equilibrium is the absence of migration. The higher frequency of the $MC1R\ d$ allele in the East lava flow population compared to other lava flow populations suggests there is a high migration rate into the East lava flow population from nearby populations of rock pocket mice living on the light-colored granite rocks.

96. Which of the following is closest to the frequency of the $MC1R\ D$ allele in the Tule Mountain population?

(A) 0.03 ✓

(B) 0.06

(C) 0.94

(D) 0.97

Answer A

Correct. Since the frequencies of each genotype are provided, and it is not certain that the population is in equilibrium, it is neither necessary nor valid to resort to the Hardy-Weinberg equation for calculations. The frequency of the $MC1R\ D$ allele is equal to the number of $MC1R\ D$ alleles divided by the total number of alleles. The total number of alleles is equal to the number of mice times two; $85 \times 2 = 170$. The number of the $MC1R\ D$ alleles is equal to the number of homozygous dominant mice (DD)

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times two plus the number of heterozygous mice; $(Dd) : 0 \times 2 + 5 = 5$. Therefore, the frequency of the $MC1R\ D$ allele is $5 \div 170 = 0.03$.

Staphylococcus aureus is a pathogenic bacterium that can infect a wide range of host species, including humans. *S. aureus* has a particular protein that binds with hemoglobin from the host organism. Hemoglobin is the iron-containing protein used to transport oxygen in the blood. Since iron is important for growth, *S. aureus* have evolved the ability to absorb the iron from the host's hemoglobin.

Different *S. aureus* strains preferentially infect different hosts and have different amino acid sequences at their hemoglobin-binding domains (Table 1; letters indicate different amino acids). In an experiment, different *S. aureus* strains were mixed with hemoglobin from macaque monkeys and their binding ability was measured (Figure 1). The differences in amino acid sequences contributed to the differential binding abilities observed.

Table 1. Selected amino acid sequences and preferred host for four strains of *S. aureus*

S. aureus Strain	Amino Acid Sequence	Host Species
1	Q Q F Y H Y A R S	Species A
2	R Q A Y H Y A R T	Species B
3	Q Q A Y H Y A R T	Macaque
4	R Q A A H Y Q L T	Species C

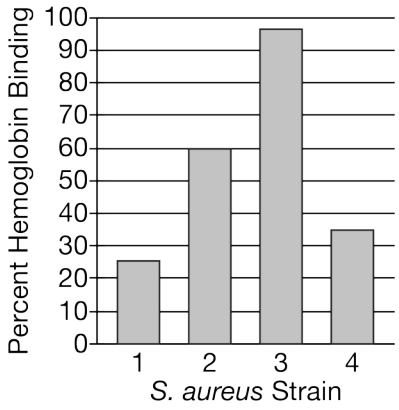


Figure 1. Macaque hemoglobin binding ability of different strains of *S. aureus*

97. Which of the following experiments would be most appropriate to determine whether populations of *S. aureus* are continuously adapting in order to obtain iron from hosts more effectively?

7 Natural Selection MCQ

(A) Grow *S. aureus* in media with macaque hemoglobin and abundant free iron. Record bacterial growth over time and compare it to growth of the original population.

(B) Culture *S. aureus* bacteria with hemoglobin from a mixture of host species. Then transfer the bacteria to media with a known effective antibiotic and record bacterial growth.

(C) Sequence the genes of different *S. aureus* strains and construct a cladogram representing their relatedness.

(D) Culture *S. aureus* bacteria with hemoglobin from a novel host species as the only source of iron for many generations, then compare protein structure from bacteria in this culture with bacteria from the original culture. ✓

Answer D

Correct. If the sequences change over time, this suggests that the bacteria adapted to better obtain iron from the hemoglobin of the novel host.

98. Which of the following processes is most consistent with the differences in the amino acid sequences listed in Table 1 ?

(A) Each host transcribes and translates this gene differently.

(B) *S. aureus* develops a mutualistic relationship with each host species.

(C) Changes in amino acid sequence are unrelated to protein function.

(D) Each strain is best adapted to a specific host species. ✓

Answer D

Correct. Since different strains of *S. aureus* preferentially infect different hosts, they are adapted to use the hemoglobin of their specific host. A strain that infects humans has a protein that binds well to hemoglobin found in humans but not as well to the hemoglobin found in other species.

---

99. The heterotroph hypothesis of the origin of life states that basic organic building blocks formed from inorganic molecules under the primitive conditions present early in Earth's development. These building blocks combined to form polymers, such as crude proteins. Dr. Sidney Fox wished to add to the heterotroph hypothesis of the origin of life by synthesizing simple proteins from a mixture of amino acids.

Which statement below best states an alternative hypothesis Fox could have used to frame his research?

7 Natural Selection MCQ

(A) Formation of crude proteins occurs by chance and occurs regardless of atmospheric O_2 levels or available free energy.

(B) Given aerobic conditions and large amounts of lightning, amino acids will spontaneously form crude proteins.

(C) Formation of simple proteins on primitive Earth only occurs spontaneously if the required building blocks were transported to Earth by a celestial event or a meteorite.

(D) In the presence of available energy and the absence of a significant quantity of atmospheric O_2 simple proteins will form spontaneously from a mixture of amino acids. ✓

Answer D

Correct. The alternative hypothesis should stipulate that the unique conditions of primitive Earth are what enabled the formation of crude protein molecules from amino acids.

100. Figure 1 illustrates a process that occurs in a certain group of organisms.

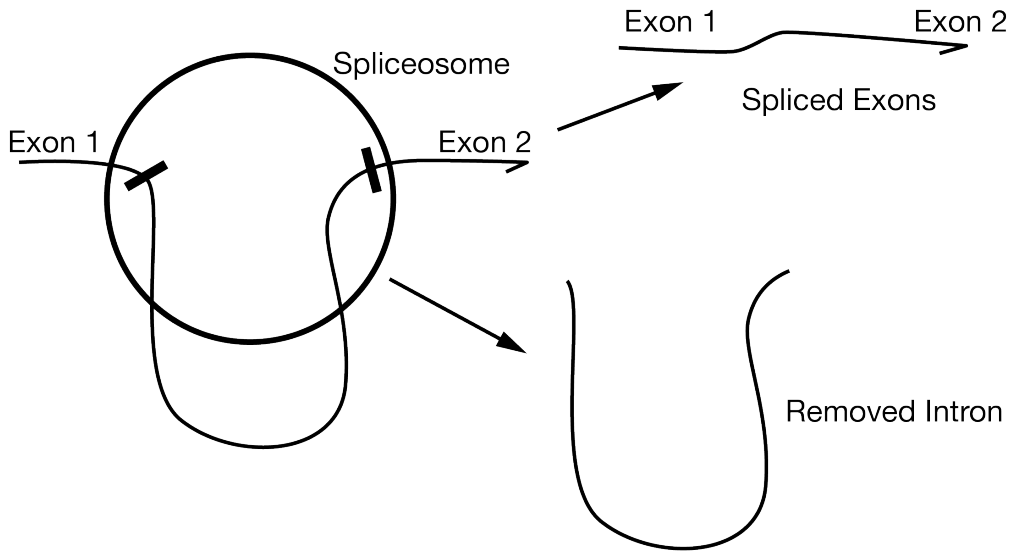


Figure 1. Model of spliceosome activity

Which of the following best predicts the importance of this process?

(A) This process is common to all members of Eukarya and allows noncoding regions of RNA to be removed so that proteins with the correct amino acid sequence are produced. ✓

(B) This process is common to all members of Archaea and allows these organisms to survive harsh environments.

(C) This process is common to all organisms and is a conserved DNA repair mechanism.

(D) This process is common to vertebrates and allows the production of large and complicated proteins.

7 Natural Selection MCQ

Answer A

Correct. All eukaryotes have introns that must be removed, and exons must be joined together in order to translate eukaryotic mRNA and produce a functional protein.

101. The Stanley Miller apparatus demonstrated that organic molecules could assemble spontaneously in an environment lacking free oxygen and containing water, methane, and ammonia in the presence of an abundant energy source, such as an electric discharge. The research was considered supportive of the organic soup hypothesis, which states that the primitive atmosphere provided inorganic precursors from which organic molecules could have been synthesized in the presence of an energy source. Based on subsequent research, the primordial atmosphere was determined to contain less methane and more carbon dioxide. The new data about the composition of the early atmosphere had which of the following effects on origin-of-life hypotheses?

(A) Miller's work was discarded as not relevant to an origin-of-life hypothesis.

(B) Miller's work was used to refute the new hypothesis about the composition of the atmosphere.

(C) The organic soup hypothesis was abandoned as no longer being supportable.

(D) A new organic soup hypothesis was proposed to account for the new data about the atmosphere. ✓

102. To prevent the loss of genes as chromosome ends wear down during DNA replication, the tips of eukaryotic chromosomes have specialized DNA "caps" called telomeres. Telomeres consist of hundreds or thousands of repeats of the same short DNA sequence, which varies between organisms. The telomeric repeat sequences for selected organisms are given in Table 1.

Table 1. Telomeric Repeat Sequences for Selected Organisms

Organism	Telomeric Repeat Sequence (5′ to 3′)
Human	TTAGGG
Mouse	TTAGGG
Paramecium	TTGGGT
Green alga	TTTTAGGG
Onion	CTCGGTTATGGG

Which of the following best predicts why humans and mice have identical telomeric repeat sequences?

(A) Humans and mice have the same number of genes.

(B) Humans and mice have the same number of chromosomes.

(C) Humans and mice share a recent common ancestor that had the 5′–TTAGGG–3′ repeat sequence in its telomeres. ✓

(D) Mice utilized the 5′-TTAGGG-3′ telomeric repeat sequence first, but the sequence independently evolved in humans as well.

7 Natural Selection MCQ

Answer C

Correct. The most likely cause of homology in telomeric repeat sequences is common ancestry between humans and mice. This is an example of structural evidence on the molecular level that indicates common ancestry.

103. One facet of the study of the origin of life focuses on the nature of the first genetic material. Figure 1 describes a possible scenario, known as the RNA World Hypothesis, that could have led to the evolution of our modern-day genetic material. This hypothesis states that the first genetic material was RNA, not DNA, and a transitional period led to the emergence of DNA as the storage molecule for genetic material.

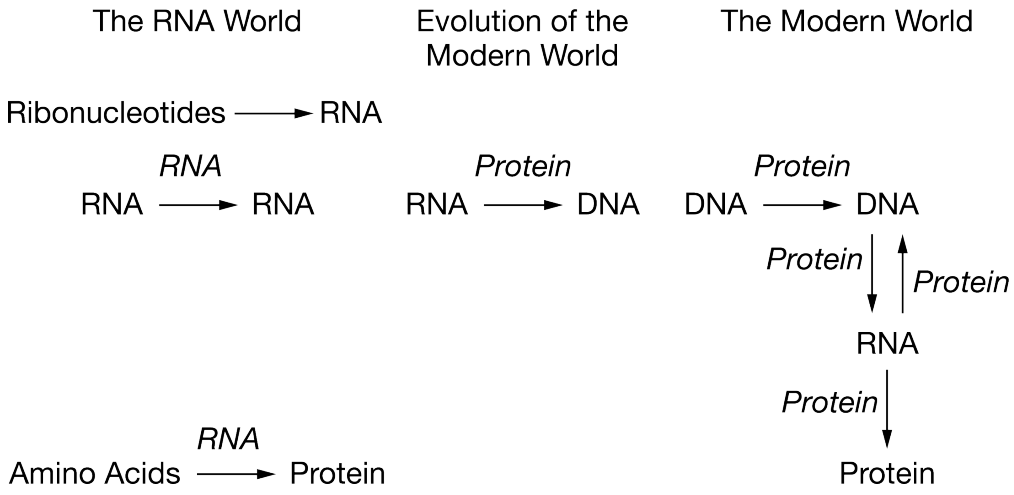


Figure 1. A model of the RNA World Hypothesis

A scientist wishes to provide experimental evidence to support the model shown in Figure 1 by demonstrating the ability to synthesize an RNA molecule. Which of the following is an alternative hypothesis that can be tested to support the RNA World Hypothesis?

(A) A protein template that matches proteins found on the primitive Earth added to a mixture of deoxyribonucleotides will produce double-stranded DNA molecules.

(B) Double strands of DNA placed in a harsh environment will unwind and form RNA molecules for long term information storage.

(C) A pool of amino acids can spontaneously form into single-stranded RNA, which would then code for proteins.

(D) A mixture of ribonucleotides such as adenine and uracil will spontaneously form single-stranded chains of RNA. ✓

7 Natural Selection MCQ

Answer D

Correct. Producing this result in a laboratory would support the claim that RNA molecules could have formed spontaneously in the primitive Earth environment.

104. Transposable elements are significant features in the genomes of almost all eukaryotic organisms. The accumulation of these elements in various species can help determine their relatedness. Table 1 indicates the presence (+) or absence (−) of 5 segments of DNA that have been moved by transposition (transposons) in 6 primate species.

Table 1. Presence or absence of 5 transposons in primate species

Species	1	2	3	4	5
Human	−	+	+	+	+
Bonobo	+	+	+	+	+
Chimpanzee	+	+	+	+	+
Gorilla	−	−	+	+	+
Orangutan	−	−	−	+	+
Gibbon	−	−	−	−	−

Which of the following best predicts how phylogenetic relationships might be revised if transposon 1 was not found in chimpanzees?

(A) Bonobos would be more closely related to humans than to chimpanzees.

(B) Chimpanzees would be more closely related to humans than to bonobos. ✓

(C) Orangutans would be more closely related to humans than to chimpanzees.

(D) Chimpanzees would be more closely related to gibbons than to gorillas.

Answer B

Correct. If chimpanzees lost transposon 1, their transposon pattern would be identical to that of humans.

7 Natural Selection MCQ

105.

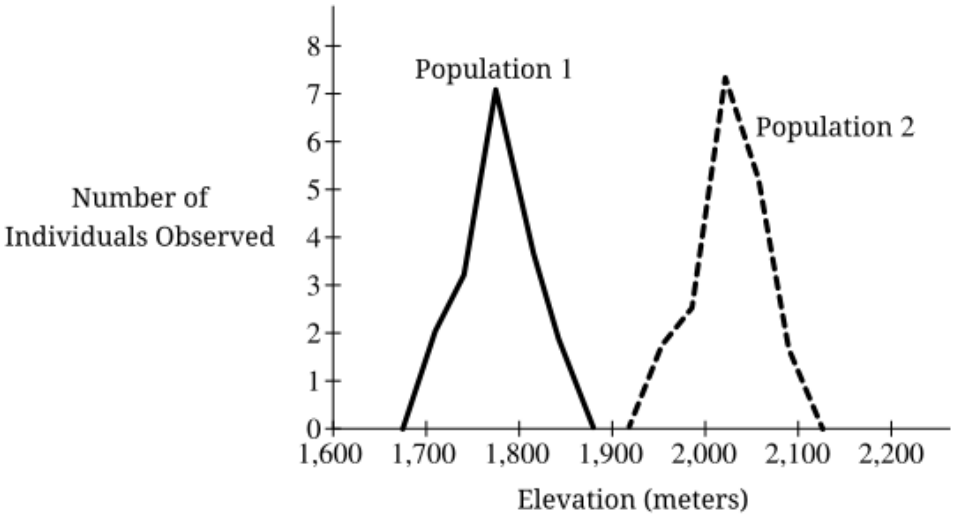


Figure 1. The distribution of individuals from two populations of the same species at different elevations

Researchers investigating two populations of the same species recorded the elevation at which individuals from each population were found (Figure 1).

Which of the following claims is best supported by the data in Figure 1 ?

(A) Gene flow is likely to occur between the two populations because they share the same habitat.

(B) Gene flow is likely to occur between the two populations because they are the same species.

(C) Gene flow is unlikely to occur between the two populations because they compete for food.

(D) Gene flow is unlikely to occur between the two populations because they are geographically isolated. ✓

Answer D

Correct. Based on the data in Figure 1, the individuals of the two populations are found in non-overlapping habitats at different elevations. Because individuals from the two populations do not meet, there will be no reproduction between members of the populations and thus no gene flow between the populations.

7 Natural Selection MCQ

106. A 100 acre woodlot was observed for the last forty years. Five plant species (1 through 5) were surveyed, with the results shown in Table 1.

Table 1. Data from Forty-Year Survey of Five Plant Species in a Woodlot

Year of Survey	Species 1 Population	Species 2 Population	Species 3 Population	Species 4 Population	Species 5 Population
1	35	30	23	32	25
20	35	42	20	2	1
40	50	50	2	0	0

Some researchers claim that the decrease in plant species richness of this woodlot is unrelated to human activities over the last forty years.

Which of the following would be an appropriate alternative hypothesis?

(A) Interspecific competition caused the extinction of species 4 and 5.

(B) The woodlot environment is arid and species 1 and 2 are well-adapted to conserve water.

(C) The woodlot is shaded, with little direct sunlight, and species 3, 4, and 5 are shade-intolerant canopy trees.

(D) During the last few decades, tall buildings have been built around the woodlot, and species 3, 4, and 5 are shade-intolerant plants ✓

Answer D

Correct. The construction of tall buildings over the last few decades that surround the woodlot has made the woodlot heavily shaded. This means that shade-intolerant species will be eliminated from the woodlot over time. Therefore, human activities have changed the environment of the woodlot and caused extinction of species 3, 4, and 5.

107. A species of snail lives in the intertidal zone along the coast of New England. The dark-colored variety of the species is more common in northern New England, the light-colored variety is more common two hundred miles away in southern New England, and both varieties are commonly found together in central New England. Which of the following best explains the observed distribution pattern of the snails?

7 Natural Selection MCQ

(A) The founder effect suggests that dark-colored snails migrated from the southern regions to the north and established the populations found there.

(B) Genetic drift at the shell-color locus caused the northern population to become homozygous for the dark-color allele.

(C) The mutation rate is higher in the south, as the longer days expose the snails to more ultraviolet radiation than in the north.

(D) Dark-colored snails absorb more solar energy and so survive more readily in the colder northern waters. ✓

108. Hunting of African elephants for their tusks resulted in a 90% reduction in the size of one particular elephant population by 1992. Since then, protections of elephant populations have improved, and the population size has begun to increase. Before 1992, tuskless elephants were rare, but now a significant proportion of the female elephants in this population are tuskless. An allele of the gene $AMELX$ appears to be responsible for the tuskless trait.

Which of the following best explains the change in the relative proportion of tuskless elephants since 1992 ?

(A) With human protections in place, tusks are less important to the protection of the herds and in particular to the females, so tusks are selected against.

(B) Female tuskless elephants were more likely to have survived than were tusked females, resulting in a greater proportion of the offspring having the genes of the tuskless elephants. ✓

(C) Female elephants that survived removal of their tusks were more likely to pass the tuskless allele than the tusked allele of the $AMELX$ gene to their female offspring.

(D) Overhunting of the elephants caused the emergence of the tuskless mutation in the $AMELX$ gene, leading to an increase in the frequency of the tuskless phenotype.

Answer B

Correct. Since tusked elephants were hunted, tuskless elephants were more likely to survive before protective measures were enacted. Traits of the tuskless elephants, such as that encoded by the $AMELX$ mutation, would have passed on to offspring, allowing the population to evolve over time to include more tuskless females.

109. Which of the following is probably the best explanation for the fact that Antarctic penguins cannot fly, although there is evidence that millions of years ago their ancestors could do so?

(A) Penguins live on land and feed in the water; therefore they have no need to fly.

(B) The Antarctic home of penguins is flat and barren; therefore there is no place to fly.

(C) Ancestral penguins without large wings were better able to swim and feed in the water; therefore they passed their genes for shorter wing structure on to their offspring. ✓

(D) Ancestral penguins did not use their wings for long periods of time; therefore today's penguins have only tiny, nonfunctional wings.

(E) The cold and wind of Antarctica make flight impossible; therefore penguins that live there have lost the ability to fly.

7 Natural Selection MCQ

110. Competition for food would probably be most severe between two

(A) closely related species in different niches

(B) closely related species in similar niches ✓

(C) unrelated species in different communities

(D) unrelated species in the same community occupying different niches

(E) ecological equivalents in different niches

111. In Darwinian terms, the fittest individuals of a species are those that

(A) attract the most mates

(B) grow to the largest size

(C) are best adapted to the temperature extremes to which they are exposed

(D) are best equipped to cope with the predators to which they are exposed

(E) leave the greatest number of reproducing descendants ✓

7 Natural Selection MCQ

112. African elephants, *Loxodonta africana*, are often hunted illegally for their tusks. Both male and female elephants have tusks, although the tusks are much larger in the males. Researchers have followed the elephant population in Gorongosa National Park in Mozambique for many years. Figure 1 shows the percent of tuskless female elephants expected in wild populations throughout the species' range, as well as the percent of tuskless females that survived the hunting between 1977 and 1992 (those over 25 years old) and the percent of female elephants born after 1992 without tusks in the park. The data are based on 200 known female elephants in the park.

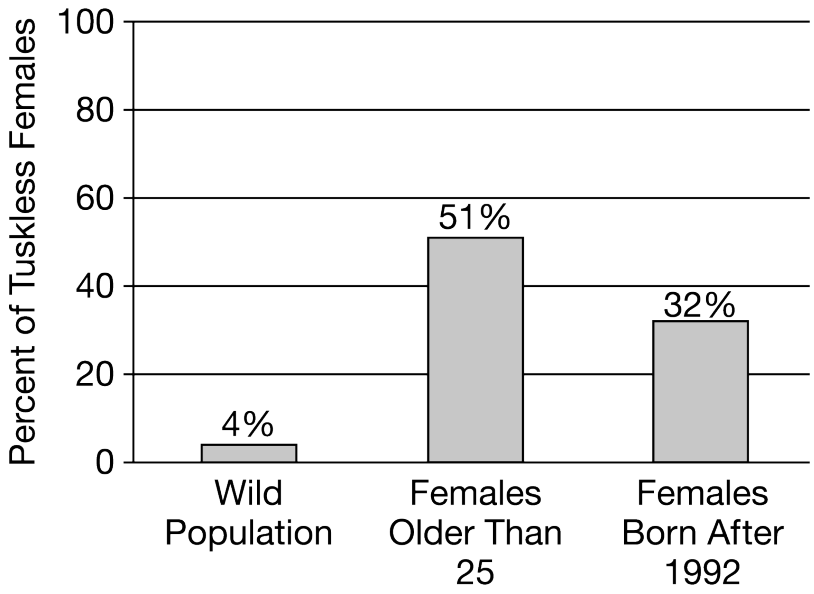


Figure 1. Comparison of the frequency of tuskless females by age cohort

Which of the following best describes the process responsible for the change in the percent of tuskless female elephants in the Gorongosa National Park population shown in Figure 1 ?

(A) Human hunting of elephants with tusks between 1977 and 1992. ✓

(B) Sexual selection for females without tusks between 1977 and 1992.

(C) Tusklessness in female elephants is a dominant trait that is only expressed in females.

(D) After 1992, female elephants from neighboring populations merged, with the population in the park adding more tuskless individuals to the population.

Answer A

Correct. The increased survival of elephants without tusks has resulted in tuskless female elephants leaving more offspring to reproduce than the female elephants that were hunted for their tusks.

113. Evolutionary fitness is measured by

7 Natural Selection MCQ

(A) physical strength

(B) reproductive success ✓

(C) length of life

(D) resistance to disease

(E) competitiveness

114. Scientists investigated the role that beak depth plays in the ability of one species of seed-eating finch to reproduce. The scientists calculated the average beak depth of finches in mating pairs and then observed whether or not the pairs produced at least one offspring that survived to the next season. The data are represented in Figure 1.

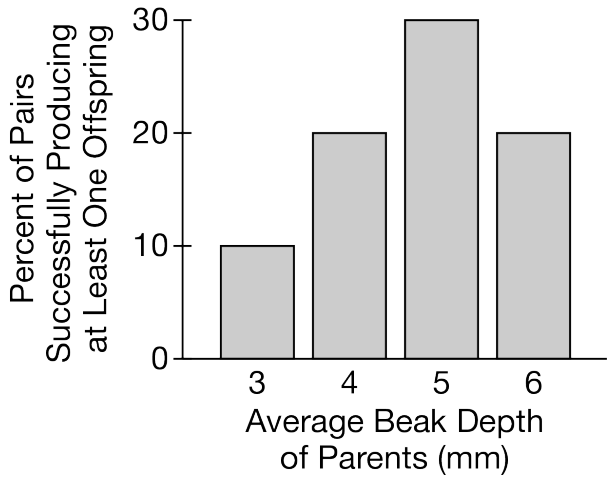


Figure 1. Beak depth and offspring survival in a species of finch

Based on the data in Figure 1, which of the following best describes the concept illustrated?

(A) Parental pairs with a specific beak depth had the highest reproductive fitness. ✓

(B) Parental pairs with a specific beak depth ate the most nutritious seeds.

(C) Finches with a certain beak depth rarely find mates.

(D) Increasing average beak depth results in increasing finch fitness.

Answer A

Correct. Darwinian fitness is the ability to successfully reproduce, meaning the finches leave surviving offspring. The parental pairs with an average beak depth of 5 mm had the greatest reproductive success.

7 Natural Selection MCQ

115. Tobacco hawk moths, *Manduca sexta*, feed exclusively on sugar-rich nectar produced in the flowers of tobacco plants. To feed, the moth inserts its proboscis (a long, tubular mouthpart) into the flower to reach the nectary, which is usually at the end of a long tube in the flower. Moths will feed on the nectar of the tobacco plant species whose nectaries they can access. A study determined the proboscis sizes in one tobacco hawk moth population. The data are shown in Figure 1. The most abundant tobacco species in the population's habitat has a nectary tube that is $5\ \text{cm}$ long.

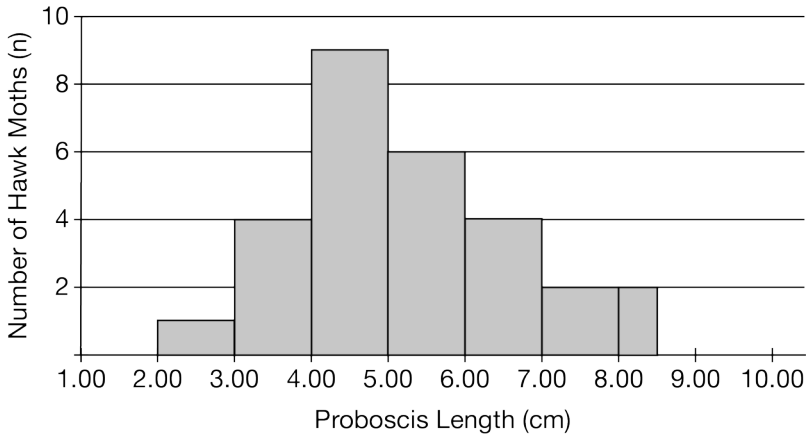


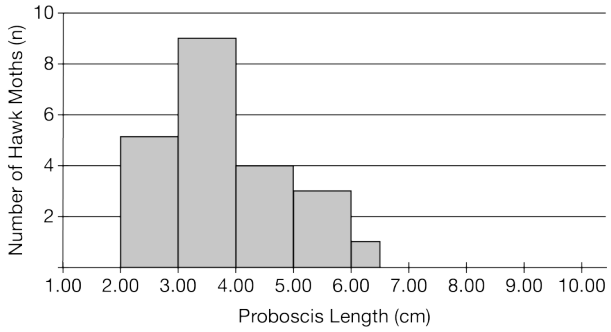
Figure 1. Proboscis length distribution in a population of tobacco hawk moths

Over the next year, a virus attacks and rapidly kills almost all plants of the tobacco species. A closely related species of tobacco that is resistant to the virus becomes established during that same year. The nectary tube in the resistant species averages $7.5\ \text{cm}$. Scientists monitored the area for the next five years to determine how this change affected the moth population.

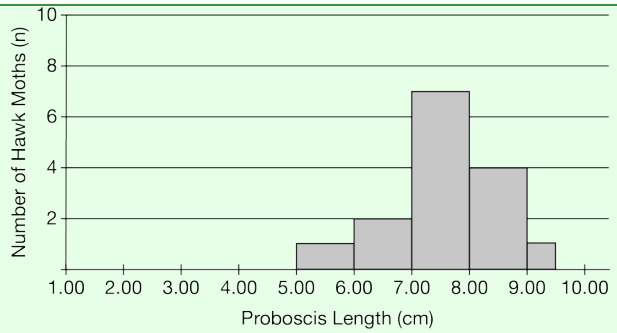
Which of the following best describes the expected distribution of proboscis size of the moths at the end of the five-year study?

7 Natural Selection MCQ

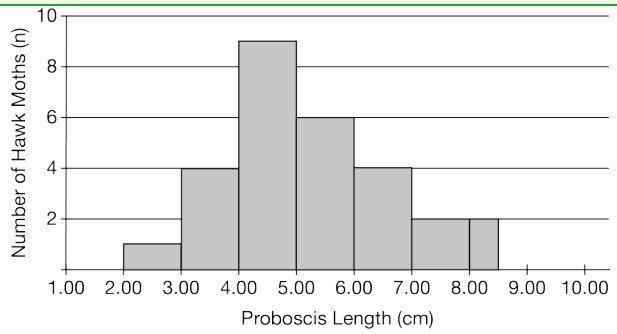
(A)



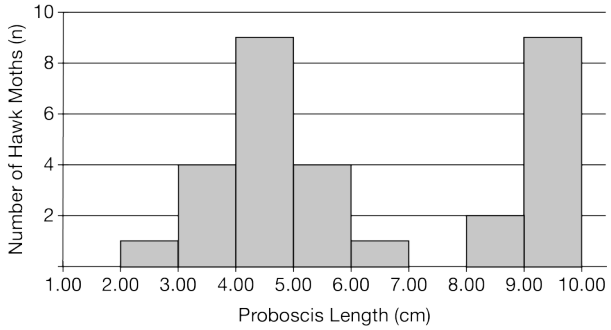
(B) ✓



(C)



(D)



Answer B

Correct. Since the moths will be relying more heavily on tobacco plants with necataries that are at the ends of longer tubes, moths with longer proboscises should have easier access to food, allowing them to survive and reproduce and pass on the genes that result in longer proboscises to their offspring.

7 Natural Selection MCQ

116. Students created a model showing one possible mechanism for speciation in which a geographic barrier separated one large population into two smaller populations, A and B. Over time, populations A and B developed into two distinct species. The model is shown in Figure 1.

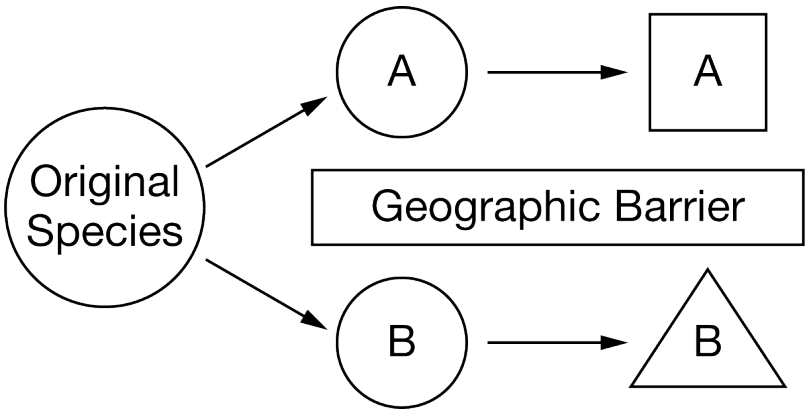


Figure 1. Model of speciation

Which of the following best describes what occurred during the periods indicated by the arrows?

(A) Natural selection acted on variation that was present in both populations A and B, and the lack of gene flow eventually led to the formation of two new species. ✓

(B) Population B was separated for a longer time from the original population than population A was, and population B evolved into a more distinct new species than population A did.

(C) Gene flow between populations A and B maintained sufficient variation in both gene pools to facilitate the development of two distinct species.

(D) In response to different environments, populations A and B experienced different mutation rates, leading to the emergence of two different gene pools and, over time, new species A and B.

Answer A

Correct. Allopatric speciation is a result of differential natural selection acting on two populations that have been separated geographically. Populations A and B became sufficiently different due to accumulated changes to their gene pools.

117. Which of the following principles is NOT part of Darwin's theory of evolution by natural selection?

7 Natural Selection MCQ

(A) Evolution is a gradual process that occurs over long periods of time.
(B) Variation occurs among individuals in a population.
(C) Mutations are the ultimate source of genetic variation. ✓
(D) More individuals are born than will survive.
(E) Individuals that possess the most favorable variations have the best chance of repro-ducing.

118. Which of the following statements best summarizes organic evolution as it is viewed by modern evolutionists?

(A) It is goal directed.
(B) It represents the results of selection for acquired characteristics.
(C) It is synonymous with the process of gene flow.
(D) It is the descent of humans from the present-day great apes.
(E) It is the differential survival and reproduction of certain phenotypes. ✓

Stickleback fish are found in both marine and freshwater habitats. The marine fish have no scales but have hardened, armorlike plates along their sides. The plates are thought to protect sticklebacks from certain predators.

In the late 1980s, sticklebacks from a marine population colonized Loberg Lake, a freshwater lake in Alaska. Starting in 1990, researchers sampled fish from the lake every four years and recorded the armor-plate phenotypes of the male sticklebacks in each sample. The armor-plate phenotypes were categorized as either complete (plates extending from head to tail), partial (plates extending from head to abdomen), or low (a few plates near the head only). The results are shown in the table below.

ARMOR-PLATE VARIATION IN THE STICKLEBACK POPULATION OF LOBERG LAKE

Percent of Males in the Sample with Each Armor-Plate Phenotype

Year	Low	Partial	Complete
1990	1%	2%	97%
1994	45%	14%	41%
1998	58%	16%	26%
2002	76%	15%	9%
2006	90%	6%	4%

119. Which of the following is an independent variable in the investigation?

7 Natural Selection MCQ

(A) Year in which the sample was collected ✓

(B) Salinity of the water in Loberg Lake

(C) Size of the stickleback population

(D) Percent of males with each armor-plate phenotype

120. To evaluate the reliability of the results, it would be best to know which of the following?

(A) The surface area of Loberg Lake

(B) The number of males in each sample ✓

(C) The average daily temperature of Loberg Lake

(D) The age of the sticklebacks in each sample

121. Which of the following best explains the changes in the phenotype frequencies of the stickleback population in Loberg Lake?

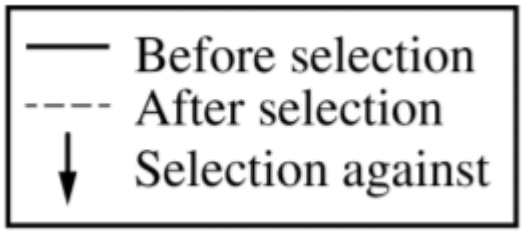
(A) Predation in the marine environment is different from predation in Loberg Lake. ✓

(B) Marine predators and sticklebacks both colonized Loberg Lake.

(C) Sticklebacks with the partial armor-plate phenotype have the highest rate of reproduction and survival.

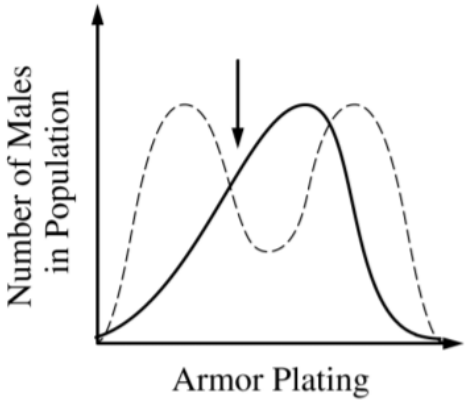
(D) The population of sticklebacks in Loberg Lake is subject to genetic drift.

122. Which of the following graphs best represents the type of selection most likely operating in the stickleback population of Loberg Lake?

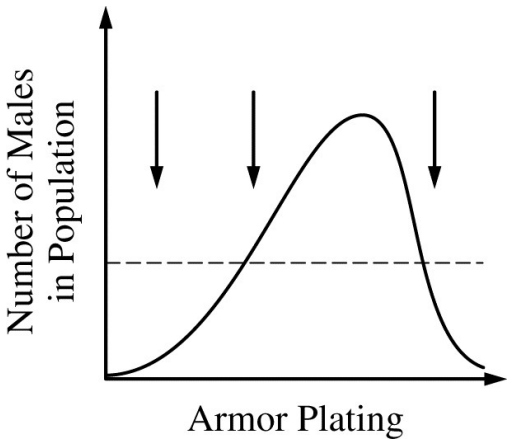


7 Natural Selection MCQ

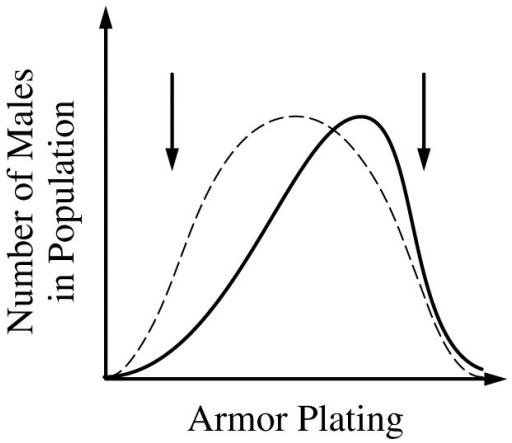
(A)



(B)

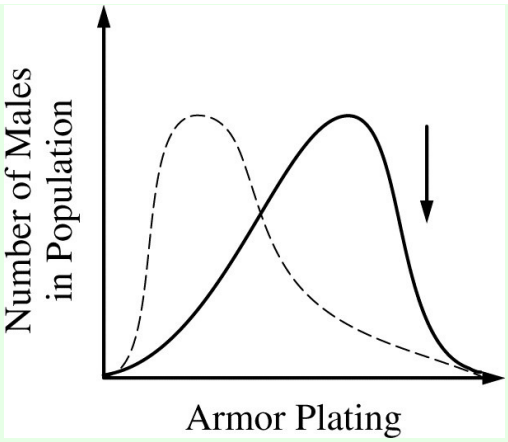


(C)



(D) ✓

7 Natural Selection MCQ



123. In a population of coyotes, there is variation in the size of individual animals. Larger animals are able to bring down larger prey but also require more food to survive. In a particular area, the primary food source for many years had been rabbits, but disease led to a sudden crash in the rabbit population. However, the deer population in the area increased at about the same time. Over time, naturalists noted that the average size of coyotes in the population had increased.

Which of the following is the best explanation for the increase in average coyote size following the crash in the rabbit population?

(A) Deer run faster than rabbits, so coyotes needed to get larger to survive.

(B) Smaller coyotes were at a hunting disadvantage, so fewer survived long enough to reproduce. ✓

(C) Since the coyotes needed to be larger to bring down the deer, they grew larger.

(D) Since smaller coyotes struggled to obtain enough food, they produced larger offspring to improve their fitness.

Answer B

Correct. The information provided indicates that larger coyotes can catch larger prey, so they would have a hunting advantage over smaller coyotes when forced to catch larger prey after the population size of the preferred smaller food source, rabbits, had decreased.

7 Natural Selection MCQ

124. In an experiment, 100 mice were released into a field to which no other mice had access. Immediately after their release, a representative sample of mice was captured, their fur color was recorded, and they were returned to the field. After twenty years, a representative sample of mice was captured and the distribution of fur color was again recorded.

Fur Color	Initial Distribution (%)	Distribution after Twenty Years (%)
Brown	80	20
Gray	10	75
Black	10	5

Which of the following could best explain the change in fur color distribution, as shown in the table above?

(A) The allele for black fur color is unstable, and over twenty years most of the black fur alleles mutated to become alleles for gray fur.

(B) The field was primarily composed of light-colored soil and little vegetation, affording gray mice protection from predators. ✓

(C) Sexual selection led to increased mating frequency of black and brown versus gray and brown.

(D) The gray mice were harder to catch, and so were underrepresented in the twenty-year sample.

125. A researcher claims that genetic variation provides organisms with the ability to survive and reproduce in different environments. To support the claim, the researcher makes the following observation: bacteria that contain plasmids (small DNA molecules) are resistant to a wider range of antibiotics than are bacteria that contain no plasmids.

Which of the following statements best establishes a connection between the observation and the researcher's claim?

(A) Some antibiotics inhibit the synthesis of bacterial cell walls.

(B) Some antibiotics inhibit protein synthesis in bacteria cells.

(C) Some plasmids cannot exist in a bacterial cell with certain other plasmids.

(D) Some plasmids contain antibiotic resistance genes. ✓

Answer D

Correct. Because some plasmids contain antibiotic resistance genes, bacteria that contain plasmids are more likely to be resistant to antibiotics than are bacteria that do not contain plasmids. The observation is evidence that genetic variation in the form of plasmids provides bacteria with the ability to survive and reproduce in different environments.

7 Natural Selection MCQ

126. A researcher claims that different metabolic pathways allow bacteria to use different molecules as sources of matter and energy.

Which of the following statements best helps justify the researcher's claim by providing a relevant example?

(A) Rhizobia bacteria form close associations with the roots of bean plants.

(B) *E. coli* bacteria reproduce in liquid media containing either glucose or galactose. ✓

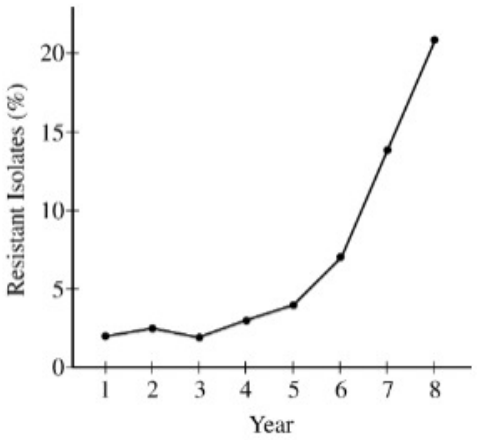
(C) The antibiotic rifampicin inhibits the growth of some bacterial strains but not of others.

(D) Some viruses that infect bacteria reproduce by either the lysogenic cycle or the lytic cycle.

Answer B

Correct. Glucose and galactose are two different monosaccharides that can be used by organisms as sources of matter and energy. The ability of *E. coli* bacteria to reproduce in liquid media containing either glucose or galactose provides a relevant example that will help justify the researcher's claim.

127.



Over several years, bacteria were isolated from members of a human population and tested for antibiotic resistance. The percent of bacterial isolates that were found to be antibiotic resistant is presented in the graph above for each year of the study. Which of the following conclusions is best supported by the information presented in the graph?

(A) The greatest increase in antibiotic resistance occurred from the first year to the third year of the study.

(B) Gene flow in the population occurred at the same rate for the entire duration of the eight-year study.

(C) Because evolution in bacterial populations occurs over millions of years, the results are due to random variation.

(D) Natural selection over multiple generations is favoring antibiotic resistance in the population of bacteria. ✓

7 Natural Selection MCQ

128. A current challenge for doctors involves the bacterial strain *Clostridioides difficile,* which no longer responds to traditional antibiotic treatments.

Which of the following best explains why this particular strain of bacteria is resistant to antibiotic treatment?

(A) Mutations creating new alleles occur more frequently in this species of bacteria than in other species.

(B) The bacteria have high genetic variability and high reproductive rates. ✓

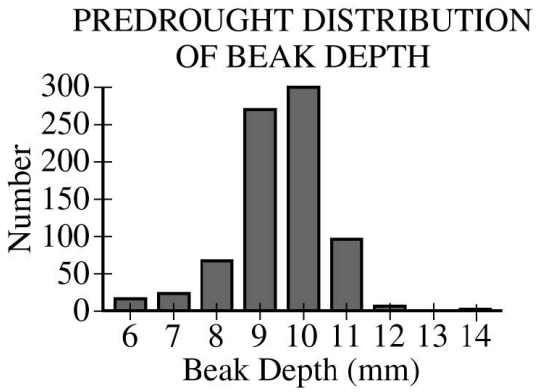
(C) Crossing over during meiosis increased genetic variability in the bacteria.

(D) The bacteria are able to recognize and destroy the antibiotics by breaking them down extracellularly.

Answer B

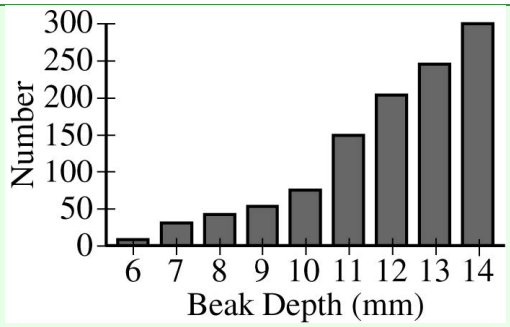
Correct. These processes promote rapid evolution of a species. Bacteria can produce several generations in a single day; each generation produced under the severe selective pressures of antibiotics is more likely to be antibiotic resistant.

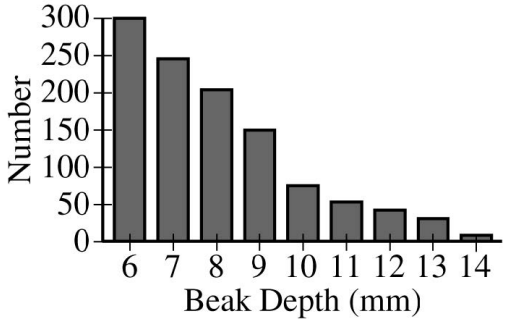
129.

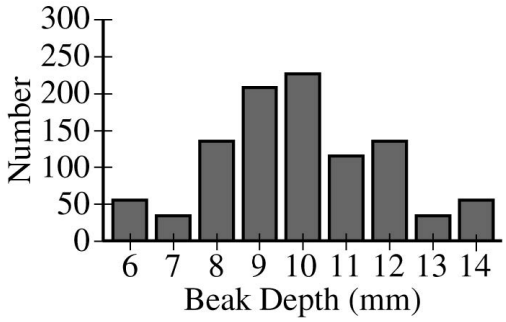


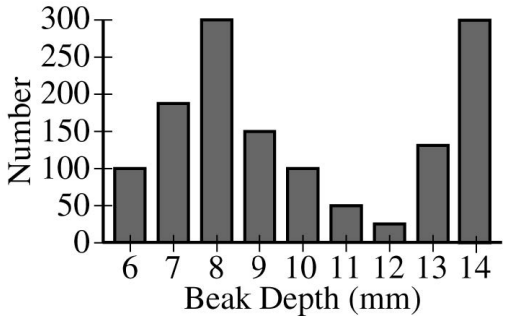
The graph above shows the distribution of beak depth in a finch population that had been living on an island under conditions of normal rainfall. During a subsequent drought, the small seeds normally eaten by the finches were less available. Most of the available seeds were large seeds that could be eaten most easily by finches with deep beaks. Which of the following graphs best predicts the distribution of beak depth in the finch population after several years of drought?

7 Natural Selection MCQ

(A) 

(B) 

(C) 

(D) 

130. The bird, *Parus major*, is commonly found in man-made habitats. During the summer these birds primarily feed on insects, and in the winter, when insects are scarce, they forage in gardens and bird feeders for seeds, nuts, and berries. Recently, climate change has led to a decrease in food available from gardens, and scientists have observed that the average beak length of the birds has increased.

Which of the following best explains the directional shift in beak length in these birds?

7 Natural Selection MCQ

(A) Longer beaks allow the birds to better fend off predators.

(B) Longer beaks increase the fitness of the birds in wild habitats.

(C) Longer beaks allow the birds to better access seeds in bird feeders. ✓

(D) Longer beaks decrease the fitness of the birds in wild habitats.

Answer C

Correct. Since there are fewer resources available in gardens, it is advantageous for the birds to be able to obtain food from bird feeders more effectively. The observed phenotypic variation is likely the result of birds with longer beaks being able to better extract seeds form bird feeders, and therefore be better able to survive and pass on the traits for longer beaks to their offspring.

131.

TABLE 1. PHOTOPIGMENTS IN CYANOBACTERIA

Pigment	Color of Reflected Light
Chlorophyll *a*	Green
Phycocyanin	Blue
Allophycocyanin	Blue
Phycoerythrin	Red

Cyanobacteria contain a variety of pigment molecules, as shown in Table 1. As a result, the color of cyanobacteria cultures can vary significantly based on the relative amount of each pigment produced.

A researcher placed a culture of cyanobacteria under green lights. Within a few weeks, the appearance of the cyanobacteria changed from green to red. The researcher claimed the color change in the culture was the result of an adaptation allowing greater photosynthesis.

Which of the following provides the best reasoning to justify the researcher's claim?

(A) In green light, more chlorophyll *a* molecules are produced, reflecting more light to other cyanobacteria to be used for photosynthesis.

(B) In green light, more phycoerythrin molecules are produced, allowing more green light to be absorbed, thus increasing photosynthesis. ✓

(C) In green light, cyanobacteria that have more phycocyanin molecules are less likely to survive and reproduce.

(D) In green light, cyanobacteria that have more allophycocyanin molecules are more likely to survive and reproduce.

7 Natural Selection MCQ

Answer B

Correct. Cyanobacteria exposed to green light would increase production of pigment molecules that can absorb green light and reflect red light, such as phycoerythrin. As a result, the color of the culture would change from green to red. In this way, the organisms are better able to survive under different conditions.

132. Ciprofloxacin is given as an antibiotic to healthy livestock to promote efficient weight gain. Strains of the bacterium *Campylobacter jejuni* naturally colonize the digestive tracts of livestock, and the *C. jejuni* can be transferred to people through the handling and consumption of contaminated meat. People infected with *C. jejuni* also are treated with ciprofloxacin.

Which of the following is the most likely consequence of adding ciprofloxacin to animal feed?

(A) Naturally occurring strains of *C. jejuni* will reproduce more rapidly in the digestive tracts of livestock, reducing the risk of food-borne illnesses in people.

(B) Ciprofloxacin-resistant strains of *C. jejuni* will have a selective advantage in the digestive tracts of livestock, increasing the risk of serious infections in people. ✓

(C) Ciprofloxacin-sensitive strains of *C. jejuni* will be consumed in meat that contains a chemical that inhibits bacterial growth, reducing the risk of food poisoning in people.

(D) Actively growing strains of *C. jejuni* will metabolize the nutrients in meat produced by livestock operations, increasing the risk of protein deficiencies in people.

133. Phycobiliproteins are a complex of accessory pigments and proteins found in cyanobacteria but not in green algae. A researcher claims that the phycobiliprotein pigments in cyanobacteria allow the cyanobacteria to survive in certain aquatic niches better than green algae can.

Which of the following statements best justifies the researcher's claim?

(A) The additional pigments allow the cyanobacteria to store light energy so that it can be used at night to continue photosynthesis.

(B) The additional pigments block light and prevent it from reaching photosynthetic organisms at greater depths, so no photosynthetic organisms can live below the surface waters containing cyanobacteria.

(C) The additional pigments require energy and cellular resources to produce, so they can be used as an energy source during times of insufficient light.

(D) The additional pigments absorb light at wavelengths that green algae cannot absorb; this may allow cyanobacteria to capture more light energy for photosynthesis than green algae can in certain areas. ✓

Answer D

Correct. Since the phycobiliprotein pigments absorb light wavelengths not absorbed by green algae, cyanobacteria are able to better perform photosynthesis and survive in areas where there is insufficient light at the needed frequencies for green algae to survive.

7 Natural Selection MCQ

Diapause is the interruption of an organism's life cycle in response to environmental cues. The soil nematode *Caenorhabditis elegans* is capable of entering adult reproductive diapause (ARD) when food is scarce. In *C. elegans*, individuals normally become reproductively mature 2 days after hatching and remain fertile for 18 days. They reproduce either by self-fertilization or by mating with another individual.

In an investigation, researchers examined the survival and reproductive success of *C. elegans* following different times in ARD. In the first experiment, groups of *C. elegans* were held in ARD without food for 0-30 days. Upon reintroduction of food, average brood sizes (average numbers of offspring per adult) were determined following either self-fertilization or mating with a well-fed male. The results are shown in Figure 1.

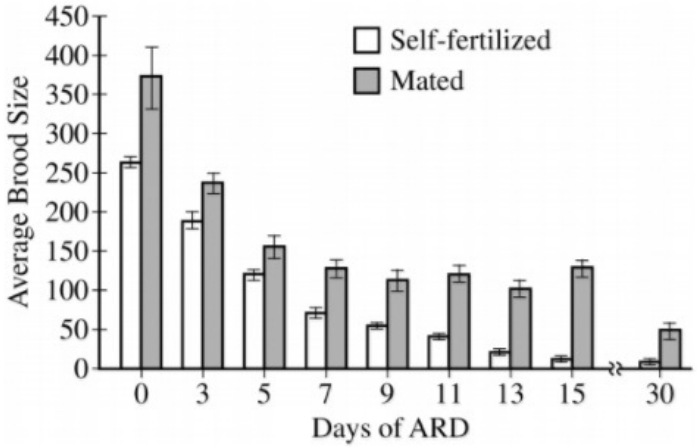


Figure 1. Mean brood sizes $\pm 2SE_{\overline{X}}$ after different times in ARD.

Individual *C. elegans* were held in ARD and subsequently

allowed to reproduce either via self-fertilization (unshaded bars)

or by mating with well-fed males (shaded bars).

In a second experiment, individuals were held in ARD without food for $0-30$ days and monitored for average survival times following reintroduction of food (Figure 2).

7 Natural Selection MCQ

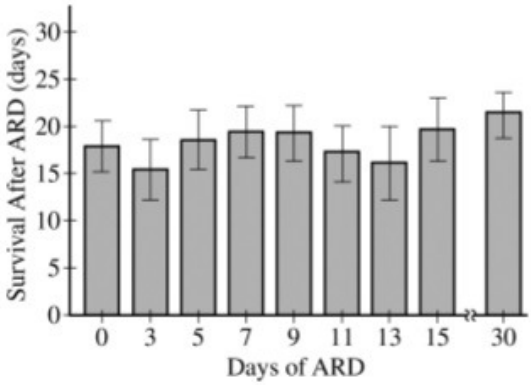


Figure 2. Mean survival $\pm 2SE_{\overline{x}}$ following different times in ARD. Individual *C. elegans* were held in ARD and subsequently given access to food, whereupon their survival times were determined.

134. Based on the experimental results, which of the following is the best evolutionary explanation for the occurrence of ARD in *C. elegans* ?

(A) The ability to enter ARD provides a strong selective advantage because reproduction can occur despite periods of food scarcity. ✓

(B) Acquiring the genes for ARD gives individuals a selective advantage because they produce more offspring than do individuals who cannot enter ARD.

(C) Individuals who can enter ARD are selected for in the population because they live longer than do individuals who cannot enter ARD.

(D) Individuals who can enter ARD have high fitness because they can reproduce even when food is scarce.

7 Natural Selection MCQ

Excess intracellular iron is toxic to cells (iron-induced toxicity). Ferritin is an intracellular iron storage protein that binds excess iron. The presence of ferritin can protect cells from iron-induced toxicity.

In an experiment to investigate the effects of dietary iron intake on ferritin synthesis, rats were given food containing different amounts of iron. Subsequently, the levels of ferritin protein in the liver were measured. The results are shown in Figure 1.

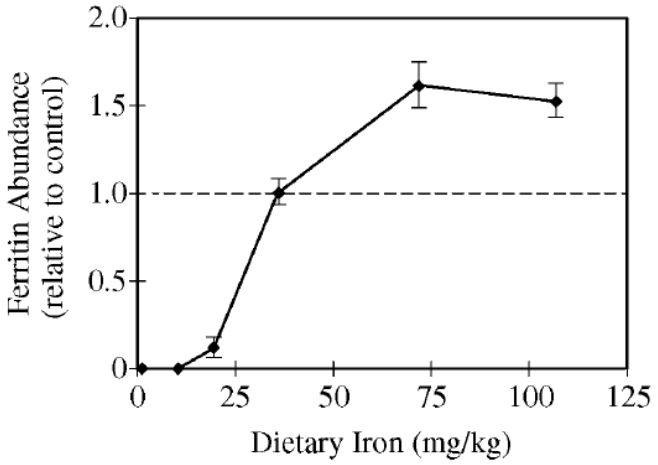
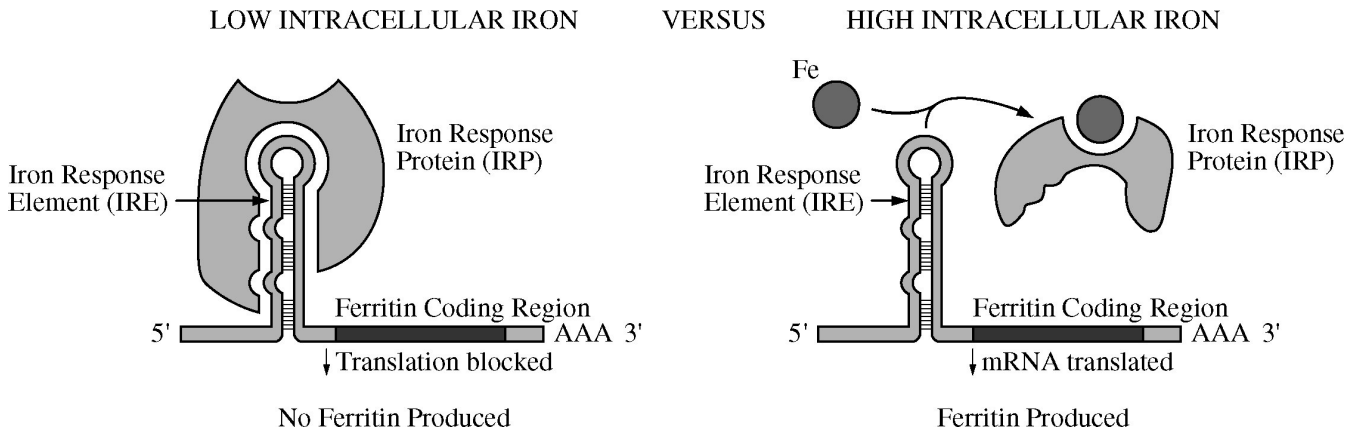


Figure 1. Effects of dietary iron on ferritin levels in rat liver

Based on these and other data, researchers have developed the following model demonstrating how ferritin synthesis is regulated by iron. When iron levels are low, a repressor of translation, iron response protein (IRP), binds to an iron response element (IRE), which is a stem-loop structure near the 5¢ end of ferritin mRNA. When iron levels are high, intracellular iron binds to the IRP, and the iron-IRP complex dissociates from the IRE, permitting ribosomes to proceed with the translation of ferritin mRNA. Figure 2 represents the model of the regulation of ferritin mRNA translation by iron.



7 Natural Selection MCQ

Figure 2. Model of regulation of ferritin synthesis by iron

135. Based on the data in Figure 1, the concentration of iron in the control diet used in the investigation is most likely within which of the following ranges?

(A) 20–25 mg/kg

(B) 35–40 mg/kg ✓

(C) 70–75 mg/kg

(D) 100–125 mg/kg

136. Which of the following conclusions about dietary iron and ferritin synthesis is best supported by the data in Figure 1?

(A) A dietary iron concentration of 25 mg/kg has no effect on ferritin gene expression compared with the control diet.

(B) At a dietary iron concentration of 50 mg/kg, rats make twice as much ferritin as rats fed the control diet do.

(C) Maximum activation of ferritin synthesis occurs at dietary iron concentrations of 75 mg/kg or greater. ✓

(D) The maximum rate of ferritin absorption from the gut occurs at dietary iron concentrations greater than 75 mg/kg.

137. After a search of nucleotide sequence databases, researchers identified an IRE in the 5¢ untranslated region of a gene encoding aconitase, an enzyme involved in the Krebs cycle. Which of the following pieces of experimental evidence best supports the claim that the synthesis of aconitase is controlled by a mechanism similar to ferritin regulation?

(A) IRP binds to aconitase mRNA in the presence of iron.

(B) The relative amount of aconitase protein increases in the presence of high levels of iron. ✓

(C) Oxygen consumption by cells increases in the presence of high levels of iron.

(D) The levels of reduced electron carriers, NADH and $FADH_2$, increase in the presence of high levels of iron.

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138. Freshwater bony fish secrete large volumes of very dilute urine. Many marine bony fish secrete small volumes of concentrated urine. Which of the following best explains the differences in nitrogenous-waste excretion between freshwater and marine bony fish?

7 Natural Selection MCQ

(A)	The differences arose during divergence, as fish from a common ancestor faced different selective pressures in environments of different osmolarity.	✓
(B)	The differences resulted from the evolution of kidneys in independent evolutionary lineages that were adapted to different environmental conditions.	
(C)	The differences are accidents of evolutionary divergence and are unrelated to the osmotic differences in freshwater and marine environments.	
(D)	The differences arise in each generation because of the physiological adjustment of individual fish to fluctuating salt concentrations in the environments in which they live.	

139. While there is only one species of Galápagos Island tortoise, there are several subspecies. Larger islands with more wet highlands have lush vegetation near the ground. Tortoises there tend to have high-domed shells and shorter necks, which restrict upward head movement. They also have shorter limbs. They are the heaviest and largest of the subspecies.

Smaller, drier islands are inhabited by tortoises with longer necks and limbs and with shells that are elevated above the neck, which allow them to browse taller vegetation.

Based on the information given, which of the following is a plausible explanation for the ancestry of the tortoise subspecies?

(A)	The subspecies share a recent common ancestor whose neck length, shell shape, and leg length were intermediate between the two subspecies.	
(B)	The tortoises with shorter legs and necks were most easily preyed on as young animals by the rats that were introduced, so they survived only on a few islands.	
(C)	Random mutations coupled with the inheritance of acquired characteristics resulted in distinct subspecies.	
(D)	Individuals with different adaptations in shell shape and leg length best exploited the food resources and left more surviving offspring on each island.	✓

7 Natural Selection MCQ

Individuals of a particular species of ground beetle are either light tan or dark brown. Light-tan beetles are predominant in habitats with light-colored sandy soils, and dark-brown beetles are predominant in habitats with dark-colored loam soils. In an experiment designed to determine the survival rates of light-tan beetles and dark-brown beetles in different habitats, 500 light-tan beetles and 500 dark-brown beetles were released in each of four habitats. Each beetle had been marked with a small spot of red paint on the underside of its abdomen before it was released. One week after the beetles had been released, any marked beetles that could be found were recaptured. The results are presented in the table below. It is assumed that differences in the numbers of beetles recaptured are directly related to differences in survival rates.

	Habitat 1 Sandy soil, no insectivorous birds present		Habitat 2 Sandy soil, insectivorous birds present		Habitat 3 Loam soil, no insectivorous birds present		Habitat 4 Loam soil, insectivorous birds present	
Color of Beetle	Light tan	Dark brown	Light tan	Dark brown	Light tan	Dark brown	Light tan	Dark brown
Number Released	500	500	500	500	500	500	500	500
Number Recaptured	130	114	123	22	65	74	13	87

140. Which of the following processes best explains why fewer light-tan beetles than dark-brown beetles were recaptured in habitat 4?

(A) Immigration
(B) Genetic drift
(C) Mutation
(D) Adaptive radiation
(E) Predation ✓

141. If all insectivorous birds and remaining beetles were removed from habitat 2 and 500 additional dark-brown beetles and 500 additional light-tan beetles were then released into habitat 2, which of the following is the best estimate of the number of additional dark-brown beetles that would be expected to be recaptured in habitat 2 after one more week?

(A) 0
(B) 22
(C) 120 ✓
(D) 220
(E) 500

142. Which of the following can be inferred from the data in the table?

7 Natural Selection MCQ

(A) Ground beetles do not emigrate from the habitat in which they live.

(B) Insectivorous birds prefer to eat light-tan beetles rather than dark-brown beetles.

(C) Ground beetles have higher rates of survival in habitats with loam soil.

(D) Insectivorous birds are predators of this species of ground beetle. ✓

(E) The reproductive success of beetles in habitats with sandy soils is greater than that of beetles in habitats with loam soils.

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143. High blood cholesterol (hypercholesterolemia, HC) can lead to cardiovascular problems such as atherosclerosis and heart attack. Exercise and monitoring of diet can often control cholesterol levels; however, in certain cases HC is inherited as an autosomal dominant disease caused by a mutation in a single gene.

A man with high cholesterol levels is about to marry a woman whose total cholesterol levels are also higher than average. A physician has suggested they get tested for the *HC allele*. Which of the following is a valid ethical question concerning the test?

(A) How important are the results of the test to the couple's health, since they already know they have HC?

(B) How can they eliminate the allele for HC from their cells?

(C) If both have the allele for HC, should an insurance company raise their rates because of the results of the test? ✓

(D) What cures are available for HC other than dietary changes and exercise?

144. Evolutionary biologists have observed variation in the average age and size of mature individuals in a population of small freshwater guppies. Guppies found in pools with pike cichlids— a predator fish that preys primarily on larger guppies—reproduce at a young age and are small at maturity. Guppies found in pools with killifish —a different predator fish that preys primarily on smaller guppies—reproduce at an older age and are larger at maturity.

Which of the following predicts the most likely outcome of moving a population of guppies from a pool with pike cichlids to a pool with killifish?

(A) The introduced guppy population will remain phenotypically similar to the original population because the killifish will not recognize them as prey.

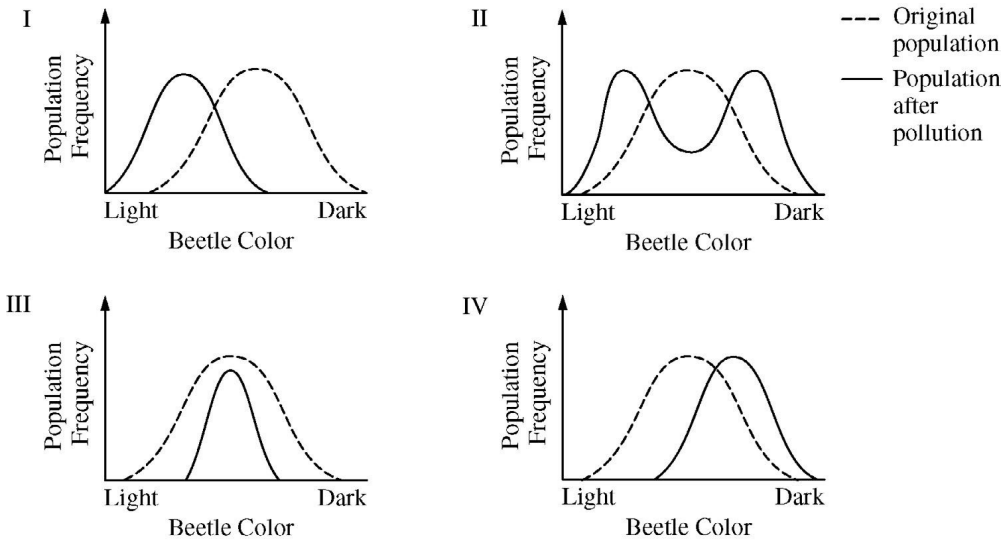
(B) Because the introduced population was in Hardy-Weinberg equilibrium, successive generations in the new population will continue to display the same allele frequencies as the original population.

(C) After many generations, the offspring of the introduced guppies will mature at an older age and larger size because of selective pressure from the new predator. ✓

(D) Within two or three generations, the introduced guppies will become extinct because the killifish prefer the juveniles.

7 Natural Selection MCQ

145. In a hypothetical population of beetles, there is a wide variety of color, matching the range of coloration of the tree trunks on which the beetles hide from predators. The graphs below illustrate four possible changes to the beetle population as a result of a change in the environment due to pollution that darkened the tree trunks.



Which of the following includes the most likely change in the coloration of the beetle population after pollution and a correct rationale for the change?

(A) The coloration range shifted toward more light-colored beetles, as in diagram I. The pollution helped the predators find the darkened tree trunks.

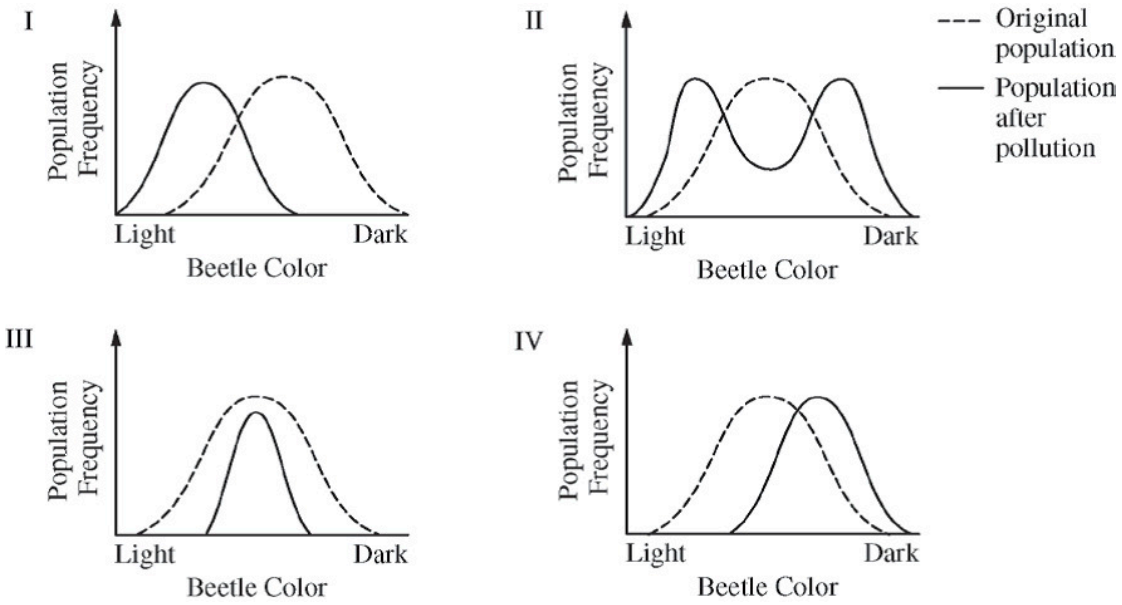
(B) The coloration in the population split into two extremes, as in diagram II. Both the lighter-colored and the darker-colored beetles were able to hide on the darker tree trunks.

(C) The coloration range became narrower, as in diagram III. The predators selected beetles at the color extremes.

(D) The coloration in the population shifted toward more darker-colored beetles, as in diagram IV. The lighter-colored beetles were found more easily by the predators than were the darker-colored beetles. ✓

7 Natural Selection MCQ

146. In a hypothetical population of beetles, there is a wide variety of color, matching the range of coloration of the tree trunks on which the beetles hide from predators. The graphs below illustrate four possible changes to the beetle population as a result of a change in the environment due to pollution that darkened the tree trunks.



Which of the following includes the most likely change in the coloration of the beetle population after pollution and a correct rationale for the change?

(A) The coloration range shifted toward more light-colored beetles, as in diagram I. The pollution helped the predators find the darkened tree trunks.

(B) The coloration in the population split into two extremes, as in diagram II. Both the lighter-colored and the darker-colored beetles were able to hide on the darker tree trunks.

(C) The coloration range became narrower, as in diagram III. The predators selected beetles at the color extremes.

(D) The coloration in the population shifted toward more darker-colored beetles, as in diagram IV. The lighter colored beetles were found more easily by the predators than were the darker-colored beetles. ✓

147. If an individual with diabetes consumes food that is high in simple carbohydrates, blood-sugar levels can rise above normal levels. Which of the following questions would provide the best direction for a researcher who wanted to study the impact of abnormally high blood-sugar levels on cellular homeostasis in diabetics?

(A) Are cells in diabetics larger in size than those in nondiabetics?

(B) Do the cells in diabetics have more potassium ion channels in the cell membrane than the cells in nondiabetics do?

(C) Does water move from cells into blood vessels more rapidly in diabetics than in nondiabetics when blood-sugar levels are higher than normal? ✓

(D) Do the cells of diabetics use simple sugars as an energy source?

7 Natural Selection MCQ

Iron is an essential nutrient that is acquired by organisms from the environment. When intracellular levels of iron are relatively high, living cells synthesize an iron-storage protein called ferritin.

The induction of ferritin synthesis by iron was investigated in rats. Figure 1 shows the results of an experiment in which cellular levels of ferritin protein were measured in the presence or absence of iron and actinomycin D, a drug that inhibits transcription.

Figure 2 shows the results of an experiment in which cellular levels of ferritin protein were measured in the presence or absence of iron and cycloheximide, a drug that inhibits translation.

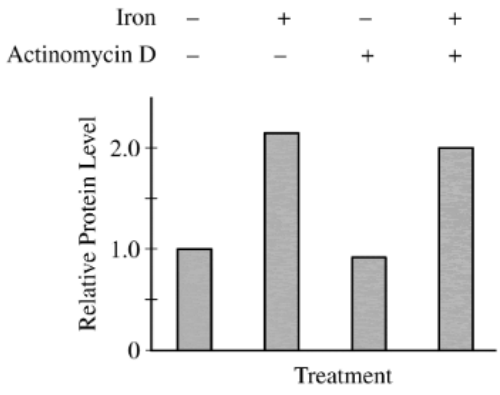


Figure 1. Ferritin Protein Levels

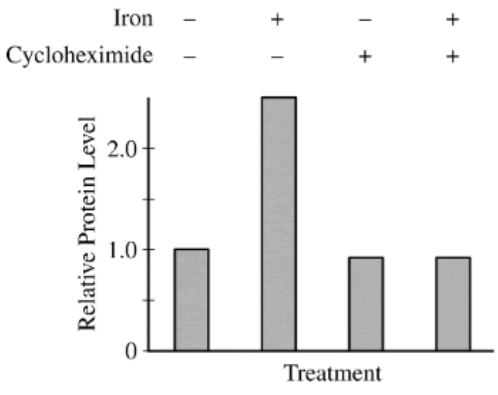


Figure 2. Ferritin Protein Levels

148. Hereditary hemochromatosis (HHC) is an iron storage disease that results in reduced uptake of iron into cells. Based on the data, which of the following is the most likely effect of HHC on ferritin protein levels?

(A) Elevated levels of ferritin protein because there is more iron in the blood of affected individuals

(B) Reduced levels of ferritin protein because there is less iron in the cytosol of affected individuals ✓

(C) Elevated levels of ferritin protein because ferritin replaces the mutant protein in affected individuals

(D) Reduced levels of ferritin protein because iron stimulates ferritin breakdown

149. The gene sequences responsible for the iron mediated changes in ferritin protein levels are highly conserved and are called iron response elements (IREs). IREs have been observed in a number of genes involved in iron transport and metabolism. The IRE sequences found in the ferritin gene are found in all multicellular organisms, whereas other nonferritin IRE sequences are found only in certain vertebrate organisms. Which of the following is the best explanation for the observations?

7 Natural Selection MCQ

(A) Ferritin IREs arose in the common ancestor of multicellular organisms, whereas nonferritin IREs arose later in specific evolutionary lineages. ✓

(B) Nonferritin IREs arose at different times in the common ancestor of multicellular organisms.

(C) Ferritin IREs arose in diverse organisms by convergent evolution, while the nonferritin IREs arose by recombination events between closely related species.

(D) Ferritin IREs have evolved independently in different evolutionary lineages, whereas nonferritin IREs arose in the common ancestor of multicellular organisms.

Mice live in territorial social groups, and each group has a single dominant male with the rest of the males in the group being subordinate. Male mice mark the area in which they live with their scent by leaving urine wherever they walk. There are two classes of proteins that are found in mouse urine, major urinary proteins (MUPs) and major histocompatibility complex (MHC) proteins. The proteins provide a unique scent for each mouse.

In an experiment, wild male mice were kept isolated in separate cages and their urination patterns were recorded. They were then combined into a single cage that was divided into two compartments by a mesh screen, and again their urination patterns were recorded (Figure 1).

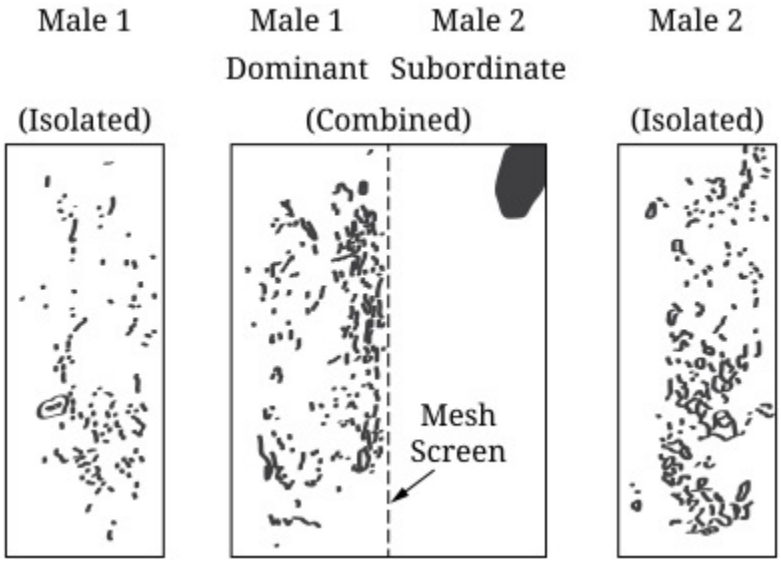


Figure 1. Urination patterns recorded during an experiment

150. A researcher claims that the life spans of dominant males are longer than the life spans of subordinate males. Based on the results of the experiment, which of the following best supports the researcher's claim?

(A) Dominant males have proteins in their urine that protect them from harmful genetic mutations.

(B) Dominant males have proteins in their urine that enable them to communicate with subordinate males in their group.

(C) Dominant males are able to move more freely throughout their territory, which gives them greater access to resources. ✓

(D) Dominant males are able to move more freely throughout their territory, which reduces the number of pathogens they are exposed to.

7 Natural Selection MCQ

Answer C

Correct. The scent patterns produced in the combined experiment indicate that the dominant male occupies his entire enclosure, whereas the subordinate male occupies only one small area of his enclosure. This would limit a subordinate male's access to resources and could limit his life span.

151.

TABLE 1. POLLINATOR VISITATION RATES TO TWO SPECIES OF *MIMULUS*

***Mimulus* Species and Flower Color**	**Bumblebees** (10^{-3} visits / flower / hour)	**Hummingbirds** (10^{-3} visits / flower / hour)
M. lewisii (wild-type, pink)	15.4	0.021
M. lewisii (mutant, orange)	2.63	1.44
M. cardinalis (wild-type, red)	0.148	189
M. cardinalis (mutant, pink)	10.9	168

Plant species in the genus *Mimulus* have evolved specialized flower structures and colors that attract pollinators. Researchers identified mutant forms of two of the species, *Mimulus lewisii* and *Mimulus cardinalis*, that have different flower colors from the wild-type forms. They measured the number of visits by two pollinators, bumblebees and hummingbirds, to the flowers of the wild-type and mutant species (Table 1).

Which of the following statements best describes the data shown in Table 1 ?

(A) Bumblebees visited wild-type *M. lewisii* approximately 13 times more frequently than they did mutant *M. lewisii*.

(B) Hummingbirds preferred wild-type *M. lewisii* over wild-type *M. cardinalis*.

(C) Hummingbirds preferred the pink flower type in both *M. lewisii* and *M. cardinalis*.

(D) Bumblebees visited mutant *M. cardinalis* approximately 70 times more frequently than they did wild-type *M. cardinalis*. ✓

7 Natural Selection MCQ

Answer D

Correct. Dividing 10.9 (the number of bumblebee visits to mutant *M. cardinalis*) by 0.148 (the number of bumblebee visits to wild-type *M. cardinalis*) produces a value of 73.6, which is close to 70.

152.

PERCENT OF MOTHS RECAPTURED IN TWO DIFFERENT ENVIRONMENTS		
Trial	Moth Color and Environment	Percent of Released Moths Recaptured
I	Light-colored moths were released in an unpolluted environment.	13%
II	Light-colored moths were released in a polluted environment.	12%
III	Dark-colored moths were released in an unpolluted environment.	7%
IV	Dark-colored moths were released in a polluted environment.	28%

A researcher released large numbers of moths into different environments in an attempt to better understand a mechanism of evolution. The moths were released in four trials as described in the table above. Each of the released moths had a small mark on the underside of a wing for identification. After an appropriate amount of time, the researcher recaptured as many of the released moths as possible. Data from the experiment are included in the table above. Which of the following claims is best supported by the data?

(A) Light-colored moths were more likely to be recaptured in the polluted environment than in the unpolluted environment, suggesting an increased chance of reproductive success.

(B) Dark-colored moths were more likely to be recaptured in the polluted environment than in the unpolluted environment, suggesting an increased chance of reproductive success. ✓

(C) Light-colored moths were less likely to be recaptured in the polluted environment than in the unpolluted environment, suggesting an increased chance of reproductive success.

(D) Dark-colored moths were less likely to be recaptured in the polluted environment than in the unpolluted environment, suggesting an increased chance of reproductive success.

153. In addition to the pigments commonly associated with photosynthesis, a certain photosynthetic species contains two additional pigment types.

Which of the following best supports the claim that this species is better adapted to environmental changes than other photosynthetic species are?

7 Natural Selection MCQ

(A) The increased pigment concentration better facilitates energy production within the cells of the species.

(B) The pigment combination allows the organism to absorb heat as well as light, making better use of available energy.

(C) The additional pigments allow the species to outcompete other species for the wavelengths of light commonly used in photosynthesis.

(D) The additional pigments allow the species containing them to harvest energy from wavelengths of light that the other photosynthetic species cannot use. ✓

Answer D

Correct. The ability of the pigment combination to absorb a wider range of sunlight allows the organism to use photosynthesis to produce energy in order to survive in environments with different ranges of available wavelengths of sunlight.

154. Which of the following factors is the most effective in changing the allele frequency in a natural population?

(A) Large population size

(B) Low rate of mutation

(C) Negligible migration

(D) Random mating

(E) Selection ✓

155. Some plant species flower in response to increasing daily temperatures in the spring. Many of these species rely on pollinators that migrate based on changes in day length and the position of the Sun. The current global warming trend is placing new selective pressures on the species involved in these relationships.

Which of the following best explains the impact of these new selective pressures on the organisms involved?

(A) If the environment for the plant species becomes too warm, the pollinators will no longer migrate to that area in the spring, continuing on to a more northern environment instead.

(B) The warmer temperatures will lead to a drier environment, so the plants will no longer produce enough nectar to attract the pollinators.

(C) The plant species will flower earlier in the spring in response to rising temperatures before the arrival of the pollinators, so seeds will not be produced. ✓

(D) Migrating pollinators will start migrating later in the year, switching from spring-flowering plants to summer-flowering plants.

Answer C

Correct. The migrating pollinators will still arrive at the same time, as their environmental cues to

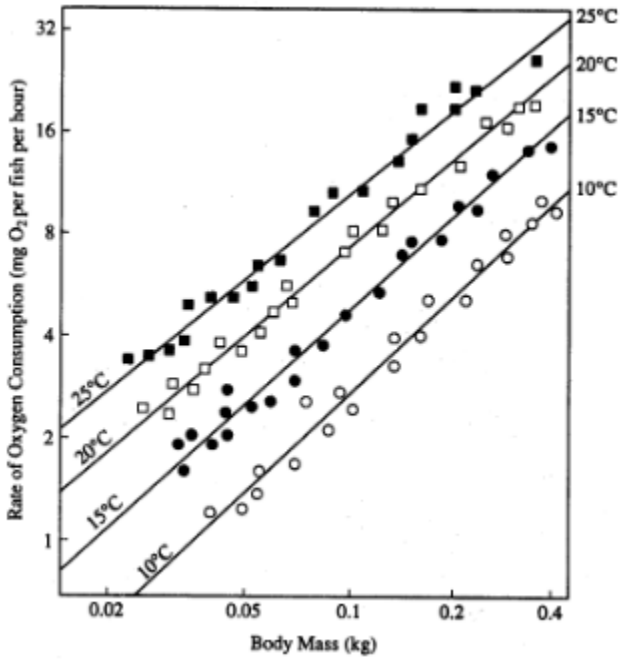
7 Natural Selection MCQ

migrate are not affected by warming. However, the plants will flower earlier, as the proper temperature cues they use will occur earlier in the spring. If the plants contain sufficient variability, they may be able to attract different pollinators.

- Directions: Each group of questions below concerns an experimental or laboratory situation or data. In each case, first study the description of the situation or data. Then choose the one best answer to each question following it.

The following questions refer to the following information and graph.

The data presented in the figure below are measurements of the rate of oxygen consumption at differing body masses in a species of fish. Each point represents measurements from a different fish. Measurements were taken at different temperatures. (○= 10°C,●= 15°C, □= 20°C, ■= 25°C.)



156. The fact that each line on the graph rises from left to right means that

7 Natural Selection MCQ

(A) higher temperatures produce higher rates of metabolism
(B) there were more large fish in the samples taken at high temperatures
(C) larger fish consume more oxygen than smaller fish at all four temperatures ✓
(D) when measurements are taken for larger fish late in the day, observed values are higher
(E) larger fish prefer to live at higher temperatures than do smaller fish

157. The best explanation for the fact that not all points lie on any given line is that

(A) the thermometer was incorrectly calibrated
(B) the scale used to weigh the fish registered 0.001 kg too little
(C) the fish grew during the course of the experiment
(D) errors were made in plotting the data
(E) organisms within populations show variability ✓

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158. The peppered moth, *Biston betularia* was generally white in color with black speckles. Over time a more solid black moth became the more common variety. Which of the following best explains the relationship between wing pigmentation and the coloration of a moth's surroundings?

(A) The frequencies of the different phenotypes in the moth population changed randomly due to genetic drift.
(B) If the surroundings of the peppered moth became darker, those backgrounds conferred a selective advantage to a black variety, as described by natural selection. ✓
(C) There was a gene flow of the black variety, spreading it throughout the moth's normal range.
(D) The black variety was a nineteenth century mutation, dominant to the peppered variety. The allele for the peppered trait, being recessive, decreased in frequency over time

Answer B

Correct. Natural selection occurred as surfaces became darker from burning coal, giving an advantage to the black variation, as they were able to be better camouflaged. The switch to cleaner fuel sources led to a return to lighter surfaces, restoring the adaptive advantage of the peppered variation.

7 Natural Selection MCQ

- **Directions:** Each of the questions or incomplete statements below is followed by four suggested answers or completions. Select the one that is best in each case and then fill in the corresponding circle on the answer sheet.

The three-spined stickleback (*Gasterosteus aculeatus*) is a small fish found in both marine and freshwater environments. Marine stickleback populations consist mainly of individuals with armor-like plates covering most of their body surface (completely plated). Approximately 10,000 years ago, some marine sticklebacks colonized freshwater environments. After many generations in the freshwater environments, the freshwater stickleback populations lacked the armor plating (low plated) typical of marine stickleback populations.

Over the period between 1957 and 2005, one freshwater population, in Lake Washington, a lake in a coastal region of the northwestern United States, changed from having a majority of individuals of the low-plated phenotype to having more individuals of the completely-plated phenotype than of the low-plated phenotype. Figure 1 shows the distribution of plated phenotypes in Lake Washington sticklebacks at four time points between 1957 and 2005.

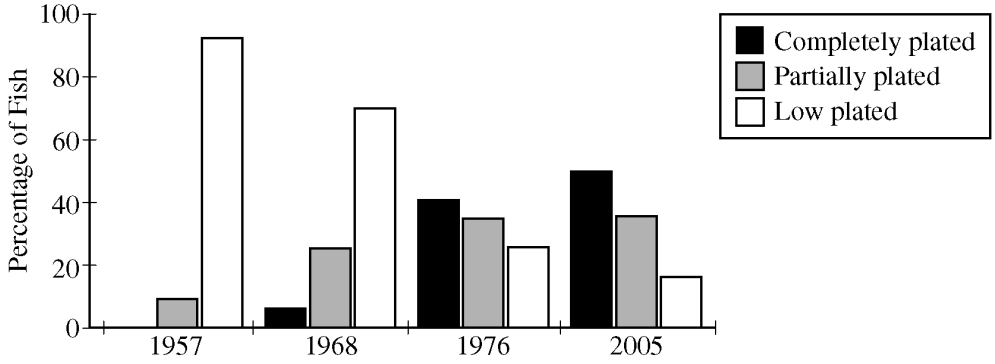


Figure 1. Armor plating phenotypes in Lake Washington stickleback population

A single gene, *ectodysplasin* (*EDA*), is thought to be responsible for the variation in the number of armor plates in sticklebacks. Figure 2 shows a phylogenetic tree constructed by comparing DNA sequences of the *EDA* gene from a number of stickleback populations with low-plated or completely plated phenotypes. Figure 3 shows a phylogenetic tree constructed by comparing the sequences of 25 genes that were randomly selected from the same populations as shown in Figure 2. In both figures, shaded populations display the completely plated phenotype.

7 Natural Selection MCQ

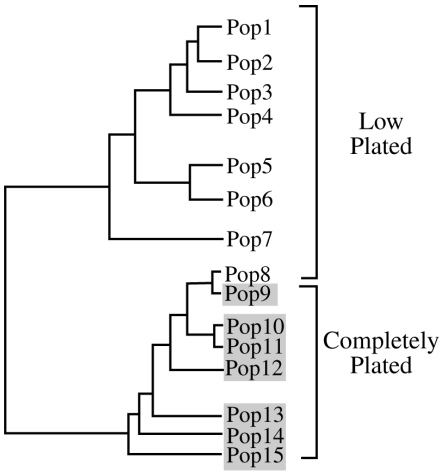


Figure 2. Phylogeny based on *EDA* gene only

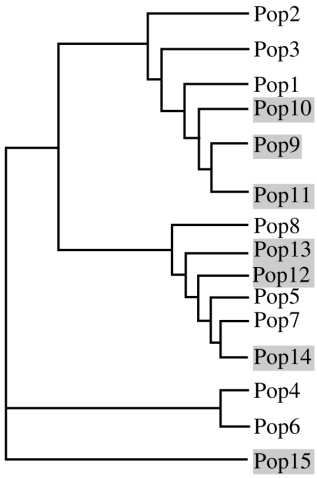


Figure 3. Phylogeny based on 25 random genes

159. Which of the following best explains the differences in the armor of the Lake Washington stickleback population summarized in Figure 1 ?

(A) Analysis of somatic cells using chromosomal staining and light microscopy indicates that stickleback fish have a diploid number of 42.

(B) Stickleback males from natural freshwater populations are typically more aggressive when competing for mates than are stickleback males from laboratory-bred populations.

(C) Fish exhibiting the low-plated phenotype were selected against in the Lake Washington stickleback population over the last 50 years. ✓

(D) Migration of individuals from other freshwater environments to Lake Washington led to gene flow between populations that were once geographically isolated.

160. Evolution of a new trait typically takes many generations. Yet a dramatic shift in the extent of armor plating in the Lake Washington stickleback population occurred in the 50 years following the cleanup of the lake. Which of the following best describes the mechanism of the rapid evolution of the armor phenotype in the Lake Washington sticklebacks?

(A) Pollutants in Lake Washington forced the sticklebacks to increase their mutation rate.

(B) The rapid change in the Lake Washington ecosystem required individual sticklebacks to evolve complete armor quickly.

(C) The increase in visibility in Lake Washington allowed sticklebacks to visually select mates with low armor.

(D) New selective pressures favored individuals with the plated phenotype, causing the plated allele frequency in the population to quickly increase. ✓

161. The phylogenetic trees in Figures 2 and 3 depict two different phylogenies of the same populations of sticklebacks. Which of the following questions will best help determine which tree represents the most accurate phylogeny?

7 Natural Selection MCQ

(A) Is the *EDA* gene as representative of the differences between the populations as the 25 random genes that were examined for Figure 3? ✓

(B) Are the low-plated populations found only in freshwater and the high-plated populations found only in saltwater environments?

(C) Is the common ancestor of the organisms represented in Figure 2 different from the common ancestor of the organisms represented in Figure 3?

(D) Is the expression level of the *EDA* gene analyzed in Figure 2 significantly greater than the expression levels of the 25 genes analyzed in Figure 3?

162.

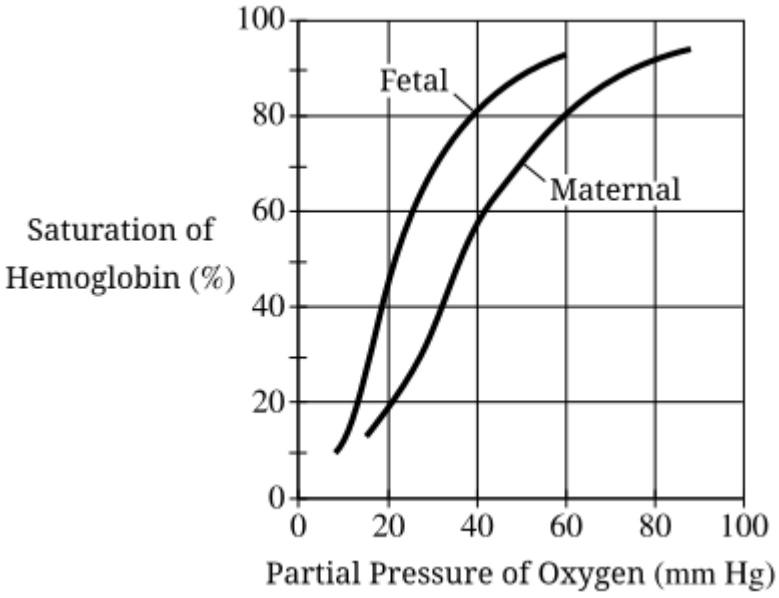


Figure 1. Percent saturation of human hemoglobins as a function of the partial pressure of oxygen

The graph in Figure 1 shows the percent saturation of fetal hemoglobin and maternal hemoglobin in humans as a function of the partial pressure of oxygen. Based on the data, which of the following values is closest to the predicted difference in the percent saturation of fetal hemoglobin and maternal hemoglobin at $40\ \text{mm Hg}$?

(A) 10

(B) 12

(C) 20 ✓

(D) 40

Answer C

Correct. Based on the data in Figure 1, at $40\ \text{mm Hg}$, the percent saturation of fetal hemoglobin is

7 Natural Selection MCQ

approximately 81% and the percent saturation of maternal hemoglobin is approximately 58%. The difference in saturation is 23%, which is closest to the choice of 20%.

163. In a region of Africa where malaria is common, 9% of the population is born with a severe form of sickle-cell anemia (ss). Carriers of the sickle-cell allele (heterozygotes) are resistant to malaria and display only mild symptoms of malaria if exposed to the malaria parasite. If an effective antimalarial program is instituted in the region, which of the following is most likely to occur over many generations?

(A) The frequency of the sickle-cell allele will increase, as there will be more heterozygous individuals.

(B) The incidence of malaria will decrease, but the incidence of sickle-cell anemia will increase.

(C) The frequency of the sickle-cell allele s will decrease as the incidence of malaria decreases. ✓

(D) The frequency of the sickle-cell allele will increase because antimalarial treatments cause mutations.

Answer C

Correct. If malaria is eradicated, there is no longer a benefit to individuals to be heterozygous, particularly because these individuals may still display some of the symptoms of sickle-cell anemia. Over many generations, the frequency of the harmful s allele will be eliminated through natural selection.

Snowshoe hares are small mammals that live in forests throughout Canada and the northern United States. Hares feed primarily on vegetation and are preyed on by lynx and owls.

In the fall, the hares shed their brown coat (molt), and a new white coat grows in over a period of 40 days. In the spring, the hares molt again and their white coat is replaced with a brown coat over a period of approximately 30 to 50 days.

In a recent study, researchers observed that the majority of hares in an isolated population began the molting process during the same week in the fall, even though the ground was not snow covered. The majority of hares also began the molting process during the same week in the spring, even though the ground was still snow covered. The researchers formulated a hypothesis that the timing of the molting process is influenced by seasonal changes in the daylight period.

164. Several climate models for the original study area predict that the length of time the ground is covered in snow will decrease over time while the daylight periods will remain unchanged.

Assuming that the researchers' hypothesis about the timing of molting and the climate models are correct, which of the following will be the most likely immediate effect of the decrease in time the ground is snow covered on the hare population?

(A) The hare population will stop molting and maintain a single coat all year.

(B) The hare population will migrate to an area with an even shorter time during which the ground is snow covered.

(C) Hares will experience longer times with coat colors that do not blend with the ground color. ✓

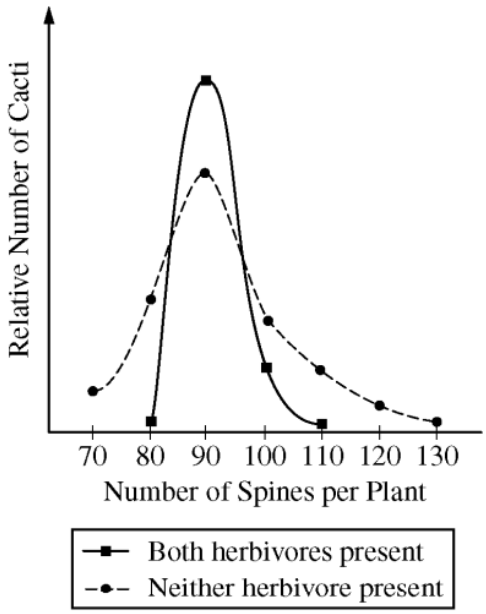
(D) The process of molting from a white to a brown coat in hares will take longer.

7 Natural Selection MCQ

Answer C

Correct. Hares will retain white coats even though the ground will not be covered by snow. This is because molting is determined by daylight length and the hares' genetic makeup rather than by the presence or absence of snow.

165.



In a species of cactus, the number of spines on a plant is genetically determined. The graph above shows frequency distributions for populations of the cactus species growing in the presence or absence of two herbivores: peccaries (a New World pig) and wasp larvae. Which of the following best accounts for the different frequency distributions in the graph?

(A) Peccaries eat cacti with the greatest number of spines, and wasp larvae show no preference with respect to spine number.

(B) Peccaries eat cacti with a smaller number of spines, and wasp larvae eat cacti with a greater number of spines. ✓

(C) Wasps have increasing difficulty depositing eggs on cacti with more spines, and peccaries show no preference with respect to spine number.

(D) Both peccaries and wasp larvae will eat the most abundant types of cacti, which are the plants with the greatest number of spines.

7 Natural Selection MCQ

166. Climate change is disrupting the length of winter in various ecosystems, resulting in less snowfall and milder winters. Scientists in Finland have been observing the tawny owl, a fairly large owl species that is commonly found in woodlands across much of Europe and Asia. Typically, the feathers of the upper body are either brown (BB) or gray (bb). Scientists studying tawny owls in Finland have found that the percentage of tawny owls that have brown feathers has increased over the past 28 years.

Which of the following best explains why the brown-feathered phenotype in tawny owls has increased over the past 28 years?

(A) Since the brown genotype is dominant, it will eventually eliminate the recessive gray genotype from the population.

(B) Darker feather color in tawny owls makes them less visible to predators as they sit in the trees.

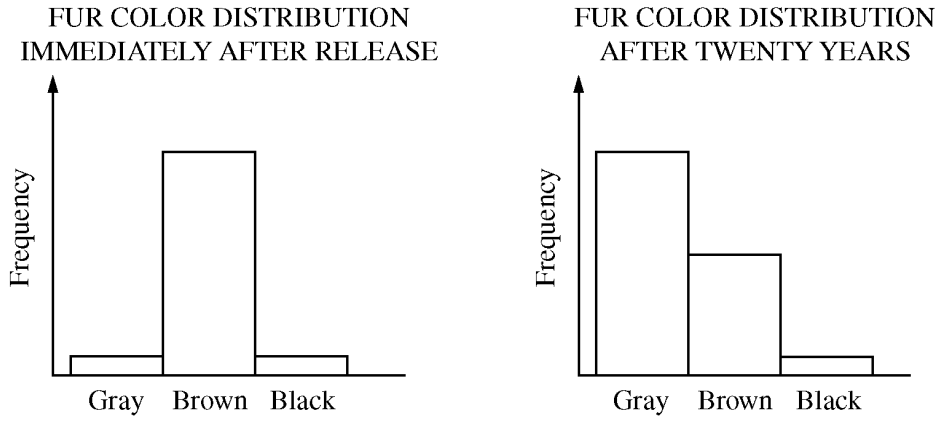
(C) With milder winters, the advantage of gray feathers that blend into a snowy environment has decreased. ✓

(D) The directional selection toward a darker phenotype is random and likely due to a bottleneck event.

Answer C

Correct. With less snowfall and milder winters, the advantages of gray feather color as camouflage has decreased.

167.



A group of mice was released into a large field to which no other mice had access. Immediately after the release, a representative sample of the mice was captured, and the fur color of each individual in the sample was observed and recorded. The mice were then returned to the field. After twenty years, another representative sample of the mice was captured, and the fur color of each individual in the sample was again recorded. Which of the following best explains the change in the frequency distribution of fur color phenotypes in the mouse population, as shown in the figures above?

7 Natural Selection MCQ

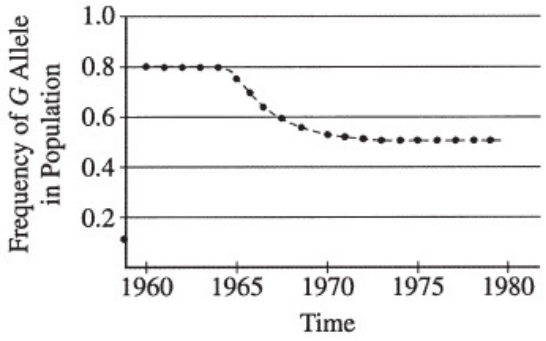
(A) The allele for gray fur color is unstable, and over twenty years most of those alleles mutated to become alleles for black fur.

(B) The field was composed primarily of light-colored soil and little vegetation, affording gray mice protection from predators. ✓

(C) Sexual selection led to increased mating frequency of black and brown versus gray and brown.

(D) The gray mice were hardest to capture and so were underrepresented in the twenty-year sample.

168. A moth's color is controlled by two alleles, *G* and *g*, at a single locus. *G* (gray) is dominant to *g* (white). A large population of moths was studied, and the frequency of the *G* allele in the population over time was documented, as shown in the figure below. In 1980 a random sample of 2,000 pupae was collected and moths were allowed to emerge.



Which of the following is the most likely reason for the observed differences in the frequency of the *G* allele between 1965 and 1972?

(A) Emigration of white moths from the population

(B) Chance

(C) Selection against gray phenotypes ✓

(D) Speciation

(E) Mutation

169. A researcher claims that budding yeast are able to survive in different environments because they produce enzymes that allow them to use different molecules as sources of matter and energy.

Which of the following statements best helps justify the researcher's claim by providing a relevant example?

(A) Yeast cells produce protein kinases, which are enzymes that catalyze the transfer of phosphate groups from ATP to protein substrates.

(B) Yeast cells produce DNA polymerases, which are enzymes that catalyze the conversion of free nucleotides into strands of DNA.

(C) Yeast cells produce invertase, which is an enzyme that catalyzes the conversion of the disaccharide sucrose into glucose and fructose. ✓

(D) Yeast cells produce catalase, which is an enzyme that catalyzes the conversion of hydrogen peroxide into water and oxygen gas.

7 Natural Selection MCQ

Answer C

Correct. The disaccharide sucrose is not transported efficiently into yeast cells. By producing the enzyme invertase, budding yeast can use the sucrose, glucose, and fructose in the surrounding environment as sources of matter and energy. As such, the production of invertase by budding yeast is a relevant example that will help justify the researcher's claim.

170. A team of scientists grew small pieces of coral in controlled tanks. Once these pieces of coral reached sufficient size, they were transferred to natural reefs to replace lost corals in the ocean. A scientist from the team suggested revising the experiment so that the water temperature in all the tanks would be increased to $0.5^{\circ}C$ above the normal ocean temperature. The corals pieces that survived the increased temperature in the tank would be separated and allowed to grow into different colonies. When these new colonies grown from those coral pieces grown in warmer water reached sufficient size, they would be transferred to natural reefs.

Which of the following statements predicts the most likely outcome of the revised experiment?

(A) The increased temperature will induce corals to create new genes, which will allow the corals to withstand future increases in ocean temperature.

(B) Stress from the increased temperature will cause corals to evolve into a new species that can tolerate future increases in ocean temperature.

(C) After being transferred to the natural reef, the new colonies of corals will return to their previous heat tolerance level and will not survive future increases in ocean temperature.

(D) The new colonies of corals are artificially selected for tolerating an increased temperature, so they will be better adapted to moderate future increases in ocean temperature. ✓

171. A researcher created an experimental chamber that allowed a population of fruit flies to mate randomly, but where female flies had to choose one of three side-chambers in which to lay eggs. All three side-chambers were the same except for the added scent of apple in side-chamber A, banana in side-chamber B, and cucumber in side-chamber C. Only eggs deposited in side-chamber C were allowed to hatch and produce offspring. Adult flies did not know which side-chamber was producing offspring. Every day for 100 days, the researcher counted the number of flies in each side-chamber. By day 100, the number of flies in side-chamber C greatly increased, while the number of flies in side-chambers A and B had greatly decreased.

Which of the following statements best explains why there were more flies in side-chamber C, than in side-chambers A or B, on day 100?

(A) More flies were entering side-chamber C because flies that prefer the scent of cucumber were artificially selected for. ✓

(B) More flies were entering side-chamber C because female flies learned which side-chamber allowed for successful reproduction.

(C) More flies were entering side-chamber C because female fruit fly offspring produced in side-chambers A and B were less fit for the environment.

(D) More flies were entering side-chamber C because male fruit fly offspring produced in side-chambers A and B were less fit for the environment.

7 Natural Selection MCQ

172. Bt crops have been genetically modified to express a gene from the bacterium *Bacillus thuringiensis*. Bt crops produce a protein that is harmful to insects that consume it but is harmless to humans and other mammals. Scientists have observed that some insects appear to be resistant to the protein produced by Bt crops. They collected data to compare the amount of Bt crops being grown with the number of cases of resistance to the Bt crop protein in insects (Table 1).

Table 1. Comparison of the Amount of Bt Crops Grown with the Number of Cases of Insect Resistance to Bt Crops

Year	Amount of Bt Crops Grown (million hectares*)	Total Number of Cases of Insect Resistance to Bt Crops
1996	0	0
2000	11	0
2004	22	2
2008	46	6
2012	70	11
2016	98	16

*1 hectare = $10,000\ m^2$

Based on the data in Table 1, which of the following best describes how the amount of Bt crops grown is related to incidence of insect resistance to Bt crops?

(A) As the amount of Bt crops grown decreases, the total number of cases of insect resistance remains stable.

(B) As the amount of Bt crops grown decreases, the total number of cases of insect resistance increases.

(C) As the amount of Bt crops grown increases, the total number of cases of insect resistance increases. ✓

(D) As the amount of Bt crops grown increases, the total number of cases of insect resistance decreases.

7 Natural Selection MCQ

173. Over many years of a breeding program, a zoo has an established population of foxes that is well adapted for living in captivity. A representative sample of wild foxes from the neighboring forest was used to start the zoo population. A study was conducted to compare the behavior of the zoo fox population with the wild fox population in the neighboring forest. The behaviors of equal numbers of foxes from each population were assessed. Each fox's behavior was scored on a continuum from docile to aggressive based on its interaction with a trained behaviorist. The data is shown in Figure 1.

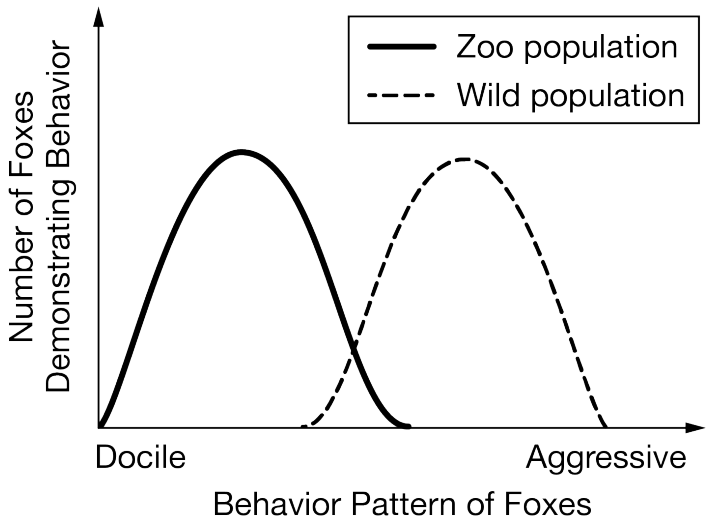


Figure 1. Aggressive behavior in zoo and wild fox populations

The phenotypic variation in behavior between the two populations can best be described as resulting from

(A) sexual selection

(B) natural selection

(C) artificial selection ✓

(D) genetic drift

Answer C

Correct. The zoo staff specifically selected the most docile foxes to produce offspring, an example of artificial selection.

7 Natural Selection MCQ

174. Table 1 shows a trend of increasing milk yield for cows. Part of the increase in milk production can be attributed to improved nutrition and health protocols for dairy cows.

Table 1. Average milk production per cow (in kilograms per cow per year)

Year	Average Annual Milk Production($\frac{\text{Kg}}{\text{Cow}}$)
1924	2,300
1937	2,300
1950	2,600
1963	3,400
1976	4,800
1989	6,400
2002	8,400
2015	10,000

Based on the information provided, which of the following best describes the source of the rapid increase in milk production from 1976 to 2015 ?

(A) From the 1970s to the early 2000s, dairy cows were not used for beef production.

(B) Starting in the 1970s, farmers increased the selective breeding of cows and bulls for offspring that produced more milk. ✓

(C) Reproduction rates in cows and bulls increased significantly in the 1970s.

(D) Cows were healthier after the 1970s, so they lived longer and produced more milk during their lifetime.

Answer B

Correct. Artificial selection by humans led to the production of offspring that produced more milk.

7 Natural Selection MCQ

175. Thoroughbred race horses are subject to intense artificial selection to produce faster-running horses. The results of breeding programs can be seen in the change in winning times of the Kentucky Derby from 1880 to the present, as shown in Figure 1.

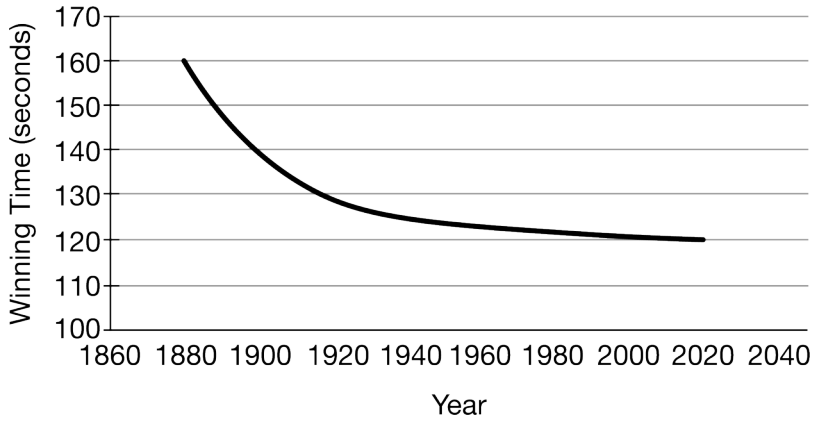


Figure 1. Best-fit curve for winning times in the Kentucky Derby since 1880

Which of the following best explains the trend shown in Figure 1?

(A) The distance the horses ran became shorter between 1880 and 1940.

(B) All of the selective breeding programs ended in 1940.

(C) After 1940, natural selection prevented any additional speed increases.

(D) By 1940, most genetic variation for speed had been lost from the gene pool. ✓

Answer D

Correct. Intense selection likely removed most of the alleles that produce slower speed by 1940, and few new combinations of the remaining alleles were possible, resulting in limited genetic variation relating to speed.

176. In eastern gray squirrels, *Sciurus carolinensis*, the allele for black fur (B) is dominant to the allele for gray fur (b). In a particular population of gray squirrels, 64% have black fur and 36% have gray fur. A researcher calculated the allelic frequencies to be $B = 0.4$ and $b = 0.6$. Five years later, the researcher returned to the location and determined the allelic frequencies within the squirrel population to be $B = 0.6$ and $b = 0.4$.

Which of the following could best explain the increase in the frequency of the B allele in the population after five years?

7 Natural Selection MCQ

(A) The frequency of the B allele increased because it is the dominant allele.

(B) The frequency of the B allele increased because squirrels with gray fur exhibited greater fitness.

(C) The frequency of the B allele increased due to the selective pressures of the environment. ✓

(D) The frequency of the B allele increased because black squirrels randomly mate with other eastern gray squirrels.

Answer C

Correct. It is most likely that the squirrels with black fur had a selective advantage, as the frequency of the allele for black fur increased.

177. In a small group of people living in a remote area, there is a high incidence of "blue skin", a condition that results from a variation in the structure of hemoglobin. All of the "blue-skinned" residents can trace their ancestry to one couple, who were among the original settlers of this region. The unusually high frequency of "blue skin" in the area is an example of

(A) mutation

(B) genetic drift ✓

(C) natural selection

(D) sexual selection

(E) heterozygote advantage

178. In humans, one allele of the $APOE$ gene, called $APOE$–$\varepsilon 2$, can result in a high tolerance of cholesterol. Cholesterol is a vital substance for humans but may lead to heart disease in an older adult with a history of high cholesterol diets. High cholesterol diets are becoming more prevalent in the United States. Currently only about 2% of humans carry the $APOE$–$\varepsilon 2$ allele.

Which of the following states a valid null hypothesis about the future distribution of $APOE$ alleles in future generations in the United States?

(A) The $APOE$–$\varepsilon 2$ allele was the result of chance mutations so it is not possible to predict how its frequency will change in the future.

(B) As high cholesterol diets become more common in the United States, individuals with the $APOE$–$\varepsilon 2$ allele will have a better survival rate from heart disease than those without the allele, so the allele will increase in frequency.

(C) The low frequency of the $APOE$–$\varepsilon 2$ allele indicates it is probably a recessive allele, so it will become less frequent as the dominant allele becomes more frequent.

(D) The variant protects an individual from a condition that is only common among humans beyond reproductive age, so the frequency of the allele will likely not change much in the future because it is not influenced by natural selection. ✓

7 Natural Selection MCQ

Answer D

Correct. Heart disease is a progressive condition that does not affect most humans during their child-bearing years. The survival rate of individuals with the $APOE$–$\varepsilon 2$ allele in later life will not affect its frequency in the next generation. The statement that there is no cause-and-effect relationship between the $APOE$–$\varepsilon 2$ allele and fitness is a null hypothesis.

179. A scientist maintains a large population of the fruit fly *Drosophila affinis* in the laboratory. After 20 generations of allowing this population of about 1000 flies to mate randomly, 10 females and 10 males were isolated from the rest of the population. Both the larger population and the isolated population were maintained under identical conditions and allowed to mate randomly for another 20 generations. After the 20 generations, DNA sequencing was used to determine the genetic composition of each population.

Which of the following predicts the expected outcome of the experiment?

(A) The two populations will show significant genetic differences because the founder effect will result in significant losses in diversity in the smaller population. ✓

(B) The two populations will show significant genetic differences because there will be much higher rates of mutation in the smaller population.

(C) The two populations will not show any differences because they were reared under identical conditions.

(D) The two populations will not show any differences because they originated from the same gene pool.

Answer A

Correct. Genetic drift, in the form of the founder effect, is likely to have much greater consequences for the smaller population than for the larger, resulting in differences between the two, including, but not limited to, lower diversity in the smaller population.

180. The different species of finches on the Galapagos Islands are believed to have arisen as a result of natural selection acting on populations of finches that had experienced

(A) convergent evolution

(B) gene flow

(C) the bottleneck effect

(D) geographic isolation ✓

(E) hybrid sterility

7 Natural Selection MCQ

181. In a large, isolated population of an insect species, a specific gene locus has one dominant allele (A) and one recessive allele (a). The genotype frequencies of the gene were collected for ten generations, as shown in Table 1.

Table 1. Genotypic frequencies over ten generations in an insect species

	AA	Aa	aa
Generation 1	0.47	0.43	0.10
Generation 5	0.32	0.60	0.08
Generation 10	0.20	0.75	0.05

Which of the following could best account for the change in genotypic frequencies over the ten generations?

(A) Because the A allele is unstable, over the course of the study most of the A alleles mutated to become a alleles.

(B) The population is in Hardy-Weinberg equilibrium, which allows for changes in genotypic frequencies every generation.

(C) The population is not exhibiting random mating between individuals. ✓

(D) The population is becoming extinct because the a allele is disappearing.

Answer C

Correct. Nonrandom mating that preferentially led to mating between AA and aa individuals could result in the observed increase in Aa genotype frequencies.

182. A group of people whose ancestors immigrated to North America 200 years ago have certain allele frequencies that differ significantly from those in surrounding populations in the United States. Which of the following is the most likely explanation for the difference in allele frequencies?

(A) Mutation

(B) Independent assortment

(C) Reproductive isolation ✓

(D) Sex linkage

(E) Gametic isolation

7 Natural Selection MCQ

183. A scientist is studying the impact of mutagens on genetic variation in mice. In an experiment, the scientist creates two groups of twenty highly inbred mice. Because the mice are highly inbred, they are essentially identical to one another. One group is given food with an added substance that is known to be a mutagen. The second group is fed the same food without the mutagen. After several generations, both groups of mice are given a new type of food that is harder to digest than the previous food. After rearing five additional generations of the mice on this new food, the two groups are assessed for how efficiently this new food is digested.

Which of the following is the most reasonable prediction about the results of the experiment?

(A) Neither group of mice will show any change in the ability to digest the food because thousands of years are required to develop such a change.

(B) Both groups of mice will show an increase in the ability to digest the new food, because they experienced the same selective pressures.

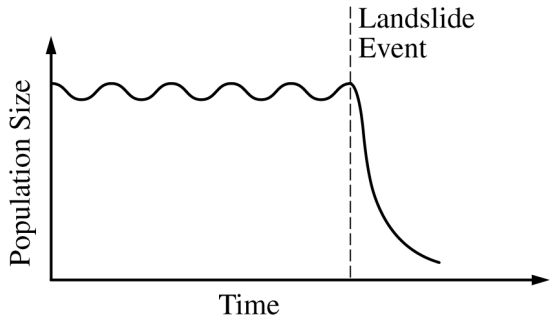
(C) The group of mice fed the mutagen will show a greater ability to digest the food, because the higher genetic variation within the group makes it more likely that some individuals will have a combination of genes better adapted to the new food. ✓

(D) The group fed the mutagen will show a decrease in the ability to digest the new food, because the mutagen prevented natural selection from acting on them.

Answer C

Correct. Inducing mutations provided the mouse population the variation necessary for a population to adapt to the new food.

184.



Undersea landslides can disrupt marine habitats by burying organisms that live on the ocean floor. The graph above shows the size of a population of a certain organism that lives on the ocean floor. The population was affected by a recent landslide at the time indicated on the graph. Which of the following best predicts how the population will be affected by the landslide?

(A) The surviving organisms will evolve into a new species.

(B) The reduced population will likely have allelic frequencies that are different from the initial population. ✓

(C) The population will adapt to deeper waters to avoid future landslides.

(D) The reduced population will have a greater number of different genes than the initial population.

7 Natural Selection MCQ

185. In addition to the ABO blood groups, humans have an MN blood type system in which the M and N alleles are codominant to one another. These alleles code for glycoproteins that act as antigens on the red blood cells. Scientists determined the frequencies of the M and N alleles in three localized and stable human populations in the Philippines. In Isabela, M and N alleles had a frequency of .8 and .21, respectively. Butuan had M and N allele frequencies of .82 and .18, respectively. Metro Manila had M and N allele frequencies of .55 and .51, respectively.

Which of the following would most affect the allelic frequencies of the populations on Isabela and Butuan?

(A) Migration of individuals between the Isabela and Butuan populations

(B) Immigration of individuals from Metro Manila into the two populations ✓

(C) Random mating within each population

(D) Increased population sizes over time in the Isabela population, but no change in the Butuan population

Answer B

Correct. Immigration of individuals from different populations can impact allelic frequencies. The Metro Manila population has a significantly different distribution of the M and Nalleles, so adding new individuals to the existing populations is likely to change the allelic frequencies in the populations.

186.

	Population 1		Population 2	
Year	Allele *R*	Allele *r*	Allele *R*	Allele *r*
1980	0.3	0.7	0.37	0.63
2010	0.0	1.0	0.75	0.25

The table shows the changes in allele frequencies of a specific gene in two populations of randomly mating small mammals after 30 years. The populations inhabit adjacent equatorial islands that have similar topography and climate. Which of the following is the most reliable conclusion that can be drawn from analysis of the data above?

(A) Genetic drift has occurred in population 1. ✓

(B) Population 2 is in Hardy-Weinberg equilibrium.

(C) Selection for allele r is occurring in both populations.

(D) The reduced frequency of allele R in population 1 will eventually lead to the extinction of population 1.

187. Retinitis pigmentosa is a rare genetic disorder that causes the breakdown of cells in the eye, resulting in vision loss. Worldwide, the occurrence of retinitis pigmentosa in the population is approximately 1 in $4,000$. However, on a small group of islands in the Atlantic Ocean, the occurrence is approximately 1 in 58.

Which of the following best explains how the occurrence of a rare, genetic disorder can be so much more prevalent on a small group of islands as compared with the rest of the world?

7 Natural Selection MCQ

(A) An unusually high number of individuals with the disease initially colonized the islands, and there has been little immigration to the islands since that time. ✓

(B) Individuals on the islands evolved adaptations to the vision loss, and because the disorder does not affect an individual's fitness, the disorder remains in the population.

(C) The population of individuals on the islands has greater genetic diversity compared with the worldwide population, resulting in a larger number of individuals with genetic disorders.

(D) The mutation rate is greater in the population of individuals on the islands than in the worldwide population, resulting in a greater incidence of rare disorders.

Answer A

Correct. The best explanation for the difference in prevalence is genetic drift, specifically a founder effect in which a population is founded by an initially small group of individuals with a large percentage of a particular trait. With very little immigration to the population, the initial percentage of the trait will remain the same over the generations.

188. Students observed the distribution of different color phenotypes in northern ravine salamanders (*Plethodon electromorphus*) before and after a spring flood. The data are shown in Table 1.

Table 1. Observed number of salamanders by phenotype before and after a spring flood

	Salamander Distribution before the Flood (n)	Salamander Distribution after the Flood (n)
Black phenotype	13	7
Dark-brown phenotype	25	14
Light-brown phenotype	6	3

Which of the following is an appropriate null hypothesis regarding the phenotypic frequencies of this population of salamanders before and after the flood?

(A) More black salamanders survived because they could better hide from predators on the dark muddy soil, so the frequency of the black phenotype significantly increased in the population after the flood.

(B) Fewer light-brown salamanders survived because they had a reduced chance of finding mates due to the high water levels, so the frequency of the light-brown phenotype in the population decreased after the flood.

(C) More dark-brown salamanders migrated to a new, drier area because of the flood and fewer of them remained in the original habitat, so the frequency of dark-brown salamander significantly decreased in the population after the flood.

(D) The proportions of black, dark-brown, and light-brown salamanders before and after the flood are not significantly different, and any observed differences are due to random chance. ✓

7 Natural Selection MCQ

Answer D

Correct. This is a null hypothesis that attributes any deviation in the phenotypic frequencies before and after flood to chance.

The three-spined stickleback (Gasterosteus aculeatus) is a small fish found in both marine and freshwater environments. Marine stickleback populations consist mostly of individuals with pronounced pelvic spines, as shown in Figure 1. Individuals in freshwater stickleback populations, on the other hand, typically have reduced pelvic spines, as shown in Figure 2.

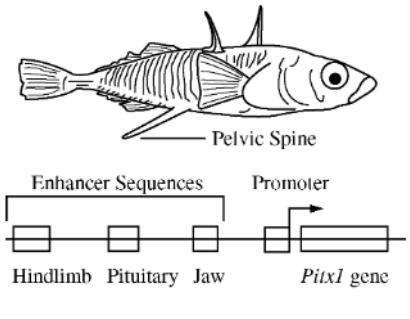


Figure 1. Marine stickleback

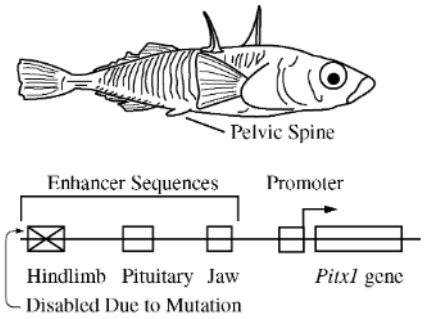


Figure 2. Freshwater stickleback

As represented in Figure 1 and Figure 2, the phenotypic difference between marine and freshwater sticklebacks involves Pitx1, a gene that influences the formation of the jaw, pituitary gland, and pelvic spine. Enhancer sequences upstream of the Pitx1 genetic locus regulate expression of the Pitx1 gene at the appropriate times and in the appropriate tissues during development. Previous studies have found that a mutation in the hindlimb enhancer interferes with the formation of a pronounced pelvic spine.

189. Which of the following explains a possible selective mechanism causing freshwater sticklebacks to typically have reduced pelvic spines?

(A) Reduced pelvic spines increase the likelihood of escaping predators in freshwater environments. ✓

(B) Reduced pelvic spines expand the range of foods that can be used as sources of energy in freshwater environments.

(C) Reduced pelvic spines result in males becoming effectively invisible to potential mating partners in freshwater environments.

(D) Reduced pelvic spines allow individual fish to pass through narrow waterways on their way to marine environments.

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7 Natural Selection MCQ

190. The California newt, *Taricha torosa,* lives in the coastal areas around Los Angeles. Which of the following is a valid null hypothesis relating fitness to survival of a bottleneck event in a coastal area where a small, isolated population of California newts resides?

(A) Only those salamanders with the lowest evolutionary fitness will be eliminated by the bottleneck effect, lowering the allelic frequencies of their traits.

(B) Only those salamanders with the highest evolutionary fitness will survive the bottleneck event, raising the allelic frequencies of their traits.

(C) Surviving the bottleneck event will be random, so any change in the allelic frequencies of the salamander population is not attributed to fitness. ✓

(D) Surviving the bottleneck event will be random, so there will be no changes to the allelic frequencies due to the bottleneck event.

Answer C

Correct. A bottleneck event is an example of genetic drift that randomly changes gene frequencies within a small population. Survival is a matter of chance, not fitness.

191. Members of one population of the striped stem borer insect engage in mating behavior in the early hours of the night, whereas members of a second population mate much later during the night. Crossbreeding between individuals from the two populations produces offspring that mate at an intermediate time.

Which of the following is the most likely cause of the different timing of mating behavior between the two populations?

(A) Genetic variation between the populations ✓

(B) The seasonal difference in day length

(C) The difference in learned behavior between the populations

(D) Gene flow between the populations

Answer A

Correct. Certain behavioral characteristics are encoded in the genome. The evidence that this is true for the striped stem borer insect is found in the crossbreeding experiment, which produced first-generation hybrid insects with a breeding time between the times of the two original populations.

7 Natural Selection MCQ

192. Cystic fibrosis is an autosomal (non-sex-linked) recessive genetic disorder in humans. In a certain geographical area, the incidence of cystic fibrosis is currently estimated at 1 in 377 live births. Which of the following is most likely closest to the frequency of the recessive allele in the human population in the area after 10 generations, assuming that the population remains in Hardy-Weinberg equilibrium?

(A) 0.003

(B) 0.052 ✓

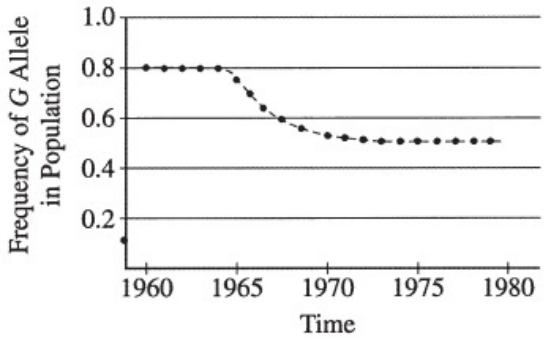
(C) 0.098

(D) 0.948

Answer B

Correct. The frequency of the recessive allele (q) in the population is determined by taking the square root of q^2 and is calculated as follows: $\frac{1}{377} = 0.00265 = q^2$, so $q = 0.051$, rounded to 0.052.

193. A moth's color is controlled by two alleles, *G* and *g*, at a single locus. *G* (gray) is dominant to *g* (white). A large population of moths was studied, and the frequency of the *G* allele in the population over time was documented, as shown in the figure below. In 1980 a random sample of 2,000 pupae was collected and moths were allowed to emerge.



Assuming that the population was in Hardy-Weinberg equilibrium for the *G* locus, what was the frequency of allele *G* in the gray moths that emerged in 1980?

(A) 0.33

(B) 0.50 ✓

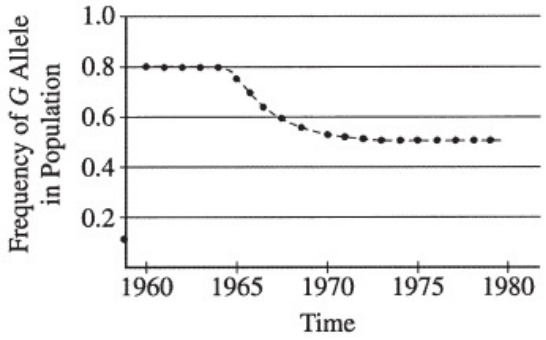
(C) 0.67

(D) 0.75

(E) 1.00

7 Natural Selection MCQ

194. A moth's color is controlled by two alleles, *G* and *g*, at a single locus. *G* (gray) is dominant to *g* (white). A large population of moths was studied, and the frequency of the *G* allele in the population over time was documented, as shown in the figure below. In 1980 a random sample of 2,000 pupae was collected and moths were allowed to emerge.



Assuming that the population was in Hardy-Weinberg equilibrium for the *G* locus, what percentage of the gray moths that emerged in 1980 was heterozygous?

(A) 0%

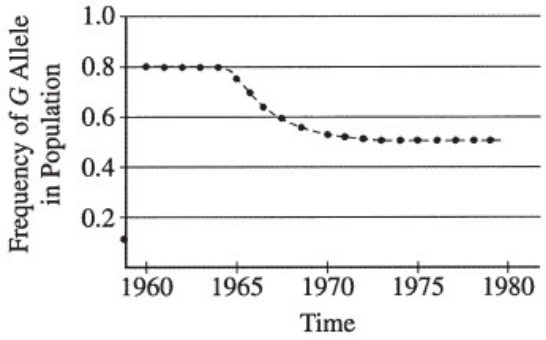
(B) 25%

(C) 33%

(D) 67% ✓

(E) 100%

195. A moth's color is controlled by two alleles, *G* and *g*, at a single locus. *G* (gray) is dominant to *g* (white). A large population of moths was studied, and the frequency of the *G* allele in the population over time was documented, as shown in the figure below. In 1980 a random sample of 2,000 pupae was collected and moths were allowed to emerge.



Assuming that the population was in Hardy-Weinberg equilibrium for the *G* locus, what percentage of moths in the natural population was white in 1962 ?

7 Natural Selection MCQ

(A) 2%

(B) 4% ✓

(C) 8%

(D) 20%

(E) 64%

For following group of questions first study the description of the situation and then choose the one best answer to each question following it and fill in the corresponding oval on the answer sheet.

A survey reveals that 25 percent of a population of 1,000 individuals have attached earlobes (are homozygous recessivefor the trait). For the following questions, assume that the population fits the parameters of the Hardy-Weinberg law.

196. What is the frequency of the recessive allele?

(A) $\sqrt{0.75}$

(B) 0.75

(C) $\sqrt{0.50}$

(D) $\sqrt{0.25}$ ✓

(E) 0.25

197. Unlike most natural populations, this population is best characterized in which of the following ways?

(A) There is genetic equilibrium. ✓

(B) There is gene flow.

(C) There is genetic drift.

(D) Mutations occur.

(E) Mating is nonrandom.

198. If p equals the frequency of the dominant allele and q equals the frequency of the recessive allele, which of the following terms represents the frequency of the individuals who show the dominant phenotype?

(A) p^2

(B) q^2

(C) $2pq$

(D) $p^2 + 2pq$ ✓

(E) $p^2 + q^2$

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7 Natural Selection MCQ

199. A population geneticist is conducting an investigation on a population of mice. The geneticist determines the genotypes for a particular trait of all of the mice in the population during generation 1 and then removes all of the mice with the genotype *aa*. The remaining members of the population are allowed to randomly mate with one another to produce generation 2 and future generations. A model of the investigation is shown in Figure 1. Each of the fifteen genotypes shown in generation 1 and generation 2 represents 100 mice.

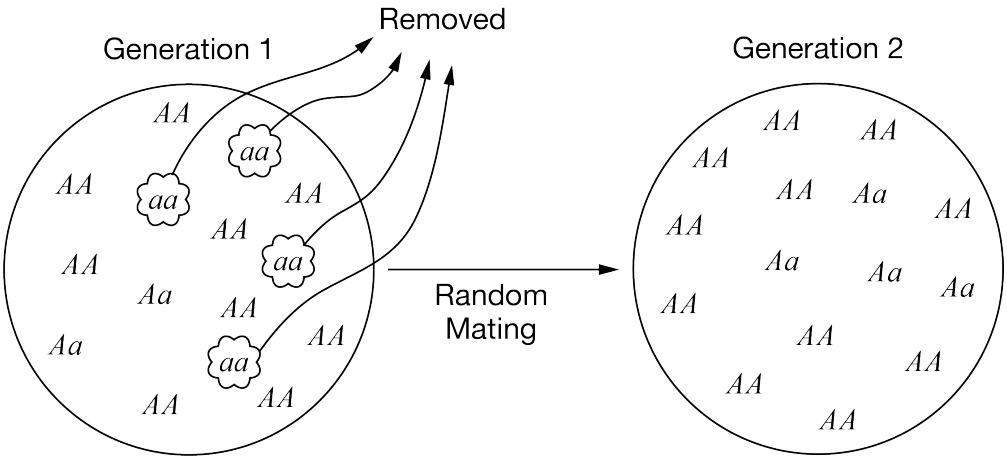


Figure 1. Model of effect of selective mouse removal on a population

Which of the following best illustrates the change in the frequencies of the *A* and *a* alleles from generation 1 to generation 2 and to future generations, assuming Hardy-Weinberg equilibrium is present after generation 1 ?

7 Natural Selection MCQ

(A)

Future generations will have the same frequencies of the A and a alleles as generation 2. Individuals with the aa genotype could be produced.

	Generation 1	Generation 2	Generation n
A	.67	.87	.87
a	.33	.13	.13

(B)

Future generations will eventually completely lose the a allele from the gene pool. There will not be any new individuals with the aa genotype.

	Generation 1	Generation 2	Generation n
A	.67	.87	1.0
a	.33	.13	0

(C)

Future generations will completely lose the a allele from the gene pool. There will no longer be individuals with the aa genotype.

	Generation 1	Generation 2	Generation n
A	.73	1.0	1.0
a	.27	0	0

(D)

Future generations will have fewer a alleles in the gene pool but will not completely lose the allele. There will be a few individuals with the aa genotype.

	Generation 1	Generation 2	Generation n
A	.73	.87	.95
a	.27	.13	.05

7 Natural Selection MCQ

Answer A

Correct. This answer is correct because in generation 1 there are 2,000 alleles and 1,000 a alleles. $\frac{2,000}{3,000} = .67 = A$ and $\frac{1,000}{3,000} = .33 = a$. In generation 2 and future generations the frequency of A is $\frac{2,600}{3,000} = .87$ and the frequency of a is .13 . The population will not completely lose the a allele because there are still heterozygotes in the population that could mate with one another. Random mating would produce a stable gene frequency similar to that in generation 2.

200. There are two species of elephant seals: northern elephant seals, *Mirounga angustirostris*, found off the western coast of North America from Baja California to the Gulf of Alaska, and southern elephant seals, *M. leonina*, found in the southern oceans. Hunting in the nineteenth century has been blamed for a drastic reduction in the northern elephant seal population. Even though the population has rebounded following restrictions on seal hunting, genetic variation is greatly reduced in the northern species compared to that in the southern species.

Which of the following conditions of Hardy-Weinberg equilibrium was most likely not met in the northern elephant seal population following overhunting?

(A) Absence of selection

(B) No new mutations

(C) Random mating

(D) Large population size ✓

Answer D

Correct. A small population typically shows decreased genetic variation, and if the population is reduced sufficiently, it will undergo a bottleneck, where some previously existing genetic variants will be lost, thus disturbing Hardy-Weinberg equilibrium.

201. Ellis-van Creveld syndrome is a recessive genetic disorder that includes the characteristics of short stature and extra fingers or toes. In the general population, this syndrome occurs in approximately 1 in 150,000 live births. In a particular isolated population, however, the incidence of this syndrome among live births is 1 in 500.

Assume that both the isolated population and the general population are in Hardy-Weinberg equilibrium with respect to this syndrome. Which of the following best describes the difference between the frequency of the allele that causes the syndrome in the general population and the frequency of the allele in the isolated population?

7 Natural Selection MCQ

(A) The frequency of the Ellis-van Creveld allele is 0.002 in the isolated population and 0.0000066 in the general population, which suggests that selection for this trait is occurring in both populations.

(B) The frequency of the Ellis-van Creveld allele is 0.0447 in the isolated population and 0.0026 in the general population, showing that the rate of genetic mutation is highest among individuals in the isolated population.

(C) The frequency of the Ellis-van Creveld allele is 0.002 in the isolated population and 0.0000066 in the general population, which demonstrates gametic incompatibility between the populations.

(D) The frequency of the Ellis-van Creveld allele is 0.0447 in the isolated population and 0.0026 in the general population, which suggests that genetic drift has occurred in the isolated population. ✓

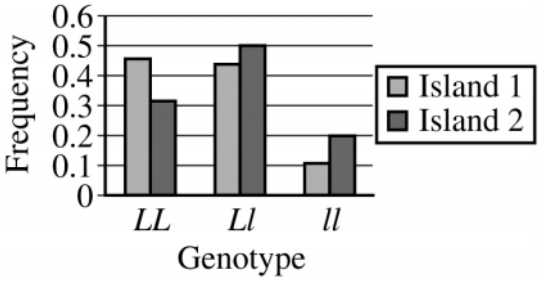
202. Scientists are studying several populations of finches on neighboring islands in the South Pacific. Previous genetic analysis has shown that a single gene controls tail-feather length in the finch populations and that the allele for long tail feathers (L) is dominant to the allele for short tail feathers (l). On two separate islands, the scientists recorded the number of finches with long tail feathers and the number of finches with short tail feathers. The results are shown in the table below.

Location	Phenotype	Number of Individuals
Island 1	Long tail feathers	1,582
	Short tail feathers	598
	Total count	2,180
Island 2	Long tail feathers	2,432
	Short tail feathers	1,110
	Total count	3,542

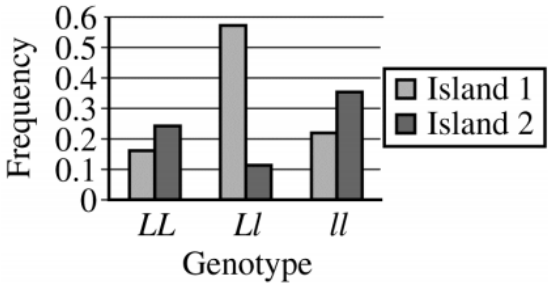
If the two finch populations are each in Hardy-Weinberg equilibrium and are isolated from each other, then which of the following graphs correctly displays the relative genotype frequencies?

7 Natural Selection MCQ

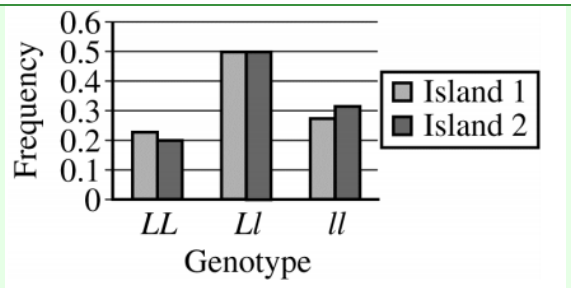
(A)



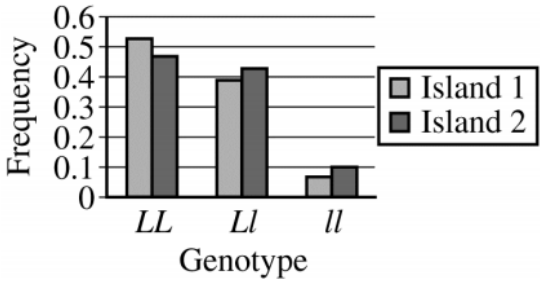
(B)



(C) ✓



(D)



203. In a certain flock of sheep, 4 percent of the population has black wool and 96 percent has white wool. Assume that the population is in Hardy-Weinberg equilibrium.

What percentage of the population is homozygous for white wool?

(A) 20%

(B) 40%

(C) 64% ✓

(D) 80%

(E) 96%

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204. Gaucher disease type 1 (GD1) is a recessive genetic disease that affects 1 in 900 individuals in a particular population. GD1 is caused by a mutation in the enzyme glucocerebrosidase.

Assuming the population is in Hardy-Weinberg equilibrium, calculate the frequency of the wild-type (nonmutant) allele for the enzyme glucocerebrosidase.

(A) 0.001

(B) 0.033

(C) 0.967 ✓

(D) 0.999

Answer C

Correct. Assuming the population is in Hardy-Weinberg equilibrium, the frequency of the dominant wild-type allele can be determined from the Hardy-Weinberg equation $p^2 + 2pq + q^2 = 1$, where p represents is the frequency of the dominant allele, q is the frequency of the recessive allele and $p + q = 1$. Since GD1 is a recessive condition, and it affects 1 in 900 individuals in the population, $q^2 = \frac{1}{900} = 0.001$. Therefore, $q = \sqrt{q^2} = \sqrt{\frac{1}{900}} = \frac{1}{30} = 0.033$. Since $p + q = 1$, $p = 1 - q$, the frequency of the dominant wild-type allele $p = 1 - 0.033 = 0.967$.

205. A large population of laboratory animals has been allowed to breed randomly for a number of generations. After several generations, 49 percent of the animals display a recessive trait (*bb*), the same percentage as at the beginning of the breeding program. The rest of the animals show the dominant phenotype, with heterozygotes indistinguishable from the homozygous dominants.

What proportion of the population is heterozygous (*Bb*) for this trait?

(A) 0.51

(B) 0.42 ✓

(C) 0.21

(D) 0.09

(E) 0.07

206. A large population of laboratory animals has been allowed to breed randomly for a number of generations. After several generations, 49 percent of the animals display a recessive trait (*bb*), the same percentage as at the beginning of the breeding program. The rest of the animals show the dominant phenotype, with heterozygotes indistinguishable from the homozygous dominants.

What is the frequency of allele *b* in the gene pool?

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(A) 0.70 ✓

(B) 0.51

(C) 0.49

(D) 0.30

(E) 0.07

207. Humans vary in their ability to taste the bitter chemical compound phenylthiocarbamide (PTC). The taster phenotype is dominant to the nontaster phenotype. Researchers tested the PTC taster phenotype of individuals from an isolated population. In the sample, 780 individuals were able to taste PTC and 70 individuals were not able to taste PTC.

Assuming the population is in Hardy-Weinberg equilibrium, what is the frequency of the non-taster allele?

(A) 0.08

(B) 0.29 ✓

(C) 0.41

(D) 0.71

Answer B

Correct. Since the population is in equilibrium, the frequency of the nontaster allele can be determined by using the Hardy-Weinberg equation, $p^2 + 2pq + q^2 = 1$, where p is the frequency of the dominant allele, q is the frequency of the recessive allele, and $p + q = 1$. In this equation, q^2 is the frequency of homozygous recessive nontasters. In this population the frequency of nontasters (q^2) is $\frac{70}{780+70} = 0.082$; therefore, the frequency of the recessive nontaster allele (q) is $\sqrt{0.082} = 0.29$.

208. In certain Native American groups, albinism due to a homozygous recessive condition in the biochemical pathway for melanin is sometimes seen. If the frequency of the allele for this condition is 0.06, which of the following is closest to the frequency of the dominant allele in this population? (Assume that the population is in Hardy-Weinberg equilibrium.)

(A) 0.04

(B) 0.06

(C) 0.16

(D) 0.36

(E) 0.94 ✓

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209. In a long-term study of a population of gray squirrels, researchers observed that most of the squirrels had gray fur. However, there were some individuals in the population with white fur (albino). Genetic analysis revealed that the albino condition is expressed by individuals who are homozygous recessive (gg). Over a ten-year period, the average frequency of albino squirrels in the population was 0.18. If the population is assumed to be in Hardy-Weinberg equilibrium, the average frequency of the dominant allele is closest to which of the following?

(A) 0.42

(B) 0.50

(C) 0.58 ✓

(D) 0.75

Answer C

Correct. If the frequency of *gg* individuals (q^2) is 0.18, then the frequency of the recessive allele (g) is equal to 0.42 $\left(q = \sqrt{.18}\right)$. Subtraction provides the frequency of the dominant allele (*p*): $1 - 0.42 = 0.58$.

210. In addition to the ABO blood groups, humans have an MN blood type system in which the M and N alleles are codominant to one another. A group of scientists found that in one population the frequency of the M allele was 0.82, and the frequency of the N allele was 0.18.

Assuming that the population is in Hardy-Weinberg equilibrium, which of the following is the expected frequency of individuals with the genotype MN?

(A) 0.148

(B) 0.295 ✓

(C) 0.640

(D) 0.672

Answer B

Correct. If the population is in Hardy-Weinberg equilibrium, the frequency of heterozygotes is determined by the expression $2pq$, where p is the frequency of one allele and q is the frequency of the other. The calculation would be $(2)(0.82)(0.18) = 0.295$.

211. Nine percent of a population is homozygous recessive (*aa*) at a certain locus. Assuming that the population is in Hardy-Weinberg equilibrium, which of the following is closest to the frequency of the recessive allele (*a*) ?

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(A) 0.09
(B) 0.27
(C) 0.30 ✓
(D) 0.90
(E) 0.97

212. Which of the following statements best explains how a condition of Hardy-Weinberg equilibrium results in a population that exhibits stable allele frequencies (i.e., a nonevolving population) ?

(A) Large populations are not subject to natural selection.
(B) Random mating prevents gene flow from changing allele frequencies.
(C) Without migration or mutation, new alleles cannot be introduced to the population. ✓
(D) In the absence of selection, allele frequencies in a population will not change.

Answer C

Correct. One of the conditions of Hardy-Weinberg equilibrium is that there is no gene flow and no net mutation, which prevent the introduction of new alleles and thus new phenotypes.

213.

Bacteria	P M V V G H E Y V G E V V G I G Q
Yeast	P L V G G H E G A G V V V G M G E
Corn	P R I L G H E A G G I V E S V G E
Horses	P V I A G H E A A G I V E S I G E
Sheep	P M V L G H E A S G T V V K V G S

Figure 1. Amino acid sequences of certain dehydrogenase enzymes in different species

Portions of the amino acid sequences of some dehydrogenase enzymes found in bacteria, yeast, corn, horses, and sheep are shown in Figure 1. Each letter represents a different amino acid in the sequence.

Based on Figure 1, which of the following conclusions about the enzymes is most likely correct?

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(A) Corn and horse dehydrogenase enzymes arose separately during each organism's evolution and now exhibit strong sequence similarity as a result of their similar catalytic roles.

(B) These enzymes all evolved from a common prokaryotic dehydrogenase enzyme, so they exhibit regions of sequence similarity. ✓

(C) Any evolutionary relationship is unlikely, because there is little sequence similarity between these enzymes.

(D) No conclusion can be drawn based on amino acid sequence, because there are twenty amino acids and only four nucleotides.

Answer B

Correct. The data in Figure 1 indicate only random amino acid differences in the dehydrogenase enzymes of the five diverse species, including a bacterium. This is expected if all enzymes evolved from the enzyme of a prokaryotic ancestor.

214.

Representative Vertebrate	Primary Nitrogenous Waste Product		
	Ammonia	Urea	Uric Acid
Fish	X		
Tadpole	X		
Frog		X	
Reptile			X
Bird			X
Mammal		X	
Property of Nitrogenous Waste Product			
Solubility in Water	High	High	Low
Toxicity	High	Medium	Low

The table above shows the types and properties of nitrogencontaining wastes produced by several vertebrates. Which of the following is the best evolutionary explanation of the data?

(A) The data support convergent evolution for nitrogen excretion in organisms occupying dramatically different environmental niches.

(B) The nitrogenous waste excreted by each vertebrate is the result of evolutionary adaptations that have decreased vulnerability to predators.

(C) Nitrogen homeostasis in terrestrial vertebrates reflects an adaptation for more frequent and higher-volume urination than that occurring in aquatic vertebrates.

(D) Ammonia secretion requires a large volume of water and was therefore selected against in terrestrial vertebrates. ✓

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215. Which of the following pieces of evidence best supports the hypothesis that birds and crocodilians (crocodiles and alligators) are more closely related to each other than they are to other organisms?

(A) Modern birds and crocodilians utilize different niches within the same geographical region.

(B) Like turtles and snakes, modern birds and crocodilians build nests in which they lay shelled eggs.

(C) The skulls of birds and crocodilians have an opening that is absent in the skulls of other reptiles. ✓

(D) Fossil evidence indicates that both modern birds and crocodilians originated during the Jurassic Period.

216. Students analyzed several photosynthetic eukaryotic organisms in the laboratory to determine their phylogenetic relationships. The organisms were *Spirogyra* (a green alga), a moss, a spruce tree, and an apple tree. Table 1 compares several characteristics in the organisms (+ indicates the trait is present, − indicates the trait is absent).

Table 1. Characteristics present in four photosynthetic species

	Green Alga	Moss	Spruce Tree	Apple Tree
Cell wall composed of cellulose	+	+	+	+
Vascular tissue	−	−	+	+
Roots	−	−	+	+
Seeds enclosed in fruit	−	−	−	+
Flagellated sperm cells	−	+	−	−

Which of the following rows of data listed in table 1 best supports the possibility of a common ancestor for the organisms listed there?

(A) Vascular tissue

(B) Seeds enclosed in fruit

(C) Cell wall composed of cellulose ✓

(D) Flagellated sperm cells

Answer C

Correct. The presence of a cell wall composed of cellulose is a characteristic of all four organisms and suggests a common ancestor.

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217. The amino acid sequence of cytochrome *c* is exactly the same in humans and chimpanzees. There is a difference of 13 amino acids between the cytochrome *c* of humans and dogs, and a difference of 20 amino acids between the cytochrome *c* of humans and rattlesnakes.

Which of the following statements is best supported by these data?

(A) Rattlesnakes apparently gave rise evolutionarily to the dog, chimpanzee, and human.

(B) Cytochrome *c* apparently has an entirely different function in rattlesnakes than in mammals, which explains the difference in the number of amino acids.

(C) Cytochrome *c* is not found universally in animals.

(D) Cytochrome *c* from a rattlesnake could function in a dog, but not in a chimpanzee.

(E) The human is apparently more closely related to the chimpanzee than to the dog or rattlesnake. ✓

218. Scientists want to determine the age of four fossil specimens. They have analyzed the fossils and gathered the data shown in Table 1.

Table 1. Data collected from four fossil specimens

	Specimen 1	Specimen 2	Specimen 3	Specimen 4
Amino acid differences*	5	21	18	11
Percent of original carbon-14 remaining	38	73	25	11
Composition	Sandstone	Sandstone	Limestone	Shale
DNA analysis	35% Guanine	28% Guanine	12% Guanine	29% Guanine

*amino acid differences of a conserved protein compared with a fifth fossil specimen

Which of the specimens indicated in the table is the oldest?

(A) Specimen 1

(B) Specimen 2

(C) Specimen 3

(D) Specimen 4 ✓

Answer D

Correct. Carbon−14 is incorporated into living organisms based on the percent of carbon−14 in the

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atmosphere. After the organism dies, carbon−14 breaks down by radioactive decay. Therefore the specimen with the lowest percent of the original carbon−14 remaining would be the oldest.

219. Which of the following explains the most accurate way to determine common ancestry?

(A) Observations of behavior provide evidence of common ancestry for organisms.
(B) Observations related to morphological homologies among organisms provide evidence of relatedness.
(C) Examination of fossil records provides evidence of organisms' evolution over time.
(D) Comparisons of DNA nucleotide sequences provide evidence of the origin of the organisms' lineages. ✓

A scientist used the amino acid sequence of cytochrome *c* in different species to consider evolutionary relationships.

The data below summarize the number of differences in the amino acid sequences of cytochrome *c* found in selected species.

Species Compared	Number of Differences
Humans–Chimpanzees	0
Humans–Rhesus monkeys	1
Humans–Horses or donkeys	7
Humans–Cows or pigs or sheep	7
Humans–Rabbits	7
Mammals–Birds and reptiles	10–15
Mammals–Fish	18–20
Animals–Plants	45–48

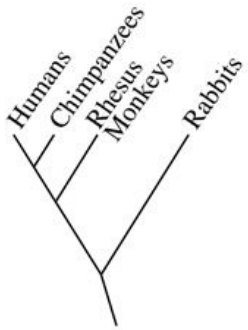
220. Interpretation of the data supports which of the following statements?

(A) Rhesus monkeys are more closely related to rabbits than they are to horses.
(B) Horses and cows have identical amino acid sequences in their cytochrome *c*.
(C) Humans are more closely related to rabbits than they are to rhesus monkeys.
(D) Plants and animals have no similarities at all.
(E) Mammals are more closely related to reptiles than they are to fish. ✓

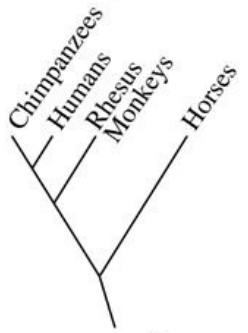
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221. Which of the following phylogenetic trees can be supported by the data?

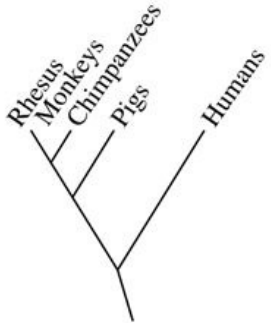
I.



II.



III.



(A) I only
(B) II only
(C) III only
(D) I and II only ✓
(E) I, II, and III

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222. Table 1 summarizes selected characteristics of organisms from each of the three domains of life.

Table 1. A Summary of Selected Characters across Domains of Life

Domain	Bacteria	Archaea	Eukarya			
Major divisions	Bacteria	Archaea	Protista	Fungi	Plantae	Animalia
Cell type	Prokaryotic	Prokaryotic	Eukaryotic	Eukaryotic	Eukaryotic	Eukaryotic
Genetic code bases	Adenine, thymine, guanine, cytosine	Adenine, thymine, guanine, cytosine	Adenine, thymine, guanine, cytosine	Adenine, thymine, guanine, cytosine	Adenine, thymine, guanine, cytosine	Adenine, thymine, guanine, cytosine
Genetic material	DNA	DNA	DNA	DNA	DNA	DNA
Metabolic pathway(s)	Glycolysis	Glycolysis	Glycolysis, citric acid cycle, oxidative phosphorylation	Glycolysis, citric acid cycle, oxidative phosphorylation	Glycolysis, citric acid cycle, oxidative phosphorylation	Glycolysis, citric acid cycle, oxidative phosphorylation
Mode of nutrition	Autotrophic, heterotrophic	Autotrophic, heterotrophic	Autotrophic, heterotrophic	Heterotrophic	Autotrophic	Heterotrophic
Number of cells	Unicellular	Unicellular	Unicellular; some colonial	Multicellular	Multicellular	Multicellular

Which of the following sets of data provides evidence that best supports common ancestry for organisms in all three domains?

(A) There are autotrophic and heterotrophic organisms found in Bacteria, Archaea, and Eukarya.

(B) There is a progression of pathways in organisms found in Bacteria, Archaea, and Eukarya.

(C) Organisms in Bacteria, Archaea, and Eukarya each have adenine, thymine, guanine, and cytosine bases forming their DNA. ✓

(D) There is a progression of cellular organization in organisms found in Bacteria, Archaea, and Eukarya.

Answer C

Correct. Genetic information is contained in DNA in all three domains, and the same nucleotides are found in the DNA in all three domains.

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223. Researchers studying the evolutionary relationships of three species of snails collected the data shown in Table 1.

Table 1. Comparison of selected characteristics of three snail species

	Species A	Species B	Species C
Location of Fossils	North America	North America	Europe
Shell Orientation	Counterclockwise	Counterclockwise	Clockwise
Shell Composition	Calcium Carbonate	Calcium Carbonate	Calcium Carbonate
Trophic Level	Carnivore	Carnivore	Detritivore
DNA Analysis	$\%\ \mathrm{A{+}T} = 43.2$	$\%\ \mathrm{A{+}T} = 43.4$	$\%\ \mathrm{A{+}T} = 43.0$
	$\%\ \mathrm{G{+}C} = 56.8$	$\%\ \mathrm{G{+}C} = 56.6$	$\%\ \mathrm{G{+}C} = 57.0$

Which of the following data sets is most likely to provide reliable information about the evolutionary relationships among these three snail species?

(A) Shell composition and shell orientation

(B) Trophic level and DNA analysis

(C) Location of fossils and shell orientation ✓

(D) Shell composition and location of fossils

Answer C

Correct. Of the characteristics given, fossil location and shell orientation are the most informative. Both species A and B are found on the same continent and have the same shell orientation, suggesting a closer relationship than either has with species C.

224. The wing of a bat, the flipper of a whale, and the forelimb of a horse appear very different, yet detailed studies reveal the presence of the same basic bone pattern. These structures are examples of

(A) analogous structures

(B) homologous structures ✓

(C) vestigial structures

(D) balanced polymorphism

(E) convergent ‘evolution

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225. Catalase is a protein enzyme that catalyzes the breakdown of hydrogen peroxide (H_2O_2) into water and oxygen gas in the cells of many living organisms. Students used an online database, which used the structure of catalase from different species, to generate the phylogenetic tree shown in Figure 1.

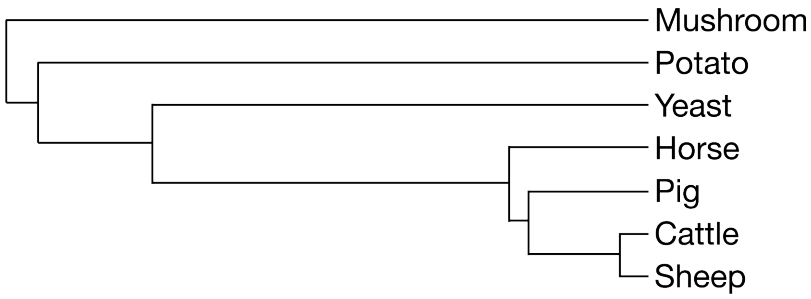


Figure 1. Relationships of selected organisms based on catalase structure

Which of the following student claims is best supported by the data provided?

(A) Mushrooms and potatoes are most closely related, since the node they share gave rise to all of the other organisms.

(B) Cattle are equally related to pigs and sheep, because the phylogeny places cattle between pigs and sheep.

(C) Yeast and mushrooms are the most closely related, because they are both fungi.

(D) Cattle and sheep are separated by only one node, and they are separated from the other organisms by more than one node. ✓

Answer D

Correct. In a phylogeny the number of nodes separating organisms indicates how closely related they are.

226. Lobe-finned fishes were present in the oceans of the world approximately 400 million years ago. The first tetrapods (vertebrates that had limbs and could move on land) date to about 365 million years ago. One hypothesis states that early tetrapods evolved from lobe-finned fishes. Which of the following is the best plan for testing the hypothesis?

(A) Determining the number of fins found on lobe-finned fishes from different geographical locations

(B) Measuring the proportion of radioisotopes found in fossils of early tetrapods

(C) Comparing the arrangements of bones in the fins of lobe-finned fishes and limbs of the earliest tetrapods ✓

(D) Sequencing DNA isolated from fossils of lobe-finned fishes to determine the size of the genome

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227. Hemoglobin is a highly conserved protein used by all mammals to transport oxygen. Table 1 was constructed by comparing the amino acid sequence in the hemoglobin molecules from five different mammal species.

Table 1. Hemoglobin amino acid differences among five different mammal species

	Species 1	Species 2	Species 3	Species 4	Species 5
Species 1	–	21	11	13	5
Species 2		–	18	17	20
Species 3			–	3	10
Species 4				–	12
Species 5					–

Which of the following best describes the importance of the number of amino acid differences indicated in Table 1 ?

(A) They reflect the different environments where the organisms were captured.

(B) They reflect the number of base pair changes in the DNA. ✓

(C) They reflect random changes due to translation errors.

(D) They reflect the number of oxygen molecules each hemoglobin molecule carries.

Answer B

Correct. Differences in the amino acid sequences are due to differences in the sequences of the DNA nucleotides of the hemoglobin genes and can be used as evidence of relatedness.

228. The placement and development of two bones of the reptilian jaw are very similar to that of two bones of the mammalian middle ear, which probably evolved from the reptile bones. The similarity of these bones is an example of

(A) structural homology ✓

(B) speciation

(C) vestigial structures

(D) convergent evolution

(E) adaptive radiation